

Murray Stahl: Selected Writings (reader draft)

Generated by `book/scripts/export_reader_markdown.py`. Do not hand-edit this file — edit chapter sources and re-run.

Murray Stahl: Selected Writings

A **reader's edition** compiled from Horizon Kinetics research publications, compendia, letters, and attributed interviews.

Copyright and attributions

Unless otherwise noted in chapter headnotes, underlying research reports, compendium sections, and letters are © **Horizon Kinetics LLC** (and, where marked in originals, related entities). This edition **does not reproduce proprietary PDFs** as embedded files; it presents **edited plain-text assemblies** with **source credits** (title, date, product line, filename) so readers can locate originals in the collector's vault.

Contributors named in originals (for example co-authors and Horizon Kinetics research signatories) are credited in each chapter headnote.

Third-party data and trademarks cited in the originals (e.g., index providers, company names, data vendors) remain the property of their respective owners.

Video and interview material is listed in Part VIII with links to the publisher-hosted pages; transcripts are placeholders until you add Stahl-only answers or third-party permissions as applicable.

You (the editor/compiler) may add an explicit copyright line for *selection and arrangement* in this reader's edition if you publish outside personal use.

Edition

See `book/EDITION.md` for versioning and how to roll forward new quarterlies.

Source volume — *Compilation of Murray Stahl's Writings*

This reader's edition is assembled primarily from **individual Horizon Kinetics PDFs** in the vault (`hk_pdfs/*.pdf`), each cited in chapter headnotes.

You also maintain a **single compiled PDF** in the same vault:

- `830350067-Compilation-of-Murray-Stahl-s-Writings.pdf` (~6.8 MB in the current vault copy)

Role in this project

- The compilation is `book_candidate=yes` in `book/sources/corpus_inclusion_manifest.csv` (tier thematic_essay) so it is **never dropped** from the corpus audit.
- A **machine extract** for search, diffing, and grep lives at:
`book/build/text/830350067-Compilation-of-Murray-Stahl-s-Writings.txt`
Regenerate with:
`python book/scripts/extract_pdf_text.py "830350067-Compilation-of-Murray-Stahl-s-Writings.pdf"`

Editorial expectation

- The compilation will **overlap** many pieces that already appear as separate chapters. It is a **convenience / provenance / completeness** anchor, not a second pass of the full book body. When a chapter's headnote cites a specific compendium or commentary PDF, that file remains the **primary** line of attribution; this volume is the **shelf copy** that bundles much of the same material under one filename.

If the compilation is updated (new edition from the publisher or a re-export), replace the PDF in the vault, re-run `audit_corpus_inclusion.py` and the extract command above, then spot-check `99-chronological-bibliography.md` (regenerate via `write_bibliography_md.py`).

Preface — How This Edition Was Built

This manuscript is a **reader's edition**, not a substitute for the original publications. The aim is to gather Murray Stahl's public and firm-attributed analytical writing in one continuous reading experience: essays, letters, commentaries, and (in later parts) interview answers and transcribed remarks, each with a clear **source line** so you can return to the primary PDF or web file.

The underlying archive lives alongside this book folder: hundreds of Horizon Kinetics PDFs, plus selected downloads mirrored from the firm's website and, where necessary, the Internet Archive when a live link was broken but a historical file could be recovered. A **single compiled PDF**, 830350067-Compilation-of-Murray-Stahl-s-Writings.pdf, also sits in the vault; it overlaps many chapters but is kept **in the audited corpus and extracted** as a reference volume (see manuscript/00-compiled-murray-stahl-writings-source.md). Text was extracted mechanically and then **lightly edited for readability**—running headers, page numbers, and obvious extraction artifacts—without changing the substance of the arguments.

Co-authored and firm-authored material. You instructed that co-authored pieces may be included. Where a chapter is a "Founders' Letter" or similar, the voice is often first-person plural and reflects Horizon Kinetics as a firm. Those chapters remain in the edition because they are part of the published record Stahl co-authors; headnotes state the genre plainly.

Interviews and video. Interview chapters will reproduce **Stahl's answers only**, with the interviewer and publication named. Video appearances are treated the same way once transcribed; embed URLs are collected separately so transcripts can be verified against the recording.

What this edition does not claim. It does not claim completeness relative to every paywalled magazine, every conference panel, or every private communication. It claims fidelity to the texts that could be located in the archive at the time of assembly, and a transparent method for adding more as they surface.

The **step-by-step assembly procedure**—extract, curate, outline, draft, index—is written out in book/00_STEPS.md so the project can continue methodically even if work pauses for months between sessions.

The first chapter proper begins immediately after this preface: a short Murray's Musings piece on bonds in an era of suppressed yields—a useful on-ramp because it is self-contained and illustrates how later, longer quarterlies will be presented.

Chapter 1 — The Transformation of the Fixed Income Asset Class

Source. Murray's Musings, published in the *Fixed Income & Cryptocurrency Compendium*, Horizon Kinetics LLC, © 2020.

PDF in vault. TheTransformationOfTheFixedIncomeAssetClass.pdf (parent hk_pdfs/ folder).

Extract used for assembly.

book/build/text/TheTransformationOfTheFixedIncomeAssetClass.txt

Minor presentation edits only: removed repeated compendium running heads, page banners, and page-break markers.

The Transformation of the Fixed Income Asset Class

Fixed income is an investment that is a type of loan. A borrower is obligated to make payments of a predetermined amount, and the principal must be paid on the maturity date. The legal predetermination of the payments is the origin of the term fixed. In other words, fixed means predetermined. This legal definition will not change.

An exceedingly low level of interest rates changes the character of this type of investment in extraordinary and not so obvious ways. It should be evident that a fixed income investment that pays a coupon of less than 2% is, both before and after taxes, a negative real (after inflation) return instrumentality. For a New York City resident of means, even a fixed income instrument that yields 4% or less, is a negative real return instrumentality. Although there is surprisingly little discussion of this reality, it should be self-evident to investors.

It is less obvious—although nonetheless true—that bonds, in this historically low interest rate period, if sold short, are actually incredibly inexpensive put options. An example will serve to illustrate this point. Consider that the Bank of America 4.44% bonds due January 20, 2048 are now priced, on a yield-to-maturity basis, at 3.31%. They trade well above par value and at maturity must be worth par value. Prior to maturity, which is almost 30 years from now, a variety of circumstances could occur in which these bonds could be worth considerably less than par.

If, for instance, U.S. government bonds for some reason yield 5% instead of the present level of 2.2%, the Bank of America bonds would probably yield 7.45%—even assuming that credit quality does not deteriorate.¹ Quite obviously, the bonds would trade well below par value.

Hence, two possibilities exist for these Bank of America bonds. The first is that interest rates will never increase. As a result, the Bank of America bonds will not be priced at par value until the maturity date of January 20, 2048. The short seller will simply lose the 3.31% annual yield to maturity. A short seller in the 50% tax bracket (Federal, State, and local) will effectively lose half that sum after the benefit of tax deductions. Consequently, one would pay 1.66%—half of 3.31%—for a long-term put option on interest rates. Which is incredibly cheap—just try to price a multi-year put option on long-term interest rates with a broker-dealer.

The other possibility is that interest rates do increase. In that event, the bonds might trade substantially below par value—depending upon how high interest rates rise. These bonds are effectively a put option for which the premium is the yield to maturity. The potential loss for a bond (absent default), or in this case for a near-30-year put option on the bond, is limited to the yield to maturity. In other words, the loss is limited to the amount of the current price premium for the bond, just as for the option premium for a conventional put option.

This similarity to a put option is problematic on a broader systematic basis, because it changes the incentives for investing in bonds. Consider the matter from the perspective of an investor who might purchase a U.S. Treasury bond due in 2048 that has a yield to maturity of 2.23%. What if, alternatively, that bond were used as collateral to establish a small short position in a Bank of America bond, say a short equal to 10% of the long position. The investor puts \$100 into the U.S. Treasury bond yielding 2.23%, then takes \$10, or 10%, shorting that amount of Bank of America debt for which they must pay a yield to maturity rate of 3.31%. In terms of that investor's new risk/reward profile, the first aspect of this hedged position is that the net yield on it is actually a positive 1.9% carry.

An investor considering such a trade must contemplate whether or not it is better to sacrifice 33 basis points of annual income in exchange for the possibility that the creditworthiness of Bank of America might deteriorate. If Bank of America creditworthiness deteriorates, as occurred in 2008, the bank's debt might trade at 50% of par value, as it did at that time. A 10% short position with a loss of possibly 60% in price would produce a portfolio-level trading gain of 6%.

One might argue that such a trade is outside of the ambit of a conventional bond investor and therefore unlikely to be made. But it is not outside of the ambit of an investor who would be inclined to sell short Bank of America common stock. The shares now yield 2.17%, so that is a cost of the short sale. Moreover, the shares can appreciate. Thus far, in 2019, the shares have increased in value by 32.67%. That would have been the loss to a short seller.

Why might a short-seller have been attracted to the Bank of America shares? The net income for the first nine months of 2019 was \$19.25 billion—a decline of 2.07% versus the comparable period in 2018, when it was \$19.657 billion.

What if a hypothetical short seller were sufficiently astute to predict that in 2019, the net income of Bank of America would decline. This hypothetical short seller would have predicted, as well, the bank's difficulty with overnight borrowing in September of that year. Although this investor might reasonably have anticipated that Bank of America shares would decline, what could not have been reasonably anticipated is that its stock would in fact appreciate by 32.67%.

Establishing a hedge against a banking-related issue with systemic dimensions can be very expensive, carry costs aside, in terms trading risk, until the problem finally manifests itself. Selling short a bank bond substantially eliminates the trading or directional risk, since the bond will ultimately mature at par value. That is to say, if an investor is worried about possible banking risks, short sale of bank stock is too blunt an instrument to use as a hedge, there being many circumstances in which bank shares could rise considerably in price, creating an extremely expensive hedge.

Surveying Both Idiosyncratic and Systemic Risks in Low-Yield Bonds

There exist many long-term bonds that, in principle, have high price convexity paired with extraordinarily low yields to maturity. One example might be San Francisco City & County Water 6.95% due November 1, 2050, which trades at a price of 159.21 with a 3.22% yield to maturity. Apart from higher interest rates, variables that could cause the bonds to decline are fluctuations in the creditworthiness of the city or an earthquake that could severely damage the water system, requiring extensive repairs.

All U.S. bonds trade in relation to Treasury bond yields. Yet, with less than a 100-basis point spread over Treasuries, it is clear that municipal bonds, project finance bonds, or corporate bonds have unique risks that are not being priced into them. For example, consider the Comcast 4.00% Notes due November 1, 2049, trading at 110.82 with a 3.41% yield to maturity. Major changes are clearly evident in Comcast's delivery of content to subscribers and in the company's ability to sell advertising. Although no investor is in a position to predict the financial conditions of Comcast over the 30-year life of this bond, the valuation is predicated upon the belief that there is no realistic possibility of a significant deterioration in its financial position.

The same statement might be made about AT&T, which is actually a fairly leveraged enterprise. Its 5.7% bonds due March 1, 2057—a 38-year life—are priced at 125.38 with a yield to maturity

of 4.32%.

Universities are now issuing debt, despite their huge endowments, because the cost of debt capital is so low, while the likelihood is high that the university will ultimately repay the debt in inflated dollars. As an example, Princeton University 5.70% bonds due March 1, 2039 trade at 139.32 with a yield to maturity of 3%. Similarly, Harvard has issued a 4.875% bond due October 15, 2040 now trading at 130.80 with a yield to maturity of 2.9%.

If one sells short the Princeton or Harvard bonds, all one needs for a substantial profit is a modest rise in interest rates. A rise in long-term Treasury yields of perhaps 150 basis points for any reason would cause the Harvard bonds to trade at par value. A 30-year U.S. Treasury now yields 2.2%; 150 basis points higher, and it would yield 3.72%. If the Harvard bond were to trade a yield spread of 115 basis points above the Treasury, it would be priced at par value almost 31 points lower. In fact, because any meaningful rise in interest rates would probably severely reduce the market value of the Harvard endowment, the assessment of Harvard's creditworthiness would probably be reduced as well, such that the spread to Treasuries would probably rise as well.

The State of Texas 5.52% bonds due April 1, 2039 are priced at 138 with a yield to maturity of 2.82%. Surely, one could envisage a diminution of creditworthiness if, for some reason, extraction of fossil fuels were to be prohibited in the United States. In a less extreme scenario, the same outcome might be achieved by a severe reduction in oil and gas prices. The bond market does not appear very concerned about this possibility, at least judging by the present yield of the State's debt.

If the production of coal were to be forbidden in the U.S., it would negatively impact the fortunes of Union Pacific Corporation (UNP). Union Pacific managed to issue a 4.375% bond due November 15, 2065 which now trades at 109.85 with a yield to maturity of 3.91%. In fact, even a severe recession could lower an investor's perception of the credit worthiness of Union Pacific. The company also issued 3.84% bonds due March 20, 2060 presently trading at 102.32 with a yield to maturity of 3.73%.

Altria Group, Inc. (MO) managed to sell 6.20% bonds due February 14, 2059, presently trading at 117.33 with a yield to maturity of 5.16%. The mere existence of such a bond is more than a statement about the long-term real rate of return on bonds. It is effectively a statement that large-scale smoking in the U.S. will continue past the year 2059.

Here is where it becomes yet more interesting.

The University of California issued 4.77% bonds due May 15, 2115, now priced at 126.36 with a yield to maturity of 3.75%. It is virtually a certainty that the person buying this bond will not live to the maturity date. Ohio State University managed to sell 4.80% bonds due June 1, 2111, now trading at 129.45 with a yield to maturity of 3.68%.

Guardian Life Insurance Company of America has issued 4.85% bonds due January 24, 2077. They trade at 124.87, a yield to maturity of 3.78%. A life insurance company is an interesting case relative to long-term bonds. If interest rates increase, it would surely have a negative impact on the market values of the debt. On the other hand, if rates decrease sufficiently from this point, the company might experience difficulty earning enough from its bond portfolio to pay its annuity liabilities. Viewed from the perspective of the bond buyer, the best-case scenario is that interest rates remain at present levels and neither materially increase nor decrease for many decades.

If an investor buys a 57-year bond from a life insurance company whose entire investment portfolio is comprised almost entirely of long-term bonds, and if one of the many long-term bonds in its investment portfolio loses value, then the collateral for your bond will lose value, as well.

These bonds are only a small sample of many that have a similar character. The common denominator is that the rate of compensation above that of Treasury bonds is less than 200 basis points per annum and sometimes even less than 100 basis points, despite the fact that many of the bonds have maturity dates far longer than the 30-year Treasury. The University of California bond will mature in 91-years or 3x the maturity of the longest-term U.S. Treasury bond. It does not appear that much, if any, marginal compensation is paid for the extra 60-year life of the bond. In the light of such pricing, there is no room for any rise in interest rates or any deterioration of credit quality. Yet, history demonstrates that such eventualities frequently occur.

The expense of selling such bonds short is small relative to the reward possible if history repeats itself. Ergo, such bonds and many others are now effectively low-cost, low-risk put options with heretofore unimaginably lengthy times to expiration.

If shorting, I prefer to sell short bonds—not stocks—because they are a much more reasonable hedge against uncertainty. There is a limit to how much money one can lose, and one knows what that limit is—as opposed conventional hedge instruments, in which the potential loss is infinite.

Q. I have a hypothetical question relative to the circumstance of a 75-year-old investor who expects to live ten additional years and who invests in long-term bonds that yield 4.8% with good credit, as with the University of California issue. Playing devil’s advocate, what would be wrong with that, if rates stay roughly the same over that period?

A. What would be wrong with that? If that investor has to live off the principal—not just the income—and the bond declines in market value, there would be plenty wrong with it. That person would be forced to sell the bond for money to live on, and the sale might not produce adequate funds. That person might die without enough money. An alternative risk is that this investor is fortunate enough to be in very good health and live longer than the expected additional 10 years, but with each passing year, the cost of living rises above the purchasing power of that bond’s coupon income. That investor would not be very happy with that outcome. That is what’s wrong with it.

End of chapter. Next: Chapter 2 — “The Achievement of Diversification” (chapter-02-the-achievement-of-diversification.md).

Chapter 2 — The Achievement of Diversification

Source. Murray’s Musings, *European Contrarian Report Compendium*, November 2020. Horizon Kinetics LLC © 2020.

PDF in vault. TheAchievementofDiversificationNovember2020.pdf

Extract used for assembly.

book/build/text/TheAchievementofDiversificationNovember2020.txt

Contributors (as in original). Murray Stahl, Chairman; Steven Bregman, President; Peter Doyle, Managing Director.

Presentation edits: removed compendium running heads, page banners, and the standard lengthy research disclaimer at the end of the PDF; merged words broken across line ends in the source extract.

The Achievement of Diversification

It would seem to be arithmetically self-evident that if various constituent members of an index have such low weightings as to be statistically irrelevant, then they cannot, arithmetically, play any role in diversification. Eighteen members of the S&P 500 index now have only a 1-basis point (0.01%) weight. There are 38 index constituents that have a 2-basis point weight, 41 that have a 3-basis point weight, 41 with a 4-basis point weight, and 46 at 5 basis points.

These 184 constituents of the 500 S&P Index are arguably statistically irrelevant. To go further, one could designate members with 6-basis point weights as also statistically irrelevant. If one draws the boundary of irrelevance in this manner, there are another 36 such positions, bringing the total number of statistical irrelevancies to 220, or 44% of the names in the index.

If one wishes to assert, as an issue of statistical irrelevance, that at least 50% of the S&P 500 names fall into this category, one has only to include the 7-basis point positions. There are 29 constituent members with that weight. Thus, the grand total of statistically irrelevant positions is 249, or 49.8% of the names in the S&P 500 index.

A summary of the index weightings of approximately half of the constituent members of the S&P 500 is shown in the following table.

Table 1: Selected S&P 500 Index Weightings

Number of companies	Weighting (basis points)	Total weight (basis points)
18	1 bp	18 bp
38	2 bp	76 bp
41	3 bp	123 bp
41	4 bp	164 bp
46	5 bp	230 bp
36	6 bp	216 bp
29	7 bp	203 bp
Subtotal	249 companies	1030 bp (~10.3% of index cap.)

Source: iShares

Roughly one-half of the number of corporations in the S&P 500 index have, in aggregate, 10.3% of the index's market capitalization.

Interestingly, there are only 17 positions with an 8-basis point weight, and another 28 9-basis point positions. If these positions, in addition to those listed above, were considered as statistically irrelevant, the total number of irrelevant positions would equal 295, or 59% of the index and would have a collective weight of 14.27%. One could continue in this manner, because there are 20 10-basis point positions.

As noted, there are 295 positions of 9-basis points weight or less that comprise 59% of the number of members of the S&P 500 index and 14.27% of its market value. To assess the importance of these firms to the index, assume that—for only the 295 companies in question—a recession occurs that is equivalent in severity to the recession of 2008. In this scenario, assume that these 295 companies as a group decline in share price by (37)%, which was the S&P 500 return for 2008. The result is a 37% loss on 14.27% of the index or (5.28)%—which is not a significant decline during a major recession.

This is hardly the same thing, though, as saying that these companies are unimportant businesses or are statistical irrelevancies within the U.S. economy. One way of understanding index weight versus economic impact is to select ten well-known firms with statistically immaterial S&P 500 weights and calculate their economic impact by a metric other than stock market capitalization. I chose these ten companies that are familiar to investors: Discovery, Inc., News Corp., Sealed Air, Newell Brands, Franklin Resources, Mohawk Industries; American Airlines, Carnival Corporation, Textron, and Lumen Technologies, otherwise known as CenturyLink, Inc.

Table 2: Ten Firms with Immaterial Weights in the S&P 500

Ticker	Name	Weighting
DISCA	Discovery, Inc. – Series A	3 bp
NWSA	News Corporation Class A	3 bp
SEE	Sealed Air Corporation	2 bp
NWL	Newell Brands	2 bp
BEN	Franklin Resources, Inc.	2 bp
MHK	Mohawk Industries, Inc.	2 bp
AAL	American Airlines Group, Inc.	2 bp
CCL	Carnival Corporation	3 bp
TXT	Textron Inc.	3 bp
LUMN	Lumen Technologies, Inc. (CenturyLink, Inc.)	3 bp

Source: iShares

Those ten firms collectively have a 25-basis point weight in the S&P 500. One might compare them to Apple Inc. (AAPL), which has a 6.45% index weight.² In other words, the weight of Apple in the index is 25.8x the weight of these ten well-known companies.

The metric I have chosen to assess these firms' economic impact is the number of their employees. This statistic is significant, because in a horrendous recession some of the employees presumably lose their employment. Together, these firms have 485,800 employees. In contrast, Apple has 137,000 employees.

Table 3: Number of Employees in Ten Firms with Immaterial Weights in the S&P 500

Ticker	Name	Number of employees
DISCA	Discovery, Inc. – Series A	9,000
NWSA	News Corporation Class A	28,000
SEE	Sealed Air Corporation	16,500
NWL	Newell Brands	49,000
BEN	Franklin Resources, Inc.	9,700
MHK	Mohawk Industries, Inc.	42,100
AAL	American Airlines Group	133,700
CCL	Carnival Corporation	120,000
TXT	Textron Inc.	35,000
LUMN	Lumen Technologies, Inc. (CenturyLink, Inc.)	42,800
	Total	485,800
AAPL	Apple, Inc.	137,000

Source: Company reports

Some argue that if the ten aforementioned firms were to have difficulties—and obviously some of these firms now face serious challenges—the economic impact would be heaviest from those firms with the largest number of employees. In industry terms, a 20% layoff rate from the diverse group with a 25-basis point collective weight would be far more damaging to the U.S. economy than if Apple were compelled to undertake a 20% layoff. If the object of diversification is to reflect economic impact, then surely one can argue that the weightings of an index should reflect the impact of a company on the economy. Market capitalization weights reflect valuation rather than economic impact.

Tesla, Inc. (TSLA) has a \$392 billion market capitalization and, viewed from that perspective, is by far the largest automobile company on the planet. General Motors (GM) has a \$52.7 billion market capitalization, and Ford Motor (F), \$32 billion. When Tesla enters the S&P 500 index, it will obviously overwhelm the weightings of GM and Ford. GM, in a reasonably good year, would build 7.7 million vehicles, Ford might produce 2.4 million vehicles, and Tesla, somewhat more than 400,000.

Thus, the question of the meaning of the term *diversification* arises. Diversification of the S&P 500 index is meant to reflect the U.S. economy, but the fact that the S&P 500 does not do so is plain to see. If diversification is based upon market capitalization, then diversification is effectively based upon valuation. The conclusion would logically follow that the index would have a bias in favor of higher valuation.

Valuation is not the only bias within a national stock market. Most large firms have become global multinational companies, regardless of their nation of origin. If the country of incorporation is a small one, then that country is probably financially irrelevant to a global multinational firm. The obvious example of this phenomenon is Switzerland. The large index members of the iShares MSCI Switzerland ETF (EWL) are surely not Swiss with respect to the geographic diversification of their businesses. The following table lists the ten largest positions in that index.

Table 4: iShares MSCI Switzerland ETF (EWL) — Ten Largest Positions

Ticker	Name	Weighting
NESN	Nestlé SA	21.07%
ROG	Roche Holding PAR AG	12.90%
NOVN	Novartis AG	9.73%
ZURN	Zurich Insurance Group AG	3.87%
ABBN	ABB Ltd.	3.70%
LONN	Lonza Group AG	3.56%
UBSG	UBS Group AG	3.47%
GIVN	Givaudan SA	3.04%
SIKA	SIKA AG	2.88%
CFR	Compagnie Financiere Richemont SA	2.82%
	Total	67.04%

Source: iShares

In 2019, Switzerland accounted for 1.26% of Nestlé's sales. For Roche, 1.94% of sales were attributed to Switzerland. The fact that these firms cannot possibly achieve an acceptable level of growth for their investors by confining their business activities to Switzerland should be apparent. For many years, Nestlé, Roche, and Novartis expanded outside of Switzerland. These three firms collectively represent 43.7% of the MSCI Switzerland index, which is still considered to be a diversified index. MSCI Switzerland is an index that cannot possibly be a successful investment without the success of Nestlé, Roche and Novartis.

The historical return of MSCI Switzerland, since inception on March 12, 1996, is 7.18% annually. Is this an acceptable return, with that level of concentration? The difficulty with weightings based on market capitalization is that eventually some company within the index will either become the best performing investment or, alternatively, achieve enormous size by raising capital via share issuance. There exists no objective limiting principle to reestablish diversification other than placing an arbitrary limit on the size of position. Most index providers would not allow a single position to become 25% or more of an index. Another reasonably popular limiting principle is to restrict the number of 5% positions to about 50% of the portfolio. All indexes, nevertheless, seem to have accumulated statistically irrelevant smaller holdings.

MSCI Switzerland has only 39 positions, therefore, concentration might be inevitable. The smallest position in the index is Banque Cantonale Vaudoise, with a 44-basis point weighting.

An alternative approach, if one wants to invest in Switzerland itself on a diversified basis, might be to purchase Swiss real estate companies, on the theory that the real estate is, after all, located

in Switzerland. Valuations of those companies must, therefore, reflect conditions in that country.

There are four primary publicly traded real estate companies in Switzerland: Swiss Prime Site, PSP Swiss Property, Allreal Holding and Mobimo Holding.

Table 5: Four Primary Traded Real Estate Companies in Switzerland

Ticker	Names	Market capitalization (CHF in billions)
SPSN	Swiss Prime Site AG	6.05
PSPN	PSP Swiss Property AG	5.15
ALLN	Allreal Holding AG	3.12
MOBN	Mobimo Holding AG	1.75

Source: Bloomberg

Of these four, only Swiss Prime Site, with a 74-basis point weighting, qualifies for inclusion in the MSCI Switzerland index. One could argue that these four real estate firms have collectively become a more representative and diversified Swiss market index than MSCI Switzerland, by virtue of the total number of properties they hold and of their diversification by geography and property type.

The following table lists the number of properties held by the four primary Swiss real estate companies.

Table 6: Holdings of Four Swiss Public Real Estate Companies

Ticker	Names	Number of buildings
SPSN	Swiss Prime Site AG	187
PSPN	PSP Swiss Property AG	160
ALLN	Allreal Holding AG	64
MOBN	Mobimo Holding AG	138
	Total	549

Source: Company filings

This table raises a profound question about the nature of indexation. If one wishes to invest in the nation of Switzerland, is this best achieved by building a concentrated portfolio of giant multinational firms that have proportionately little business activity in Switzerland? Or, alternatively, can one devise a weighting system among the four publicly traded Swiss real estate firms that have their entire investment located within Switzerland? The fact should be obvious that a portfolio of Swiss real estate firms is a superior means of investing in Switzerland.

The limiting factor is that if specific Swiss diversification is to be achieved in this manner, this diversification is not scalable. The aggregate market capitalization of these four Swiss real estate companies is only 16.07 billion CHF. Nestle, alone, has a market cap of 300 billion CHF. Mass-scale indexation must necessarily be global multinational investing, because only global-scale firms have sufficient equity liquidity to support indexation at the trillion-dollar level.

Concentrated—not diversified—investing can be rewarding during a period of declining interest rates and increasing globalization. If these economic variables are not dominant in the future, an entirely different form of indexation will probably take their place.

Original PDF includes Horizon Kinetics standard research disclosures and reproduction restrictions.

Next: Chapter 3 — Inflation and Economic Thought.

Chapter 3 — Inflation and Economic Thought

Source. Murray's Musings, *Contrarian Research Report Compendium*, November 2020. Horizon Kinetics LLC © 2020.

PDF in vault. `InflationandEconomicThought_Nov2020.pdf`

Extract used for assembly. `book/build/text/InflationandEconomicThought_Nov2020.txt`

Contributors (as in original). Murray Stahl, Chairman; Steven Bregman, President; Peter Doyle, Managing Director.

Presentation edits: removed compendium running heads and page banners where redundant; merged words broken across line ends in the source extract; omitted the lengthy standard research disclaimer at the end of the PDF.

Inflation and Economic Thought

Historians write about inflation in a very different way than economists.

Historians view inflation as a destabilizing force, whatever the cause. For instance, historians agree that a salient cause of demise of the Song Dynasty of China (960–1279) was the fact that it issued paper money, the first government to do so. The successor Yuan Dynasty lasted until 1368, and the inflation caused by fiat currency issuance is generally cited by historians as a factor in its demise. Similarly, the inflation caused by the influx of gold and silver into Spain from Spanish colonies in the New World in the 16th and 17th centuries is known to historians as “the price revolution.” The repercussions of the inflation caused by this dramatic rise in the supply of gold and silver undermined Spain’s international economic standing. There were other historical debasements, such as King Henry VIII of England continually reducing the gold and silver content of the coinage to pay for his lavish lifestyle. In British history, this is known as “the Great Debasement.”

None of these events or any of the many other historical debasements have seemed to attract the interest of even the greatest economists. Many financial experts regard John Maynard Keynes as the greatest economist of all time, yet in his magnum opus, *The General Theory of Employment, Interest and Money* published in 1935, the index shows only seven mentions of inflation in the entire book. Of these mentions, one simply states the fact of inflation in the Central Europe of the postwar era. The other references are largely definitional and used as foundations upon which the theory of employment is placed. Inflation basically is defined as the point beyond which monetary expansion does not produce increases in output. The word *debasement* is never used in the book. Keynes might be forgiven for such a view, since *The General Theory of Employment, Interest, and Money* was written in a period of history during which the challenge was to increase output and thereby increase employment.

In the magisterial three-volume biography of John Maynard Keynes by Robert Skidelsky, there are no references to inflation in the index to Volume I, only six references to inflation in the index to the 635-page Volume II, and a single reference in the index to Volume III, which is about as long as Volume II.³ Ironically, Keynes became very wealthy by speculating on the decline of the German currency in the aftermath of the First World War. In other words, Keynes became wealthy because of German inflation and his understanding of it. He was very interested in inflation as an investor. He appears not to have been vitally interested in the subject as a scholar.

A similar phenomenon is observed in *A Monetary History of the United States 1867–1960* by Milton Friedman and Anna Jacobson Schwartz.⁴ Since the book is about monetary issuance, one might expect some discussion of the subject of debasement. The word is not mentioned once in the index to this book that is 700-pages long excluding the appendixes. There are five references to inflation in the index. This should not be surprising. The object of the book is to study the optimal rate of currency issuance with a view to economic stimulus. Some degree of currency issuance makes government and individual debt more manageable, so used properly and in moderation, inflation is just a policy tool to increase economic growth.

In neither the Keynes book nor the Friedman book is there any advice as to what an investor might do to protect wealth from inflation’s destruction of purchasing power. Of course, neither book is designed for an investor. These books were written for policymakers and other economists. It would never have occurred to either economist that the policymakers be deprived of the power or policy tool to create money and, thereby, inflation.

Friedrich Hayek is unique among the major economists of the 20th century in that he advocated that national governments be deprived of the money creation function, which he describes in *The Denationalization of Money* (1976).⁵ The basic idea of the Hayek book is that governments cannot be trusted not to abuse the money-creation function. Therefore, Hayek advocates the creation of “private money.” The problem is that one cannot necessarily trust those in control of private money either, and as a consequence of this obvious objection, the Hayek book remained in obscurity until rediscovered by the proponents of cryptocurrency.

In the world of cryptocurrency and blockchain, it is possible that users of the currency can continually verify its issuance status. As an example, if bitcoin will issue slightly less than 2.5 million coins between the time of this writing and the year 2140, that is objectively verifiable as a fact by means of the blockchain, which forms the permanent public record of every bitcoin transaction since its creation. Bitcoin ease of transfer has yet to rise to the level of being accessible to the average person, but that function is constantly improving. Private money might be the solution to the inflation problem. Unfortunately, it is not yet ready for mass usage. In any case, the community of academic scholars tends to avoid the subject of cryptocurrency.

In 1975, the economist John Kenneth Galbraith wrote a popular account of the history of money entitled *Money: Whence It Came, Where It Went*.⁶ Given the era in which the book was written, the author could hardly avoid discussing the subject of inflation. Galbraith strongly advocates for government wage and price controls or, alternatively, wage and price guidelines that are informal requests with which government pressure might force compliance even if they are not formal controls. Even a book that is directed towards the general public is really a call for

action by policymakers.

The common factor among economists of a variety of philosophical orientations is that inflation is something that must be controlled by government action of some sort. The underlying assumption is that the inflation problem has nothing to do with existing government policy. Moreover, inflation is not considered to be a problem that can be confronted by the individual on behalf of the individual.

The problem with a government policy is that its effects on individual circumstances differ from person to person. As an example, consider a pension adjusted annually to a national consumer price index. Even if it is calculated in a manner to properly measure and account for inflation, it might not properly measure the increase in expenses of a given pensioner if that individual's expenditure is weighted heavily towards items experiencing above average inflation, such as health insurance or education.

The economic consequences of inflation, the historical documentation of which spans well over 2,000 years, are such that it merits its own asset class in investment planning. It should be self-evident that one cannot simply assume that common stocks *sui generis* are efficient or are a sufficient hedge against inflation. Labor is the most significant operating expense for most members of the popular stock indexes. Many people assume that a business can maintain profit margins by increasing wages at the same rate as general inflation. However, a wage increase of X% in a progressive tax system does not necessarily become an after-tax wage increase of that same X%, because in a progressive system, higher wages are taxed at a higher rate. The U.S., of course, follows a progressive tax structure. As a generalization, one requires a pre-tax wage increase above inflation to obtain an after-tax increase equivalent to inflation. Most businesses, under a condition of persistent inflation, are not able to maintain their margins once wage rates begin to inflate.

A good argument can be made for gold as an inflation hedge in a portfolio, but it must be remembered that gold can be highly idiosyncratic for very long periods of time. For example, from 1980 to 2000, gold actually declined, although inflation clearly existed during this period; the price of gold was unchanged from 1934 to 1967, as a consequence of government action, although price levels increased by 2.5x.

The Federal Reserve Bank of Minneapolis calculates inflation in the United States back to 1800 and the data illustrates the same point.⁷ The First World War years in the U.S. were highly inflationary, as is evident in the following figures.

Table 1: U.S. Inflation Rate, 1916–1920

Year	Rate
1916	7.7%
1917	17.8%
1918	17.3%
1919	15.2%
1920	15.6%

Source: Federal Reserve Bank of Minneapolis

The gold prices for those same years are as follows; clearly, the gold price did not keep pace with inflation at that time:

Table 2: Gold Prices, 1916–1920

Year	Price
1916	\$18.99
1917	18.99
1918	18.99
1919	19.95
1920	20.68

Source: piketty.pse.ens.fr

As a matter of historical curiosity, in 1717, during Isaac Newton's tenure as Warden and Master of the Royal Mint of the United Kingdom, he set the gold price at 3 pounds 17 shillings. It was still at that price in 1914, on the eve of the First World War.

On the other hand, one could have purchased gold in the year 2000 for \$279.11 an ounce and it now trades at \$1,908.07 an ounce. This is about a 10% annualized increase, considerably above most estimates of inflation for this period. If one can tolerate the idiosyncratic behavior of gold, it can be a good protection against inflation.

Far better than gold are the royalty companies that finance gold production by the purchase of

revenue interests, or some effective equivalent, in precious metals mines. The two most successful are Franco Nevada Corp. (FNV) and Wheaton Precious Metals Corp. (WPM). Once the investment is made, the return is greatly enhanced if the price of the commodity rises, since the royalty companies experience no increase in expenses; theirs is simply a contractual relationship, and they require none of the capital equipment or physical field operations of a mining company. Other examples of gold royalty companies, are:

Table 3: Additional Gold Royalty Companies

Ticker	Company
RGLD	Royal Gold, Inc.
SAND	Sandstorm Gold Ltd.
OR	Osisko Gold Royalties Ltd.
MMX	Maverix Metals Inc.
MTA	Metalla Royalty & Streaming Ltd.

This is an example of a “hard asset” business model. The revenue is a contractual commitment by the producer of the commodity to the company providing the financing. The inflation beneficiary aspect is that the royalty company’s operating expenses do not increase in the event of inflation; in the event of inflation, a revenue increase is essentially a pre-tax profit increase.

The challenge to asset allocators and other investors is that an insufficient number of such hard asset companies exist to form an index. Also, the aggregate market capitalization is very small in relation to the great amount of assets in the world that require some inflation protection.

Many investors in the practical world, therefore, are compelled to use leverage as an inflation hedge, on the premise that the cost of debt capital is low and, in any case, even modest inflation will eventually erode the purchasing power of the obligations that must one day be repaid. That is, for investors employing leverage, paying back “in cheaper money”: the amount of the bond or loan due on the maturity date in five years or 10 remains the same as on the first day of the loan, but the nominal income of the borrower has presumably inflated over time.

John Maynard Keynes did not write extensively about inflation, but his personal portfolio was structured in such a manner to benefit from consistent inflation inasmuch as he made use of copious leverage. The following table, taken from *Keynes’ Way to Wealth* by John F. Wasik, will illustrate the point.⁸

Table 4: John Maynard Keynes — Leverage and Net Worth, 1919–1945

Year	Loans	Net assets	Year	Loans	Net assets
1919	£15,498	£16,315	1933	£78,859	£55,222
1920	20,837	(1,837)	1934	165,343	146,007
1921	N/A	N/A	1935	188,271	220,619
1922	2,720	21,558	1936	299,347	506,522
1923	1,200	34,364	1937	190,035	215,244
1924	1,200	63,797	1938	106,470	181,547
1925	1,200	43,610	1939	109,136	199,495
1926	2,200	40,800	1940	60,655	171,090
1927	46,900	44,000	1941	28,753	205,281
1928	25,790	13,060	1942	19,720	254,073
1929	14,000	7,815	1943	31,643	312,271
1930	65,000	12,525	1944	46,167	355,310
1931	11,965	15,100	1945	38,886	411,238
1932	19,774	21,722			

Source: John F. Wasik, *Keynes’ Way to Wealth* (New York: McGraw Hill Education, 2014), 119.

There were years where Keynes was enormously leveraged. For example, in 1920, Keynes had £20,837 of borrowings, and his net assets were negative £1,837. One could say technically he was insolvent. Similarly, in 1928 on the eve of the stock market crash, he had £25,790 of leverage and £13,060 of net assets. In the year 1929, it was £14,000 of leverage and £7,815 of net assets. One can see this involves a fairly large multiple. In 1930, he had £65,000 of loans, and £12,525 of net assets, he was over 5x leveraged. It was only near the end of his life, during the war years, that Keynes had fairly modest leverage in relation to his net assets. The last full year of his life, 1945, he had only £38,886 of borrowings against net assets of £411,238.

It should be noted that Keynes drew living expenses from his personal account, so that the years

of decline in net assets are not entirely from investment depreciation. The net asset values are not meant to illustrate investment performance. In any case, it is obvious that Keynes frequently made use of enormous leverage relative to his net assets.

This process has continued among investors and business people to this day. The contemporary world might be the most leveraged society in history. That means that the constituency supporting debasement of debt and money is very large. Inflation is the potential savior of the debt leveraged. Therefore, it is prudent to develop inflation hedges.

Original PDF includes Horizon Kinetics standard research disclosures and reproduction restrictions.

Next: Chapter 4 — Diamonds as an Investment (chapter-04-diamonds-as-an-investment.md).

Chapter 4 — Diamonds as an Investment

Source. Murray's Musings, *Contrarian Report Compendium*, December 2022. Horizon Kinetics LLC © 2022.

PDF in vault. [Contrarian-Compendium-December-2022_Diamonds-as-an-Investment.pdf](#)

Extract used for assembly. [book/build/text/Contrarian-Compendium-December-2022_Diamonds-as-an-Investment.txt](#)

Contributors (as in original). Murray Stahl, Chairman; Steven Bregman, President; Peter Doyle, Managing Director.

Conflict / offering note (from original). An affiliate of Horizon Kinetics Asset Management LLC acts as co-sponsor to a private fund that invests in diamonds; the fund is available only to certain qualified investors.

Companion essay in this edition. *Wealth and Crisis: The Role of Diamonds in Asset Allocation* (European Contrarian, January 2023) → Chapter 8; vault PDF [European-Contrarian-Compendium-January-2023_Wealth-and-Crisis-The-Role-of-Diamonds-in-Asset-Allocation.pdf](#).

Presentation edits: removed compendium running heads and page banners; merged words split across line breaks; omitted lengthy standard research disclaimer; light punctuation cleanup only.

Diamonds as an Investment

Although diamonds were known and gathered 4,500 years ago, diamond mining on a systematic basis could only proceed with the development of the modern sciences of geology and crystallography. This was possible only at the end of the 19th century, when systematic diamond extraction in South Africa commenced. This ultimately gave rise to the De Beers monopoly. De Beers controlled the diamond market until the end of the 20th century.

The De Beers monopoly was an obvious barrier to price discovery and transparency. Thus, the evolution of diamonds as an asset class only became possible in the 21st century. Even still, diamonds present valuation problems not ordinarily present in commodities. Gold and silver of given levels of purity, for example, are homogeneous commodities, while all diamonds are heterogeneous; no two are identical. Accordingly, there developed a detailed grading system. Diamonds are commonly divided by the four criteria of color, cut, clarity, and carats, each with its own measures.

That grading system has existed for over 50 years, but it is only lately that modern computing technology has enabled the ability to assemble a diverse set of diamonds into packets that each have equivalent value in much the same way that ETFs are designed to combine sets of individual companies to be representative of an industry exposure. Such packets, then, are homogeneous, like other commodities. That is a significant change in the investment characteristics of diamonds as an asset class.

This section of the Compendium is intended to serve as a reference guide for and answer certain questions about what might be regarded as basic diamond investing. One such question might be: Where is historical data on diamond prices kept? Another might be: Is historical diamond price data accessible to the public?

Historical price data is available via the Rapaport Group, which is a large Belgian broker of diamonds. This is the only database that I have seen that makes diamond prices readily available. Rapaport publishes the price lists, and one can purchase them from the company. For instance, one can buy the price list for April 27, 2007 for \$150.

Rapaport also publishes and sells a Diamond Price Statistics Annual Report for each year, starting with 1994, which is approximately the time that the De Beers monopoly was effectively dismantled. It might be recalled that the U.S. government charged both General Electric Company (GE) and De Beers, in 1994, with industrial diamond price fixing.

It is fair to assert that diamond prices rarely decline, at least according to the data. A notable exception was during the 2008 financial crisis, but even then the decline was modest in comparison with other assets, such as stocks and bonds. A brief drop in diamond prices also occurred when the World Trade Center was destroyed in 2001.

Constructing a diamond price index would require making certain assumptions regarding index weightings, and such assumptions can be subjective. It will not be attempted in these pages. In order to comprehend the complexity of building an index of diamond prices, one might parse the problem by considering the impact on price just of the differences in diamond cut.

Petra Gems publishes a diamond buying guide that illustrates the complex nature of this challenge. Oval or round cut diamonds are the most desirable and therefore the most expensive diamonds if all other factors are held constant, which of course, is rarely the case. In order of desirability:

- Oval cut diamonds are priced 5–15% lower than round cut diamonds.
- Princess cut diamonds are priced 17–26% lower than round cut diamonds.
- Cushion cut diamonds are priced 22–30% lower than round cut diamonds.
- Radiant cut diamonds are priced 20–27% lower than round cut diamonds.
- Emerald cut diamonds are priced 19–27% lower than round cut diamonds.
- Pear cut diamonds are priced 15–22% lower than round cut diamonds.
- Marquise cut diamonds are priced 23–32% lower than round cut diamonds.
- Asscher cut diamonds are priced 21–31% lower than round cut diamonds.

Another complexity is that the Rapaport Price List is based on ask prices rather than actual realized prices. A dealer frequently will demand and receive a 15% or even 20% discount to the Rapaport ask price. The market still lacks a certain level of transparency, since it is difficult to know actual realized prices.

A further problem is that even realized prices are semilogarithmic with respect to certain attributes, such as carat size. To illustrate this relationship, let us assume the existence of a round cut diamond of 1 carat, with color D (highest color grade) and clarity FL (flawless). If the selling price is X , a 1.5 carat round cut diamond with color D and clarity FL would sell for about 2.14x the price of the comparable 1 carat diamond. To be clear: if one compares the 1.5 carat diamond to a 1 carat diamond with these characteristics in common, the price would not be 50% higher just because it has 50% more weight; it actually would be 2.14x as expensive as the 1 carat diamond.

The only difference between the two diamonds in this example is that the latter has 1.5x the weight of the former. The 2.14x greater price is because a 1.5 carat diamond is rarer than a 1 carat diamond. In other words, the value density of a 1.5 carat diamond is more than 1.5x as great as a 1 carat diamond's value density.

Value density will ultimately become extremely important as the concept of physical commodities increasingly dominates the world of commodity ETFs. A 3 carat round cut diamond with D color and FL clarity could sell for \$170,000.

To be clear about the value density, a 3 carat diamond weight is only 6 grams, and there are 31.1035 grams in a troy ounce. Thus, a diamond with the aforementioned characteristics and price is selling for $\$170,000 \div 6$ grams, or \$28,333 per gram. One troy ounce of diamonds of this caliber would theoretically sell for $\$28,333 \times 31.1035$, or \$881,255. In contrast, one troy ounce of gold sells for about \$1,757. Therefore, one ounce of diamonds with the aforementioned quality features is 501.6x more value dense than gold ($\$881,255 \div \$1,757$).

The largest physically backed gold ETF is SPDR Gold Shares (GLD), with \$50.736 billion of assets under management. Using the current \$1,757/ounce gold price for the calculations, this fund holds roughly 28,876,494 ounces of gold. If confidence in the financial system were to be lost, the investors ostensibly could demand to collect their 28,876,494 ounces of gold. What would that entail?

A troy ounce of gold has the following dimensions: 0.95 inches long, 1.65 inches wide, and 0.08 inches thick, which equals 1.254 cubic inches. An ounce of gold is easily transported and easily hidden in a crisis. However, transporting and hiding 29 million ounces is a far more complex logistical challenge—that would be 1.999 million pounds. One Honda Accord weighs about 3,300 pounds.

Contrast that value density with diamonds. If an ounce of diamonds is worth \$881,255, achieving a valuation of \$50,736,000,000, the market capitalization of SPDR Gold Shares, would require only $\$50.736 \text{ billion} \div \$881,255$ or 57,572.43 ounces, or 3,985 pounds. A human being could easily transport or hide over \$8 million by holding a mere 10-ounce diamond packet.

One of the reasons investors wish to hold physical gold is that it cannot be easily debased like the U.S. dollar or other fiat currencies. However, all of the large central banks have extensive gold reserves, and it might be necessary for central banks to sell those reserves if a time comes when the fiat currency is not trusted and might not even be an acceptable medium of exchange. In that case, large gold sales might depress the price of gold. On the other hand, central banks do not have reserves of diamonds to sell.

Gold has been readily available in sufficient supply at the requisite level of purity to use as a medium of transaction for thousands of years. On the other hand, a market for diamonds could not exist that was fair in any meaningful sense of the word until the modern science of crystallography developed.

A factor in the performance of diamonds is the evolution of India's diamond industry, which is where much of the cutting and polishing of rough diamonds is now done. The Indian diamond industry employs 850,000 people. In fact, the Indian city of Surat in the western state of Gujarat has become known as Diamond City. Diamond cutting is a highly skilled art, and people who are expert in this field are well compensated in Western nations. Indian skilled labor is far less expensive. The Indian diamond export trade is centered almost entirely in the Bharat Diamond Bourse in Mumbai, which handles 98% of the export trade. It has a constructed area of 2 million square feet, with two basement complexes containing an extra 1 million square feet.

The transition of a rough diamond to a polished cut diamond is a time-consuming process. Indian labor has greatly reduced friction in that area. That inexpensive labor has served to restrain diamond prices in recent years.

The global market for diamonds is expanding, as a greater proportion of the world population enters the middle class and becomes part of the diamond consumers' universe. Existing diamond exchanges include the London Diamond Bourse, the Diamond Federation of Hong Kong, the Israel Diamond Exchange, the Dubai Diamond Exchange, the Amsterdam Diamond Bourse, the Bharat Diamond Bourse, the Tokyo Diamond Exchange, the Shanghai Diamond Exchange, the Antwerp Diamond Exchange, the Diamond Dealers Club of Australia, Borsa Diamanti d'Italia, and the Guangzhou Diamond Exchange.

A number of new diamond bourses are in the process of being established. These include the New York Diamond Dealers Club, Borsa Istanbul, Diamant Club Wien, Bangkok Diamonds and Precious Stones Exchange, the Diamond Exchange of Singapore, and the Korea Diamond Exchange. This does not include the World Federation of Diamond Bourses itself.

It is important to recall that De Beers had a global monopoly, for all practical purposes, and was involved in price fixing that was only effectively challenged in 1994. This was over 300 years after the Buttonwood Agreement of 1792 that became the foundation of the New York Stock Exchange. The first important revision of that agreement did not occur until 1817. The U.S. Securities and Exchange Commission did not commence operations until 1934. The Depository Trust Company (DTC), which does clearing, was only founded in 1973, as the central securities depository of the U.S. market system, 181 years after the New York Stock Exchange. Bitcoin began its existence as a white paper published on October 31, 2008, and was first used in 2009. In 2022, it still has no regulatory structure.

In other words, the modern diamond market is evolving more rapidly than the securities market.

The first recorded use of a diamond engagement ring was in 1477, when Archduke Maximilian of Austria proposed marriage to Mary of Burgundy. Diamonds were a rarity even for engagement rings until 1947, when De Beers engaged the now-defunct advertising firm N.W. Ayer & Son to create the campaign entitled "A diamond is forever." This was one of the most successful advertising campaigns of all time and created the modern diamond market. The universal diamond grading system was only created by the Gemological Institute of America (GIA) in the 1950s. The Gemological Institute of America was only founded in 1931.

The diamond asset class is developing very rapidly. However, it is very unlike ordinary asset classes like securities, in which creation can be infinite—which is to say, if a certain kind of "paper" is appealing to the public, issuers can and will produce, with virtual immediacy, as great a quantity of it as the market will bear. The global production of rough diamonds is in decline, and probably in inexorable decline. The Facts & Figures section in this Compendium contains more data on that.

An investable asset class is in the process of being created and, moreover, with an asset manifesting declining production supply. According to Statista, rough diamond production was 177 million carats in 2005 and 116 million carats in 2021. The most alluring returns from an asset class occur before it actually becomes an investable asset class.

Original PDF includes Horizon Kinetics standard research disclosures and reproduction restrictions.

Next: Chapter 5 — Trends in Healthcare.

Chapter 5 — Trends in Healthcare

Source. Murray's Musings, *Devil's Advocate Report Compendium*, February 2023. Horizon Kinetics LLC © 2023.

PDF in vault. Devils-Advocate-Compendium-February-2023_Trends-in-Healthcare.pdf

Extract used for assembly. book/build/text/Devils-Advocate-Compendium-February-

Contributors (as in original). Murray Stahl, Chairman; Steven Bregman, President; Peter Doyle, Managing Director.

Presentation edits: removed compendium running heads and page banners; merged words split across line breaks; omitted lengthy standard research disclaimer; tables rebuilt in Markdown.

Trends in Healthcare

According to Statista.com, in the calendar year 2022, there were 4.76 billion retail prescriptions filled in the U.S. The current U.S. population is 334.299 million. This amounts to about 14.24 prescriptions for every human being in the nation. This does not include pharmaceuticals dispensed in hospitals. It does not include pharmaceuticals dispensed in ambulatory care clinics. It does not include so-called over-the-counter drugs sold without a prescription.

This is one of the reasons that healthcare is a growth industry. The proportion of GDP devoted to healthcare is generally increasing. A recent calculation by the World Bank tabulates that in 2019, healthcare spending in the U.S. amounted to 16.8% of GDP. The Statista calculation for 2020 indicates that healthcare spending accounted for 18.8% of U.S. GDP. Obviously, the 2020 figure was impacted by extraordinary measures necessary to deal with the pandemic.

COVID-19 is a worldwide phenomenon, yet the United States is unique even among industrialized nations as for its healthcare spending as a proportion of GDP. Apart from the U.S., the highest such expenditures as a percent of GDP are by the following ten nations, the highest of which is Canada at 12.9%.

Table 1: Healthcare expenditure as a percentage of gross domestic product (2020)

Country	Healthcare expenditure / GDP
Canada	12.9%
Germany	12.8%
France	12.2%
United Kingdom	12.0%
Switzerland	11.8%
Austria	11.5%
Sweden	11.5%
Norway	11.3%
Netherlands	11.2%
Japan	11.1%

Source: Statista

According to Statista, Costa Rica spends 7.9% of its GDP on healthcare. Hungary spends 7.3% of its GDP on healthcare. Luxembourg spends 5.7% of its GDP on healthcare. Mexico spends 6.2% of its GDP on healthcare.

One measure, although certainly not the only measure, of the effectiveness of these levels of spending is life expectancy. The World Health Organization (WHO) calculates life expectancy figures for nations annually.

Table 2: Life expectancy by nation (2022)

Country	Years
United States	78.203
Canada	82.847
Germany	80.989
France	83.229
United Kingdom	82.156
Switzerland	84.255
Austria	82.412
Sweden	83.505
Norway	83.393
Netherlands	82.451
Japan	84.820

Country	Years
Costa Rica	77.320
Hungary	74.958
Luxembourg	82.591
Mexico	74.832

Source: World Health Organization

When comparing national expenditures as a proportion of GDP, it is important, for meaningful comparison, to take account of any wide differences in per-capita GDP. One can say that the U.S. spends 2½ times more on health care than Costa Rica, based on spending 18.8% of GDP on healthcare versus Costa Rica spending 7.3%. However, the per-capita GDP of the U.S. is much higher than that of Costa Rica. The estimated per capita GDP of the U.S. in 2022 is \$76,027, while for Costa Rica it's \$13,089.⁹ Those figures allow one to calculate the per capita dollar amount of spending on healthcare:

- U.S.: $18.8\% \times \$76,027 = \$14,293.08$
- Costa Rica: $7.3\% \times \$13,089 = \955.50

That reveals that the per capita expenditure of the U.S. on healthcare is 15.0 times that of Costa Rica. Nevertheless, the U.S. life expectancy is 78.203 years and the life expectancy of Costa Rica is 77.320 years. Viewed on a per capita basis, the vast excess of U.S. expenditure does not appear to be productive.

A problem with this type of calculation is that the various data sources do not agree on the life expectancy figures. The aforementioned life expectancy figure for Costa Rica is compiled by the World Health Organization. The Population Division of the Department of Economic and Social Affairs at the United Nations calculates the life expectancy of Costa Rica at 79 years. Macrotrends.net lists it at 80.56 years. The peer reviewed journal known as *Proceedings of the National Academy of Sciences* is the source of the macrotrends.net life expectancy figure.

In any case, in the U.S., national healthcare expenditure routinely expands more rapidly than GDP. According to CMS.gov (Centers for Medicaid and Medicare Services), the annual spending figures for the past 10 years are as follows.

Table 3: U.S. national healthcare expenditure

Year	Annual expenditure (in trillions)
2011	\$2.676
2012	2.783
2013	2.856
2014	3.002
2015	3.165
2016	3.207
2017	3.448
2018	3.604
2019	3.757
2020	4.114
2021	4.256

Source: CMS.gov

The compound annual expansion rate of spending is 4.75%. It is, of course, needless to say that the U.S. population does not grow at 4.75% per annum, and the GDP does not grow at that rate, either. The logical consequence of the trend is that the healthcare expense burden on the economy will become increasingly heavy.

In common investment parlance, the healthcare industry is frequently used as a synonym for pharmaceuticals. However, the U.S. healthcare sector is far more diverse. This is evident in the sectoral diversification of the largest healthcare ETF, known as The Health Care Select Sector SPDR Fund (XLV). This ETF is divided into five sectors, as detailed in the accompanying table.

Of these, biotechnology is an interesting category. It is obvious that the sector commands significant market capitalization. However, it does not produce large quantities of revenue. In any analysis of healthcare expenditure, it is more meaningful to determine the respective shares of that expenditure exclusive of biotechnology due to its limited revenue relevance. If this were done, the proportionate share of each of the four sectors would be as follows. While the pharmaceuticals sector accounts for somewhat over a third of the total, the balance is made up of health care providers and services, and two separate categories of companies providing health care equipment and supplies, and analytical tools, instruments, clinical trial services and research services.

Table 4: XLV industry allocation

Sector	Weight
Health Care Equipment & Supplies	18.77%
Health Care Providers & Services	22.36%
Life Sciences Tools & Services	12.95%
Pharmaceuticals	30.28%
Biotechnology	15.64%

Source: ssga.com

Table 5: XLV industry allocation excluding biotechnology

Sector	Weight
Health Care Equipment & Supplies	22.25%
Health Care Providers & Services	26.51%
Life Sciences Tools & Services	15.36%
Pharmaceuticals	35.89%

Source: ssga.com

In order for this system to continue to expand at its historical rate, the typical human being would have to devote more time to medical examination and analysis in order to determine if medical intervention is required and, should such intervention be required, the degree of intervention and type of treatment. Ultimately, the growth proposition for health care spending can be reduced to the internet/technology growth proposition that has previously been propounded in Compendium discussions: that in order for the historical rate of industry growth to be sustained, the average person must spend more time on the internet. In the medical sphere, the average person would have to spend increasing amounts of time interacting with the medical system.

The OECD (Organisation for Economic Co-operation and Development) maintains data on the length of the average hospital stay for most nations. It will be recalled from a prior table that the U.S. spends more on healthcare, as a proportion of GDP, than other industrialized nations. The following table compares the length of the average hospital stay in the U.S. versus the other industrialized nations.

Using length of hospital stay as a metric indicates that the U.S. uses its healthcare system less intensively than all nations on the list, excepting Sweden and the Netherlands. Even in these two instances, the intensity of hospital use is only marginally less than that of the U.S. Nevertheless, the U.S. spends a far greater proportion of its GDP on healthcare than does either Sweden or the Netherlands. The relevant figures are that Sweden spends 11.5% of its GDP on healthcare, the Netherlands spends 11.2% of its GDP on healthcare, and the U.S. spends 18.8%. If one were to use life expectancy as a measure of healthcare outcomes, the U.S. obtains a less satisfactory result than either Sweden or the Netherlands, or even any other industrial nation.

Table 6: Length of average hospital stay (2021)

Country	Days
Canada	7.7
Germany	7.4
France	5.5
United Kingdom	6.2
Switzerland	6.9
Austria	6.4
Sweden	5.3
Norway	5.9
Netherlands	5.2
Japan	16.4
Costa Rica	6.8
Hungary	5.4
Luxembourg	7.5
Mexico	N/A
United States	5.4

Source: OECD

If U.S. healthcare expenditures were to remain at 18.8% of GDP and not rise beyond this point, it must logically follow that the healthcare industry would be growing at the rate of GDP. This would be the nominal figure and not the real figure. Thus, if real GDP were to average 2% growth per annum and inflation were held at the level of 2%, and the healthcare industry maintains its current share of GDP, it would be growing at a 4% rate.

At the time of this writing, The Health Care Select Sector SPDR Fund (XLV) trades at 16.32x earnings and 4.67x book value, and the 3–5 year estimated annual growth rate, as stated on the most recent fund factsheet, is 7%.

Those valuations make it fairly obvious that the stock market participants in this sector presume that the current circumstance, which is to say the historical rate of expansion, will continue. Implicitly, that embodies the presumption that the healthcare industry will represent an ever-increasing share of U.S. GDP. It presumes that the general populace will not be troubled by the fact that other nations have a healthcare industry that consumes a much smaller proportion of GDP; that foreign healthcare systems appear to be used more intensively by the population and with a far lower financial burden than is the case in the U.S. Moreover, the general populace will not be troubled that the outcome, at least as measured by life expectancy, however imperfectly, is lower in the U.S. than in other industrialized nations.

The alternative view to that of the investment community as expressed in the valuation of the health care sector, is that the current circumstance is unsustainable and that the unsustainability will have negative consequences for equity indexes, the U.S. GDP, and the national government solvency. In any case, the consensus does appear to be untroubled, an additional measure of that sentiment being the \$40.987 billion in assets under management of XLV, as of the most recent reckoning.

Original PDF includes Horizon Kinetics standard research disclosures and reproduction restrictions.

Next: Chapter 6 — Bitcoin v. Bonds.

Chapter 6 — Bitcoin v. Bonds

Source. Murray's Musings, *Devil's Advocate Report Compendium*, November 2022. Horizon Kinetics LLC © 2022.

PDF in vault. Devils-Advocate-Compendium-November-2022_Bitcoin-vs.-Bonds.pdf

Extract used for assembly. book/build/text/Devils-Advocate-Compendium-November-2022_Bitcoin-vs.-Bonds.txt

Contributors (as in original). Murray Stahl, Chairman; Steven Bregman, President; Peter Doyle, Managing Director.

*Presentation edits: removed compendium running heads and page banners; merged words split across line breaks; omitted lengthy standard research disclaimer; tables in Markdown. One obvious stray "\$" before a percentage in the source ("0.87%") is rendered as **0.87%**.*

Bitcoin v. Bonds

The bond market thus far in 2022 has produced results that generate questions about the nature of the asset class itself. These questions arise from the year-to-date returns for salient aspects of the asset class. The returns can be visualized using popular exchange traded bond index funds (ETFs), as follows:

Table 1: Returns for selected bond ETFs, Dec. 31, 2021–Oct. 21, 2022

Ticker	Fund	Return
TLT	iShares 20+ Year Treasury Bond ETF	(36.11)%
IEF	iShares 7–10 Year Treasury Bond ETF	(18.08)%
AGG	iShares Core U.S. Aggregate Bond ETF	(16.62)%
HYG	iShares iBoxx \$ High Yield Corporate Bond ETF	(14.09)%
TIP	iShares TIPS Bond ETF	(13.42)%

Source: iShares.com

The obvious question to be posed when noting the negative returns experienced this year is whether the long-term return prospects of the bond asset class are sufficiently large to compensate the investor during the inevitable declines in those periods when the Federal Reserve must increase interest rates.

One might begin to answer this question by studying the returns since inception for the aforementioned funds. This data appears in the table below.

Table 2: Annualized returns, selected popular bond ETFs, inception through 9/30/22

Ticker	Fund	Inception date	Return
TLT	iShares 20+ Year Treasury Bond ETF	July 22, 2002	4.50%
IEF	iShares 7–10 Year Treasury Bond ETF	July 22, 2002	3.52%
AGG	iShares Core U.S. Aggregate Bond ETF	Sept. 22, 2003	2.90%
HYG	iShares iBoxx \$ High Yield Corporate Bond ETF	April 4, 2007	4.08%
TIP	iShares TIPS Bond ETF	Dec. 4, 2003	3.42%

Source: iShares.com

It should be pointed out that the inception dates of all these funds to the most recent date coincide with the greatest bond bull market of all time by far. It is an observation to keep in mind as one reflects on these results.

An obvious problem with this type of analysis is the question of subjectivity. One might choose to compare these funds to the prevailing rate of inflation during the relevant time period, in order to understand what the real, inflation-adjusted returns have been. Unfortunately, the calculation of the rate of inflation has itself become controversial and no agreement on a universal inflation measure appears to exist. Nevertheless, the returns can certainly be compared to one another. Even this approach, however, is not very promising because the returns since inception are for different time periods and the periods may not be sufficiently long to arrive at meaningful conclusions. Also, the returns for the different periods and the different classes of bonds are not radically different.

One can say that the highest return is for long term U.S. Treasury bonds, at least for the time period in question. If duration is a primary measure of risk, then it is reasonable that the highest return category is long term U.S. government bonds. But, we cannot know if the greatest overall risk is to be found in the high-yield bond sector, which in prior eras would have been the natural presumption due to the inherent credit risk. That can no longer be presumed, because the Federal Reserve has pursued an extremely aggressive policy of monetary accommodation for most of the time period in question. The alteration of this stance only commenced in mid-June 2022. Some time must pass until the withdrawal of monetary stimulus results in a more historically normal incidence of credit defaults or near defaults.

Credit spreads will likely widen and higher yields will be available in the sector if defaults once again become a primary risk factor for high-yield bonds. In that case, after a period of low returns or quite possibly negative returns, the best performing bond segment might ultimately become high-yield bonds.

On a year-to-date 2022 basis, high-yield bonds slightly underperformed inflation-protected government bonds. Credit problems have not yet made an appearance.

Another means of viewing these funds is through the prism of after-tax returns, since individual taxpaying investors frequently make use of ETFs.

Table 3: After-tax annualized returns, selected bond ETFs, inception through 9/20/22

Ticker	Fund	Inception	Return
TLT	iShares 20+ Year Treasury Bond ETF	July 22, 2002	3.06%

Ticker	Fund	Inception	Return
IEF	iShares 7–10 Year Treasury Bond ETF	July 22, 2002	2.36%
AGG	iShares Core U.S. Aggregate Bond ETF	Sept. 22, 2003	1.79%
HYG	iShares iBoxx \$ High Yield Corporate Bond ETF	April 4, 2007	2.07%
TIP	iShares TIPS Bond ETF	Dec. 4, 2003	2.20%

Source: iShares.com

The calculations in Table 3 were made by iShares, not Horizon Kinetics. To make it universally applicable to all U.S. residents, only federal taxes are assessed. Residents of various states with state income taxes or even municipalities with local income taxes will have different rates of return. However, the returns are at least standard in that none include state and local taxes. State and local income taxes would not apply to any of the three Treasury ETFs. For the iShares Core U.S. Aggregate Bond ETF and iShares iBoxx High Yield Corporate Bond, the yields in the table are shown without the impact of state and local taxes.

If one presumes merely a 2% annual inflation rate, the Core U.S. Aggregate Bond ETF did not provide a positive real rate of return after taxes. This is significant because this particular ETF is designed to represent the entirety of the bond market. The fund itself has \$75.64 billion of assets under management. It is the second largest bond ETF and the tenth largest of all ETFs in the United States. The largest bond ETF is the Vanguard Total Bond Market ETF (BND), with \$79 billion of AUM and an inception date in 2007. In composition and returns, it is very similar in character to the iShares Aggregate U.S. Bond ETF.

If the after-tax rate of return of the entire bond market is lower than inflation over a long period of time during which inflation was mostly at a low rate and relatively benign—in this case covering periods of 15 to 20 years—one must logically rethink one's approach to bond investing.

The composition of the Aggregate Bond ETF is as follows:

Table 4: Composition, iShares Core U.S. Aggregate Bond ETF

Bond type	Percentage of AGG
Treasury	41.72%
MBS	27.05%
Industrial	14.41%
Finance	8.16%
Utilities	2.06%
CMBS	1.85%
Agency	1.22%
Supranational org.	0.92%
Local Authorities	0.84%
Cash	0.48%
ABS	0.35%

Source: iShares.com

Segments of the bond market clearly did provide a positive real after-tax rate of return when measured against the conventional CPI inflation statistic. If such is the case, yet the broad index underperformed inflation after taxes, this would constitute the first relatively long-run empirical argument against broad indexation in favor of an actively constructed portfolio.

The market value of all publicly traded fixed income securities outstanding as of year-end 2021, is \$53.017 trillion.¹⁰ Bear in mind this is not all debt; it is the value of all publicly traded debt securities, which is lower than the value of all debt. This is the largest asset class in the U.S. and it is considerably larger than the market capitalization of the S&P 500. In fact, fixed income issuance has continued to be robust in 2022 even though, at the moment, the market for initial public offerings of equity is almost nonexistent.

In principle, the best decision that could have been made over the course of the past two decades within the broad bond asset class to maximize returns would have been a 100% allocation to long-term treasury bonds. The after-tax annualized return is 3.06%. Assuming roughly 2% annual inflation, which, again is subject to some degree of dispute, the real rate of return was

1.06% per annum. In order to earn this rate of return, one had to accept a standard deviation of 14.03, an effective duration of 17.17 years, and a weighted average maturity of 25.58 years.

By way of comparison, the first reliable price quote for bitcoin was \$0.0495 on July 17, 2010, a bit less than 5 cents.¹¹ Bitcoin now sells for \$19,189. The current interim bitcoin inflation rate is 1.73% per annum and at the time the notes for this Compendium were prepared, 19,186,625 bitcoin were in existence. By the time this is published, there will be even more. A total of 1,813,375 bitcoin will be issued from the time the notes were prepared for this Compendium until the year 2140. The number yet to be issued will continuously drop. In 556 days, the bitcoin interim inflation rate will decline to 87 basis points per year. With each passing day, of course, there will be one less day until the bitcoin interim inflation rate drops to **0.87%**. Moreover, the bitcoin interim inflation rate will decline by 50% every four years in accordance with the bitcoin issuance protocol. The projected 0.87% annual inflation rate will decline to 0.44% after four years.

Profits on bitcoin are taxed at the capital gains rate, not the statutory federal income tax rate that applies to interest on debt. This is because the IRS considers bitcoin to be property. Also, unlike interest on debt, capital gains tax is paid after sale at a profit.

What if a holder of a bond ETF, such as iShares Core U.S. Aggregate Bond ETF (AGG), were to have made the most marginal change possible to the investment allocation plan by investing 10 basis points annually into bitcoin? Structured that way, in the worst instance, the after-tax rate of return would have declined from 1.79% annually to 1.69%. That basically assumes a complete failure of the Bitcoin project, that bitcoin would have gone to zero.

An investment in bitcoin made in July 2010 would have increased 387,797x in value by today. In other words, a \$100 investment would now be worth \$38,779,797. If the investment had been made with only \$100 of a \$1 million bond portfolio even once in 2010—that would be a 1 basis point investment, one-hundredth of 1%—the remaining \$999,900 that was invested in bonds compounding at an after-tax 1.79% per annum rate for 12 years would now be worth \$1,237,137. If the entire \$1 million had been invested in bonds for the 12 years under discussion, the total capital sum held at this point would be \$1,237,261, or \$124 greater. That quantifies the risk of investing \$100 in bitcoin in a bond portfolio as a loss of about \$124. That \$124 difference would be the amount of loss in principal value after 12 years if bitcoin had failed completely.

In actuality, those bitcoin returns are somewhat understated since the calculation gives no credit for the distribution value of the various bitcoin forks such as Bitcoin Cash and Bitcoin Gold.

The market capitalization of Bitcoin is now \$368.3 billion. Bitcoin is worth less than 7/10ths of 1% of the U.S. publicly traded bond market value. The value of the bond asset class is constantly expanding due to new issuance. If the bond asset class did actually protect investors against debasement caused by inflation, there would be no need for this instrumentality.

The bitcoin return should be genuinely extraordinary if, at some point in the future, the investment community comes to the conclusion that bitcoin will better protect an investor against inflation than the bond asset class which appears to provide no such protection.

Original PDF includes Horizon Kinetics standard research disclosures and reproduction restrictions.

Next: Chapter 7 — The Importance of the Oil Service Industry.

Chapter 7 — The Importance of the Oil Service Industry

Source. Murray's Musings, *Spin-Off Report Compendium*, October 2022. Horizon Kinetics LLC © 2022.

PDF in vault. Spin-Off-Compendium-October-2022_The-Importance-of-the-Oil-Service-Industry.pdf

Extract used for assembly. book/build/text/Spin-Off-Compendium-October-2022_The-Importance-of-the-Oil-Service-Industry.txt

Contributors (as in original). Murray Stahl, Chairman; Steven Bregman, President; Peter Doyle, Managing Director.

Presentation edits: removed compendium running heads and page banners; merged words split across line ends; omitted lengthy standard research disclaimer.

The Importance of the Oil Service Industry

The energy sector weighting of the S&P 500 at the time of this writing is 5.04%. Some two years ago, it was in the vicinity of 2.5%.

Almost one-third of that 5.04% is accounted for by Exxon Mobil Corp. (XOM), with a 1.35% weight. Chevron Corporation (CVX) accounts for 92 basis points. The ConocoPhillips (COP) weight is 47 basis points.

The largest energy service company weighting in the S&P 500 index is Schlumberger Ltd. (SLB) at 19 basis points. The next oil service company to make an appearance is Halliburton Co. (HAL), which comprises 9 basis points. It is the 255th largest company in the S&P 500. Baker Hughes Co. (BKR) is an 8-basis-point position. This group is the extent of the S&P 500 exposure to oil service. In total it amounts to 36 basis points.

The oil service industry obviously can be ignored as a purely statistical factor in the S&P 500 Index. However, without the seismic fracking and well construction services offered by these companies, oil could not be extracted, and the oil extraction companies are important in the index. The oil service industry, as a generalization, has earned very poor returns on capital since 1981. From 2013 until very recently, it was in crisis, with numerous bankruptcies and substantial losses by virtually all industry participants.

Schlumberger, by example, had \$39 billion plus of shareholders' equity in 2003, whereas by the end of 2020, it had dropped to modestly in excess of \$12 billion. By year-end 2021, book value was slightly in excess of \$15 billion. As to fixed assets less accumulated depreciation, generally understood to be plant, property and equipment, this amounted to \$15.096 billion at the close of 2013. By the end of 2021, this figure had declined to \$6.429 billion. It was even lower at the end of June 2022: \$6.386 billion.

In the first six months of 2022, Schlumberger produced a 17.9% annualized return on shareholders' equity. Nevertheless, the company's \$664 million of capital expenditures in the first six months of 2022 is only 62.4% of the \$1.065 billion of depreciation and amortization recorded in this period, which means the balance sheet value of PP&E is still declining at a rapid rate. Schlumberger is not investing in itself even though it is the leading oil service company, with services that are critical to the global extraction of oil and natural gas.

The figures for Halliburton are similarly profound. It had \$15.581 billion of shareholders' equity at the end of 2013, and PP&E net of accumulated depreciation was \$11.322 billion. At the end of 2021, shareholders' equity had declined to \$6.713 billion, and net PP&E had declined to \$4.326 billion. By June, this last measure had declined to \$4.165 billion. The annualized return on shareholders' equity thus far in 2022 is 10.69%. Nevertheless, despite its profitability, Halliburton is making capital expenditures of only about 85% of depreciation expense. This is exclusive of the fact that the company actually sold \$116 million of equipment, and it also excludes yet another \$366 million of impairments in the first half of 2022.

In other words, Halliburton is also disinclined to invest in its business. All of this is quite understandable in the context of Halliburton history. At the end of 2013, the company had debt of \$7.816 billion against shareholders' equity of \$13.581 billion. At the end of 2021, long term debt was a higher \$9.12 billion against half as much shareholders' equity of \$6.713 billion. At the end of June 2022, debt had declined to \$8.525 billion against shareholders' equity of \$7.12 billion. Halliburton is still a leveraged company, although it should be noted that net current assets amount to \$5.496 billion.

Schlumberger stock trades at more or less the price reached in the summer of 2005. The same can be said for Halliburton, even though its shares had a slightly better outcome for the time period in question.

It might be argued that modern oil rigs are more productive than the oil rigs of the past, so that fewer rigs are needed to produce a given quantity of hydrocarbons. The U.S. Energy Information Administration (EIA) calculates drilling rig productivity over time using "new well oil production per rig" as the key metric. The data is contained in a drilling productivity report per region. The data is not so clear on this question.

The EIA data shows that the Anadarko region of Western Oklahoma's new well oil production per rig has more or less quintupled since 2012. On the other hand, the productivity increase since 2012 in a region such as Appalachia has been minimal. This contrast might be compared to the roughly 8x productivity increase in the Bakken region of North Dakota, but with minimal increases in the Haynesville area of East Texas and Western Louisiana.

The Permian Basin is perhaps the most prolific energy producing region in North America, and new well oil production per rig has advanced approximately tenfold there since 2012. Productivity reached a peak in 2020, with significant shutdowns of producing wells in all regions. Only the most productive wells remained in operation. This phenomenon is known as high grading. Since 2020, productivity has been in decline. This is not necessarily indicative of a loss of productivity per rig; it is likely that less productive land is now being brought into production.

Both Schlumberger and Halliburton arguably have now achieved an acceptable level of profitability. This is not true for many oil service firms. Baker Hughes, for example, is profitable on an operating basis, with an operating margin of 6.68%. Unfortunately, the company still reports impairments in non-operating losses. The balance sheet has \$14.88 billion of shareholders' equity against \$6.625 billion of debt, and balance sheet cash amounts to \$2.928 billion. However, goodwill and other intangible assets account to \$5.741 billion and \$4.049

billion. This total of \$9.79 billion can be subtracted from shareholders' equity to arrive at a hard book value of \$5.09 billion. Viewed in this light, Baker Hughes might still be regarded as a leveraged company, so it still displays signs of the damage caused by the long energy depression from June 2014 to June 2021.

The contraction in the number of operating offshore oil rigs has been far more extreme, and the duration of the contraction now exceeds four decades. There were 250 U.S. offshore crude oil and natural gas rotary rigs in operation at the end of 1980.¹² By the end of 2007, just before the financial crisis of the following year, the number of offshore rigs in operation had dropped to 62. In the next 24 months, this figure shrank to 37. The number of rigs in operation offshore then began a gradual increase to 58 in June 2014, on the eve of what was to become an energy depression that lasted for eight years. Data for August 2022 is the most recent, and at that time 18 oil and natural gas rotary rigs were in operation in the offshore U.S. waters.

A shortage of oil may be developing right now. The U.S. Strategic Petroleum Reserve has been used to balance supply in relative to the embargo on Russian oil. The SPR has declined to 416.4 million barrels as of September 30th, 2022, after a drawdown of 201.4 million barrels in the past 12 months. This rate of diminution is not sustainable. One possible solution is a resolution of the Ukraine conflict. That does not seem likely in the foreseeable future.

The other obvious solution is a meaningful increase in U.S. oil production, but in order to accomplish that, the remaining oil service companies must increase investment in their facilities to facilitate that production increase. The government plans, though, are still to replace hydrocarbons with other forms of renewable energy once the current crisis is resolved. Effectively, the proposition to the oil service companies is that they would need to make long term investments to solve what the government hopes is a short-term problem. If the Ukraine conflict is resolved, energy commerce might return to normal, in which case any oil service company that made investments to handle temporary U.S. oil production increases would be left with enormous amounts of redundant equipment.

The oil service companies have experienced about four decades of excess oil service supply and endured enormous losses in order to establish low break-even operating levels. Significant capital expenditure would inevitably raise break-even points for oil production levels that world governments would rather not be sustained. In fact, it is the stated intention of those governments to reduce hydrocarbon production as soon as it can be practically achieved.

The surviving oil service companies, therefore, have absolutely no incentive to increase investment in their output capacity. In fact, the entire industry continues to disinvest in their plant and equipment. It will be difficult, if not impossible, for hydrocarbon production to increase while the oil service companies are reducing their capital investments.

This makes oil service a very interesting industry from an investment perspective. It is the only industry that is currently aligned operationally to endure a depression, simply because it has just experienced a depression. It is therefore a potential non-correlated asset in an equity portfolio. The reluctance of the industry to invest will eventually serve to greatly increase typical oil service company return on equity to very alluring levels.

Original PDF includes Horizon Kinetics standard research disclosures and reproduction restrictions.

Next: Chapter 8 — Wealth and Crisis: The Role of Diamonds in Asset Allocation (European Contrarian, January 2023).

Chapter 8 — Wealth and Crisis: The Role of Diamonds in Asset Allocation

Source. Murray's Musings, *European Contrarian Report Compendium*, January 2023. Horizon Kinetics LLC © 2023.

PDF in vault. `European-Contrarian-Compendium-January-2023_Wealth-and-Crisis_The-Role-of-Diamonds-in-Asset-Allocation.pdf`

Extract used for assembly. `book/build/text/European-Contrarian-Compendium-January-2023_Wealth-and-Crisis_The-Role-of-Diamonds-in-Asset-Allocation.txt`

Contributors (as in original). Murray Stahl, Chairman; Steven Bregman, President; Peter Doyle, Managing Director.

Conflict / offering note (from original). An affiliate of Horizon Kinetics Asset Management LLC acts as co-sponsor to a private fund that invests in diamonds; the fund is available only to certain qualified investors.

Companion essay in this edition. *Diamonds as an Investment* (December 2022) appears as Chapter 4.

Presentation edits: removed compendium running heads and page banners; merged words split

Wealth and Crisis: The Role of Diamonds in Asset Allocation

The modern understanding of asset allocation is based on the premise of social stability. If the assumption of social stability is removed, the conceptual foundation of modern asset allocation vanishes along with it.

For example, a basic building block of asset allocation is the notion that short-term government securities represent a risk-free asset. This is obviously untrue in an inflationary environment, since the asset will lose purchasing power. It is also untrue in an environment of war or civil war.

Inflation has not usually been a problem in the developed world since World War II, with the great exception of the 1970s decade, although inflationary episodes did occur in the aftermath of World War II, as one might expect.

Globally, however, inflation is not uncommon. Trading Economics recently produced a list of 10 nations with very serious inflation:

Table 1: Nations with Highly Problematic Inflation

Country	Annual Inflation Rate
Venezuela	1198.0%
Sudan	340.0%
Lebanon	201.0%
Syria	139.0%
Suriname	62.3%
Zimbabwe	60.7%
Argentina	51.2%
Turkey	36.1%
Iran	35.2%
Ethiopia	33.0%

Source: *TradingEconomics.com*

This list is only comprised of the nations with the highest recorded rates of inflation. If problematic inflation is defined as a recorded rate that exceeds 10% per annum, this list would double. Twenty countries, or about 10% of the world's nations, are experiencing serious inflation.

Inflation, though serious, is not the most grave threat to wealth preservation. The most serious threats are war, civil war, and social unrest.

Civil war can last surprisingly long. In the Western world, investors generally ignore such conflicts. It is astonishing to learn that the Myanmar civil war has been raging since 1948.

The country, once known as the British colony of Burma, achieved independence that year. Since then, seven states of that nation—representing a variety of ethnicities—have engaged in a continuous armed struggle for either autonomy or independence.

This is not the only example. A left-wing armed revolt against the government of the Philippines has been continuing since 1969. An armed struggle between the government of Colombia and the drug cartels has been ongoing since 1964. The civil war in Syria began in 2011. In Sudan, the war in Darfur has been continuing since 2003. The civil war in Yemen has been happening since 2014. Conflict still roils Afghanistan, a nation that's been in a continuous state of armed conflict since 1978.

One might be tempted to dismiss these conflicts as irrelevant to industrialized nations, but one must note that the low-intensity Ukrainian civil war that commenced in 2014 has evolved into an actual Russo-Ukrainian war.

None of the long-term studies of wealth and investment returns really take into account the influence of war and social unrest. Few studies of the impact of war upon wealth appear to exist. Many books have been written on the French Revolution and the Napoleonic wars, but the focus is on military strategy and social change. Similar remarks might be made about the First and Second World Wars. One reason for this is that the period from the Battle of Waterloo in 1815 to the beginning of the First World War in August 1914 is usually considered to be a peaceful era, but it is only peaceful from a Eurocentric or American perspective. In fact, even the American perspective ignores the wealth impact of its own Civil War.

At least fifty 19th-century civil wars have been almost entirely ignored by historians. One

example is the Seinan Senso or Southwestern War, so called because it was largely fought on the island of Kyushu in Southwestern Japan. Essentially, the samurai class was not initially opposed to the Meiji Reform that implemented imperial rule in Japan in 1868, but it became increasingly concerned with the direction the government was taking and established an independent military academy. This was viewed as a serious challenge to governmental authority, and eventually a series of incidents quickly deteriorated into civil war. The government eventually prevailed, but the cost of the conflict nearly bankrupted the country. The war forced Japan off the gold standard and effectively tripled the country's national debt.

This very brief survey of conflict should be sufficient to establish that the probability of destabilizing conflict is not zero. Therefore, an investor's asset allocation profile should contain one or more assets that could maintain value and possibly even increase in value in the event of such a conflict.

Cash, bonds, stocks, private equity, and real estate—the current standards in asset allocation strategies—are vulnerable in the event of war. Cash and bonds are vulnerable since they are obligations of the government, and the creditworthiness of the government is in question during such a crisis. Equities represent business enterprises. Amid conflict, an enterprise might be subject to much higher taxation. Its business prospects could be greatly diminished during war, and might even be seriously damaged by the ensuing violence. The same points would apply to private equity. Real estate can obviously be damaged during an armed crisis, as well. It is also a highly visible asset that is not movable and could be expropriated.

Given the problematic features of conventional assets during a true crisis, the obvious and conventional alternatives are gold and silver.

One could readily dismiss silver as a crisis asset on the question of value density—that is, the physical quantity of an asset needed to afford protection. To demonstrate, let us assume that the basic unit of account for wealth is \$1 million. Using the current silver price of \$23.56/ounce, 42,444.82 ounces of silver would be required to store \$1 million in value. Each troy ounce is 0.18 cubic inches, so this equals 7,674.8 cubic inches, or 4.44 cubic feet, of volume. The challenge of transporting 4.44 cubic feet of silver is not trivial; it would weigh 2,910 pounds, or about that of a Nissan Sentra 4-door sedan.

This is not to assert that silver wouldn't appreciate in a crisis, but the primary wealth management problem in a crisis is not the maximization of return, but the prevention of seizure.

In the case of gold, seizure can be prevented if it is held outside of the regulated custodial system. In comparison with silver, a gold price of \$1,788 per ounce requires only 559.2841 ounces of gold for \$1 million of value, thus mitigating the transportation problem to some extent. Although this is far simpler than transporting 42,444.82 ounces of silver, weighing 38 pounds it still requires a transportation conveyance and is not easily concealed.

Art, in the form of a painting, is potentially value dense. The transportation problem is not necessarily complex and it does not necessarily require specialized conveyance. However, a painting is very fragile. It could be concealed, but maintaining the proper storage conditions to preserve the painting's value in a crisis is not a trivial problem.

This leaves diamonds as an important crisis asset that has not yet become an accepted asset class. Diamonds are light. One carat weighs one-fifth of a gram. Stated in other terms, one carat is equivalent to 0.007 ounces, and there are approximately 2,265 carats to a pound. The average person should have no problem locating a one-carat diamond worth \$2,000 for purchase, although much more expensive stones can be found. Even with a \$2,000 average per-carat price, one pound of diamonds—or 2,265 carats—is worth \$4,530,000.

In contradistinction, gold's value density is much lower. A pound contains 14.5833 troy ounces. At \$1,788 per troy ounce, the value of one pound of gold is $14.5833 \text{ (troy ounces)} \times \$1,788 \text{ (price/troy ounce)}$, or \$26,075. Assuming the purchase of relatively inexpensive \$2,000 diamonds as the basis for the calculations, one pound of diamonds is worth \$4,530,000, whereas one pound of gold is worth \$26,074. The ratio of the two assets, \$4,530,000 versus \$26,074, means that diamonds are 172.58 times as value dense as gold. It should be noted that this exercise greatly understates the potential value density of diamonds.

Investors might not understand the importance of the concept of the value density of diamonds, but the idea has been explored thoroughly in popular literature and film. For instance, in the short story titled "The Adventure of the Blue Carbuncle" by Arthur Conan Doyle, Sherlock Holmes investigates a theft of a priceless gemstone from the hotel suite of the Countess of Morcar. The story would have been impossible to write, had the premise been that the equivalent value of gold were stolen: there would be no reason to have such a quantity of gold in a hotel suite. Moreover, Doyle would've had to explain the logistics of removing a large quantity of gold from a hotel suite without being observed. Similarly, in the Alfred Hitchcock movie *To Catch a Thief*, a cat burglar steals jewels, not gold. The jewelry theft is the MacGuffin, or plot device, upon which Hitchcock weaves the story based on David Dodge's novel of the same name. A burglar's escape by rooftop while carrying a large quantity of gold simply couldn't happen.

Accepting the value density aspect of diamond investing leads to the logical conclusion that diamonds should be the asset of choice for the circumstance when social cohesion is in question. Social cohesion has been in question many times in history, and it is now unraveling in at least

10% of the world's nations. Nevertheless, diamonds have yet to attract institutional interest.

This is probably because most investing in the contemporary era is institutional investing, and institutional investing can only exist with sufficient social cohesion for the institution to continue to function. If a financial institution's investment committee were ever to turn its attention to the potential loss of social cohesion, it almost necessarily follows that the committee also must reflect on the possible inability of the committee and the institution to function. For such a committee considering that possibility, it would essentially be preparing for its own demise, in which case it might have an enhanced interest in a diamond portfolio to access in a genuine state of emergency. That would be much like planning for beneficiaries while simultaneously imagining the demise of the management.

As difficult as it might be to plan for the loss of social cohesion, the great number of breakdowns of the institutional order in many nations just in the first part of the 21st century makes an allocation necessary to an asset that can endure, is universally accepted, is transportable, can be hidden and protected from seizure, and appreciates over the course of time. Such an asset is diamonds.

Original PDF includes Horizon Kinetics standard research disclosures and reproduction restrictions.

Next: Chapter 9 — The Rise of the Artificial Asset Class (Part II).

Chapter 9 — The Rise of the Artificial Asset Class

Source. *The Global Spin-Off Report Compendium*, June 2016. Horizon Kinetics LLC © 2017 (as printed in the compendium PDF).

PDF in vault. 20210622195755_29f44b80_June-2016_The-Rise-of-the-Artificial-Asset-Class.pdf

Extract used for assembly. book/build/text/20210622195755_29f44b80_June-2016_The-Rise-of-the-Artificial-Asset-Class.txt

Placement. Opens **Part II — Indexation, concentration, and “the artificial asset class”** in this reader's edition.

Presentation edits: removed compendium running heads and page footers; merged words split across line breaks; replaced one garbled formula block with a short editorial note (see below); light punctuation cleanup only.

The Rise of the Artificial Asset Class

The Limits of Idiosyncratic Risk Avoidance in Indexation

Modern asset allocation is practiced as if it were a science. The objective is to balance risk and reward in accordance with individual risk tolerance. Careful attention is paid to individual goals and time horizon. This is accomplished by investing in different genres of equity and fixed income, and even cash equivalents. Thus, the risk of U.S. large capitalization versus emerging market equities is considered. The system considers the risk/reward ratio of high yield versus U.S. Treasuries. One now commonly finds foreign currencies invested alongside U.S. dollar cash equivalents.

One result of this systematized approach to portfolio construction is the Robo-advisor, which makes these allocations via computer algorithms. Hence, in practice, Wall Street has achieved a self-driving portfolio before Google has achieved a self-driving car. In fact, Wall Street has achieved a self-driving portfolio before scientists at the MIT Artificial Intelligence Laboratory have invented a robot that will properly clean your home.

Is the “rule set” of portfolio construction far simpler than the “rule set” for driving an automobile? This is a good question, actually. In each case, the obvious problem is to map all possible idiosyncratic circumstances and program a proper response. In the case of driving an automobile, this is quite difficult, but there are definable conditions that precede accidents so that a sufficiently fast computer could undertake avoidance strategies. In contrast, in accordance with Modern Portfolio Theory, news of idiosyncratic circumstances such as earnings misstatements, corruption, terrorist attacks, oil spills, and many other calamities are assumed to be instantaneously reflected in stock prices. How can asset allocation programs react, since individual securities change the character of the asset class?

For instance, if Exxon had spilled oil into the Gulf of Mexico rather than British Petroleum, one could argue that the risk of holding the S&P 500 would be higher, given that Exxon is the third-largest position in the S&P 500. However, the S&P 500 portfolio beta would remain 1, by definition. Upon what basis, if any, could an asset allocation program react?

Indexation itself is designed to mitigate the idiosyncratic security risk problem by owning so

many securities that no individual security, however idiosyncratic, can alter the volatility characteristics of the index. Indeed, at the moment, Exxon is only 2% of the S&P 500 Index.

Yet, as asset allocation becomes more specialized, there is a corresponding increase in specialized indexes used to craft just the right risk/reward balance. An example is the iShares U.S. Energy ETF (IYE), which holds in excess of \$1.2 billion in assets under management. Exxon is a 25.84% position in the index. Surely no one would care to defend the proposition that this index would be immune to the idiosyncratic aspects of Exxon if it were to become involved in a disaster similar to that which befell British Petroleum.

It is extraordinarily difficult to create an index that is sufficiently liquid to accommodate trillions of dollars of investment without being dominated by the weight of a relatively small number of huge global multinational companies. The paradox is that the scope of operations of such firms is so vast across the globe that most idiosyncratic developments on the international stage have some bearing upon the potential profits of these giant enterprises.

The Rise of Artificial Asset Classes

Consequently, parallel to the development of indexation, there has been a movement to devise artificial cash-settled investment vehicles in which the outcome, positive or negative, rests upon the calculation of a completely transparent number. This is made possible in real time by enormous advances in computers, telecommunications, and semiconductor design.

An example of such an artificial index is the VIX Index. The VIX is merely a weighted average of near-term options on the S&P 500. VIX futures were first listed on March 26, 2004. The open interest is now 479,510 contracts with a contract multiplier of 1,000. This is no doubt a considerable sum of money. In any case, considered within the context of contemporary indexation, which accommodates trillions of dollars, VIX is a mere rounding error. Moreover, it is ultimately a derivative of the S&P 500 and, therefore, intrinsically contains individual security risk.

Given the direction of advances in computational power and the need for new asset classes, it was inevitable that a wholly artificial asset class would be created. In the case of the Satoshi white paper in 2008, that artificial asset class was Bitcoin. It is perhaps coincidental that all equity asset classes, all commodity classes, and many classes of fixed income securities simultaneously collapsed in value in 2008.

The basic idea of Bitcoin as an asset class, leaving aside the questions of Blockchain technology, is that it would be accepted as a store of value like money, with the singular difference that the units of money would be fixed so as not to exceed 21 million units. The rate of issuance until the maximum is reached is expressed by a geometric series in the original publication.

[Editorial note: the PDF extract garbled the displayed series and summation notation; the narrative below restates the schedule in prose.]

The 210,000 in the formula is the number of starting coins; the 50 quantity in the formula is the starting Bitcoin proof of work reward. This figure is halved every four years. It will be 12.5 Bitcoin on or about July 11, 2016 insofar as can be calculated. Ultimately, an issuance of 21 million will be attained.

This is structurally different from all other currencies, which are so-called fiat currencies. In the latter instance, the central banks, not infrequently under great pressure from the central government, expand the amount of currency to the degree considered necessary.

The consequence is that money as a store of value — or even in many instances bank deposits, government bonds, and corporate bonds — generate negative real rates of return. This has not been true in the United States and most industrial developed nations since 1981 because interest rates declined. However, it may well become true in the foreseeable future due to the low level of bond rates as well as taxation of whatever little interest income is generated. In most nations of the world, fixed income securities or cash deposits have experienced negative rates of return in the past. As of April 30th, Tradeweb has reported that there are \$7.85 trillion of negative yielding sovereign debt in the world, including the majority of the debt of Switzerland, Japan, Germany, the Netherlands, Austria, France and Denmark.

In essence, Bitcoin is a permissionless instrument. In other words, it can function as money or a store of value if a sufficient number of people accept it as such. In principle, anything can be accepted as money. For example, in the Roman Republic, 20 sesterii, or 5 denarii, was legally equal to 18.22916 and two-third grains of gold. In the time of Nero, the denarius contained 99% silver. Under Trajan and the Antonines, the denarius first contained 5% copper, and eventually it contained 25% copper. Under the Emperor Severus, the denarius was 50% copper.¹³

In history, accepted instruments of exchange and store of value have included barley in Mesopotamia, cowry shells in Aboriginal Australia, tobacco leaves, and other commodities. In post-World War II Germany, cigarettes assumed this function for several years. Of course, gold and silver have been used as money and stores of value.

The common feature of every medium of exchange or store of value used thus far throughout history has been that the supply would generally increase. Sometimes, this was within the ability

of the government to control; sometimes, it was left to circumstance. For example, historians believe that the impact of the California Gold Rush of 1849 was such that, in effect, the supply of money was suddenly increased and this stimulated economies all over the world, not just in the United States. Some would argue that the Transcontinental Railroad was built, in part, by demand for transportation services stimulated by the Gold Rush.

On the other hand, the inflationary pressures of the 1970s were certainly responsible for the increase in the price of gold during that decade. This, in turn, stimulated exploration, which then enhanced supply. Eventually, the price of gold declined precipitously.

Bitcoin, in contradistinction, is fixed in supply. If it were accepted as an asset class and, therefore, as a store of value, why should not this store of value be equivalent in capitalization to other stores of value? In other words, if supply is fixed, the only rationing mechanism remaining is price.

It should be recognized that almost all serious students of central banking and national monetary policy consider Bitcoin to be something akin to a Ponzi scheme. However, as an instrument, Bitcoin offers no promise of profit, which is the animating element of a Ponzi scheme.

Bitcoin has a market capitalization of roughly \$9 billion as of this writing. On its World Debt Clock, *The Economist* calculates that the total global government debt that trades has a U.S. dollar value of \$58.891 trillion.¹⁴ This is just government debt. What if Bitcoin were worth an amount equal to this ever-increasing sum? This would represent a coefficient expansion of 6,654x, or 665,400%.

It is perhaps worthy of note that in March of 2006, the Federal Reserve ceased publishing M3, the broadest money supply measure. It still publishes M2, which was about \$12.652 trillion in April 2016. In the first four months of 2016, U.S. M2 expanded by 2.61%, or an annualized rate of 8.04%. Should one add the value of M2 to world government debt as a somewhat better measure of store of value? If one were to do this, it would still be hopelessly inadequate as a calculation, since it does not account for all the other national M2 calculations.

In any event, if one merely added U.S. M2 and global government debt, the total is \$71.54 trillion, or 7,949x the current market value of Bitcoin. Euro M2 is another €10 trillion. Yen M2 is ¥934.765 trillion, or about \$8.5 trillion. No one has ever needed to calculate the worth of every instrumentality with a store of value function. It would surely be a huge number.

If the various stores of value in the world are engineered in such a manner as to generate negative real rates of return, surely the holders of such instruments would find this objectionable. Historically, if all central banks simultaneously enacted similar real purchasing power destruction policies, the investors had no recourse other than the political process.

Now there is an alternative that is accepted by very few and rejected by most investors. If the predilection of the majority of investors changes to be favorable to Bitcoin, a market capitalization of \$9 billion would hardly be adequate to express those preferences. Indeed, a market capitalization of \$1.227 trillion might not be adequate. After all, this figure is merely the aggregate market capitalization of the four “FANG” stocks: Facebook, Amazon, Netflix, and Google. If an artificial asset class such as a crypto-currency, of which Bitcoin is only one example, is accepted generally, this is a once-in-a-century event in terms of rate of return.

Original PDF includes Horizon Kinetics / PCS distribution boilerplate and reproduction restrictions.

Next: Chapter 10 — Russell 2000® index construction white paper; then Chapters 11–12 (S&P float / bond substitute commentary).

Chapter 10 — Russell 2000® Index Construction: When Small Caps Became a Big Problem

Source. Horizon Research Group white paper (dated January 2013 on cover; December 2013 in filename). Horizon Kinetics LLC © 2013.

PDF in vault.

20160321035257_1cdcd84f_R2000_20Index_20Construction_20White_20Paper_Dec2013.pdf

Extract used for assembly.

book/build/text/20160321035257_1cdcd84f_R2000_20Index_20Construction_20White_20Paper_Dec2013.txt

Contributors (as in original). Research team listing on cover: Murray Stahl, Steven Bregman, Therese Byars, Ryan Casey, James Davolos, Derek Devens, Peter Doyle, Michael Gallant, Matthew Houk, Utako Kojima, Eric Sites, Fredrik Tjernstrom, Steven Tuen.

Presentation edits: removed page banners and repeated copyright lines; normalized paragraph

Russell 2000® Index Construction: When Small Caps Became a Big Problem

The Russell 2000® Index, a subset of the Russell 3000® Index, measures the performance of the small-cap segment of the U.S. equity universe and includes approximately 2000 of the smallest securities based on a combination of their market cap and current index membership. The index is unmanaged and cannot be invested in directly. Russell Investments is the owner of the trademarks, service marks and copyrights related to its indexes. Past performance is not a guarantee of future results.

This report is based on information available to the public; no representation is made with regard to its accuracy or completeness. This document is neither an offer nor a solicitation to buy or sell securities. All expressions of opinion reflect judgment at this date and are subject to change. Horizon Asset Management LLC, and/or its related entities may acquire or maintain positions in securities of companies mentioned within. Reproduction of this report is strictly prohibited. © Horizon Kinetics LLC® 2013.

Introduction

Passive investing via exchange-traded funds (“ETFs”) has risen at an enormous rate over the past several years, as evidenced by the total assets under management held by ETFs. From 2001 to 2011, ETF assets have grown from \$82.9 billion to \$1.0 trillion, representing a compound annual growth rate of 28.8%. During 2012, assets rose an additional 25% to \$1.3 trillion¹, which, as we will see, has deep implications not only for equity ETF investors, but for all equity investors.

This change occurred even as total assets invested in U.S. equity mutual funds have declined since the 2008 Financial Crisis, in favor of bonds. It might be surprising that the net outflows from U.S. equity mutual funds during the first 10 months of 2012 were \$99.3 billion, exceeding the \$80.1 billion outflows during the same period in 2011 (for the full year of 2011, total net outflows reached \$129.6 billion). At this rate, the amounts withdrawn from equity mutual funds over the 2011-2012 time frame (during which the S&P 500 Index (“S&P 500”) produced positive returns) will easily eclipse the \$228.5 billion of assets withdrawn during the 2008 crisis year. Yet, within a net outflow market for equity mutual funds, equity ETFs have received large scale, ongoing net inflows. One might be especially surprised that even while equity mutual funds experienced large outflows in 2008, equity ETFs enjoyed inflows of over \$143 billion.

This may be one of the greatest flows of funds on record. This paper will examine whether this phenomenon is of sufficient scale to alter security and sector clearing prices and distort the character of the critical benchmarking tools upon which investors rely: equity indexes. If so, then the very tools that investors employ to measure risk and return have become unreliable and perhaps, in the extreme, dangerous. We will commence with two of the largest, most prominent equity indexes: the Russell 1000 and Russell 2000 Indexes (“Russell 1000 and Russell 2000,” respectively), although distortions within smaller and more sector-specific indexes will be shown to be far more dramatic.

This phenomenon is not strictly a result of retail investor demand. In fact, many institutional investors, such as pension funds and endowments, have increasingly utilized ETFs as a means to gain exposure to multiple asset classes, including fixed income, commodities, and more esoteric ones such as volatility. These highly sophisticated investors utilize quantitative tools to establish asset allocation policies in accordance with Modern Portfolio Theory, which is based on a belief in the applicability of the Efficient Frontier.

Those familiar with the process of establishing institutional investment policy will understand that a portfolio’s allocation into various strategies, such as “large cap equity,” “small capitalization equity (‘small cap’),” and “foreign equity,” among others, is established only after careful consideration, often utilizing complex calculations that consider data on risk, returns, and volatility. Thus, it fundamentally follows that the investment vehicles chosen to represent a desired asset class must actually own securities from that asset class; otherwise, the investment characteristics expected of the investment would not actually be represented. For example, if an investment vehicle were chosen because it purported to invest in “value” stocks, but instead held “growth” stocks, the entire investment policy would be invalidated.

¹ Source: Investment Company Institute With regard to active managers, the discrepancy described above is referred to as “style drift.” To institutional investors, style drift is a serious offense and is grounds for dismissing the investment manager responsible for perpetrating it. However, what if the style drift is committed by indexes and, by extension, index funds? What would the recourse be if the index is a nationally-recognized standard for scores of billions of dollars of investment capital? Indeed, given that investors typically do not scrutinize index funds in the same way they do active managers, one must assume that the issue of style drift is never mentioned when index funds are utilized.

Consider the Russell 1000 and Russell 2000. These are a subset of the Russell 3000 Index (“Russell 3000”), which contains the largest 3,000 companies in the United States and captures approximately 98% of the U.S. equity market. By definition, the Russell 1000 consists of the 1,000 largest companies as measured by market capitalization (“market cap”), and is universally accepted as an appropriate representation of large cap (“large cap”) companies. The following 2,000 companies, that is, the 1,001st largest capitalization company to the 3,000th largest capitalization company, make up the Russell 2000, which is considered a proxy for small cap stocks. The construction of these indexes is fairly transparent and easily understood.

So far, so good. Except for one problem: the Russell 1000 doesn’t actually represent the largest 1,000 companies and the Russell 2000 doesn’t hold the 2,000 smallest companies. For example, the last company in the Russell 1000 by index weight is Groupon Inc. (Nasdaq: GRPN), which has a market cap of \$3.14 billion.² The largest security in the Russell 2000 by index weight is Ocwen Financial (NYSE: OCN). Its market cap is \$4.72 billion, or 1.5x as large as Groupon. On a float-adjusted basis (which is the market cap excluding shares held by company insiders), the discrepancy is even more glaring. Groupon’s float-adjusted market cap (i.e. the total market cap of its equity not held by insiders) is \$1.25 billion, while Ocwen’s is \$2.83 billion, or about 2.25x as large. Can one persuasively argue that Groupon deserves to be in the large cap index and Ocwen Financial in the small cap index?

Lest one suspect that this discrepancy is an isolated event, expanding the observation to the bottom 50 companies in the Russell 1000 and the top 50 companies in the Russell 2000 yields a similar conclusion. The combined market cap of the bottom 50 companies in the Russell 1000 totals \$110 billion. In contrast, the top 50 companies in the Russell 2000 have an aggregate market cap of \$136 billion.

How, one might reasonably ask, can the aggregate market cap of the top fifty companies in the small cap index exceed the aggregate market cap of the bottom fifty components of the large cap index?

Clearly, a distortion exists in the construction of the Russell indexes (and, we suspect, in many others) that is not recognized by the investment community. Although the construction of these indexes is simple in theory, it is more complex in practice. The result is that investment products are not quite what they purport to be. Given the vast sum of investment dollars affected, which measures in the trillions,³ this is a serious issue that requires study.

² References to market prices are as of December 14, 2012. ³ Russell reports assets under management of \$159.1 billion, and \$3.9 trillion of assets invested in products that are benchmarked to the Russell U.S. indexes.

Therefore, this paper attempts to provide greater clarity with regard to the risks lurking in index-based passive investment products—risks that are not immediately obvious even to professional investors. However, because this paper is intended to be introductory, and the subject matter is complex, we will only touch on the most glaring problems, with less breadth or depth than is truly required. We will explore the central theme—distortions in index construction—as well as highlight a number of other, no less serious, issues arising from the ETF phenomenon.

Index Construction and Methodology According to Russell Investments In order to highlight the distortions in index construction, one must first be familiar with their stated methodology. In the case of Russell Investments (“Russell”), the creator of the Russell indexes, the Russell 3000 (and by extension, the Russell 1000 and Russell 2000) methodology is disclosed in the paper “Russell U.S. Equity Indexes Construction and Methodology.”⁴

According to Russell, the indexes “are constructed to be unbiased and comprehensive, with each index incorporating the entire market segment it is designed to represent.” Additionally, the process is intended to be transparent and rule-based, in that all rules regarding the construction and maintenance of the indexes are published and publicly available. The indexes are designed to be a comprehensive representation of the investable U.S. equity market and to serve three main purposes: to act as performance standards for active managers, to serve as proxies for asset allocation purposes, and to become purchasable and replicable vehicles for passive investment strategies.

To ensure that the market segments are represented accurately, the indexes are reconstituted—that is, rebuilt—annually. This allows for the inclusion of securities in the appropriate market cap (and style) index as they change in size and style over time. We will see that the annual reconstitution is the starting point of the distortion. One will also observe that most of the steps outlined below seem entirely straightforward and logical.

Basic Methodology

Index construction begins with the Russell 3000 Extended (“Russell 3000E”), which is intended to be the broadest U.S. index and is comprised of the largest 4,000 U.S. companies. It is from this comprehensive universe that the various domestic Russell indexes are created, including the similarly-named Russell 3000, which holds the largest 3,000 securities. All companies that are determined to be part of the U.S. equity market are included in the Russell U.S. indexes. Those determined to be non-U.S. are included in the Russell Global ex-U.S. Index.

Inclusion in the U.S. index can be determined based on a number of criteria. These are:

☐ Home-country indicator—if a company incorporates in, has a stated headquarters location in, and also trades in the U.S., it is assigned to the U.S. index; however, ADRs and ADSs are not eligible. ☐ Trading requirement—all securities for inclusion must trade on a major U.S. exchange; bulletin board, pink sheet or over-the-counter traded securities are not eligible for inclusion. ☐ Minimum closing price—a stock must have a closing price at or above \$1.00 on the last trading day in May.

4 http://www.russell.com/indexes/data/US_Equity/russell_US_indexes_methodology.asp

☐ Minimum total market cap—companies with a total market cap of less than \$30 million are not eligible. ☐ Minimum float requirement—companies with less than 5% available float (those shares available to the public; i.e. excluding shares owned by insiders/management) will not be eligible for inclusion. ☐ Corporate structure—companies with the following structures are excluded from inclusion: royalty trusts, U.S. limited liability companies, closed-end investment companies (although business development companies are eligible), blank-check companies, special-purpose acquisition companies, and limited partnerships. ☐ UBTI—companies that produce unrelated business taxable income (“UBTI”) are restricted from inclusion in Russell U.S. indexes. ☐ Share exclusion—the following share types are not eligible for inclusion: preferred and convertible preferred stock, redeemable shares, participating preferred stock, warrants, rights, and trust receipts.

Once eligible members are identified, Russell calculates the market cap of each security and sorts them in descending order. Companies number 1 through 1,000 are then included in the Russell 1000, which is the company’s large cap index offering. Companies number 1,001 through 3,000 are allocated to the Russell 2000, the company’s small cap index. Numbers 2,001 through 4,000 become members of the Russell Microcap Index (i.e. the smallest 1,000 securities in the Russell 2000 plus the next 1,000 securities).

The market cap level that separates the 1,000th and 1,001st company is deemed the “breakpoint” between the Russell 1000 and Russell 2000. Given that the breakpoint is based upon the demarcation line between companies 1,000 and 1,001, and not a predetermined market cap level, the breakpoints vary year to year, as market caps invariably fluctuate. This is evident from the following table.

Exhibit 1: Historical Breakpoints for Russell 1000/Russell 2000 Year 1,001st Breakpoint 2012 1,943,992,920\$ 2011 2,066,827,500\$ 2010 1,714,702,220\$ 2009 1,296,536,150\$ Source: ITG

Accordingly, Russell’s definition of large cap stocks, as expressed in the Russell 1000 has ranged from as low as \$1.29 billion to as high as \$1.94 billion within the past four years. Investors typically define large cap stocks as companies with market caps over \$10 billion. If this definition is used, then the Russell 1000 represents a deviation from the commonly accepted notion.

The Russell indexes follow market cap weighting rules, which means that the index weighting of each member is based upon the size of its market cap relative to the entire universe. In such a scheme, the largest companies (i.e. Exxon Mobil Corp. and Apple Inc.) have a commensurate weighting in the indexes, with an additional proviso that the weightings be adjusted to reflect free float.⁵ The method by which this is achieved is as follows:

☐ Sort the Russell 3000 members in descending order by total market cap. ☐ After membership is determined, adjust a security’s shares to include only those shares available to the public. ☐ Calculate the total market cap of the Russell 3000 by summing all members’ float-adjusted market caps. ☐ Calculate the percentiles for each company in the Russell 3000 by dividing the float-adjusted market cap associated with each member by the total market cap of the Russell 3000.

Within the index world, the adoption of float-adjusted ranking is relatively new and represents a material methodological change. Although correlation is not necessarily indicative of causation, it is interesting to note that several major index providers moved to float-adjusted weightings within a few years of one another, just as the flow of capital into ETFs began to surge. For example, FTSE, the British-based provider of global indexes, moved to a free float methodology in 2000. In 2002, the investment analytics and index company MSCI made the change. This was followed by Standard & Poors, which made the decision to move to a free float system for its U.S. indexes, including the S&P 500, S&P MidCap 400, and the S&P Small Cap 600 Indexes, in 2005. Between 2000 and 2005, ETF assets under management swelled from \$65.5 billion to \$300.8 billion, a 4.5-fold increase. There are important implications of such methodological changes that are not immediately obvious.

One will therefore find that the companies at the top of the indexes virtually dominate the rest of the index members. For example, the top 10 companies in the Russell 1000, combined, represent 17% of the index, while the bottom 100 have an aggregate weight of only 0.8%, which equates to less than 1 basis point, or 1/100th of 1%, each.

Reconstitution The process by which the indexes are constructed occurs once every year, which is called the Russell Reconstitution, a rebalancing in layman’s terms.

Reconstitution begins in May. On the last trading day in May, all of the eligible securities are ranked by their total market cap, in accordance with the process delineated above. In the

beginning of June, preliminary lists for the additions and deletions to the indexes are communicated to the marketplace. In late June (June 22 nd in 2012), the changes go into effect.

The exception to the reconstitution process involves corporate actions that occur between the rebalancings. For example, securities that leave an index for any reason (i.e. mergers, acquisitions, bankruptcy, among others) are not replaced. Specifically in mergers, the company that survives remains in the index, with the acquired company's market cap moving to the acquiring company's stock. Additionally, newly-created companies following spin-offs and initial public offerings are added as they occur, provided they meet the requirements of the index.

5 The purpose of this adjustment is to exclude the capitalizations that are not available for purchase and therefore not part of the investable opportunity set.

Overall, the methodology that Russell follows in constructing its indexes is fairly straightforward. However, it did introduce a slight change—yet one with great implications—in 2007. Prior to that year, Russell followed a “hard cutoff,” in that the ranking was strictly arbitrary. In other words, the 1,001 st stock in the ranking became the first stock in the Russell 2000.

In 2007, Russell began to use a “capitalization band,” in which the company retained the flexibility to override the process. It works as follows: Russell calculates a range of five percent around the newly determined market cap breakpoints, this being a range of 2.5% above and 2.5% below the breakpoints. Those companies that would automatically be deleted from an index due to a change in their ranking (e.g., from #995 [Russell 1000] to #1,050 [Russell 2000]), would stay in the current index. Such banding, according to Russell, is intended to reduce “unnecessary turnover.”

Theory versus Practice

“In theory there is no difference between theory and practice. In practice there is.” -Yogi Berra

In theory, if one desired to replicate the Russell 1000/2000/3000, it would be a relatively straightforward exercise, given the rules that must be followed. Each company would be listed in order of its market cap (float-adjusted), and one could easily discern where the breakpoint should be.

However, if one were to examine Russell's own data, one would find that the theoretical list and the actual list diverge markedly, as shown in the following table:

Exhibit 2: Russell Index Historical Market Capitalization Ranges (1995-2012) Hi Lo Hi Lo Hi Lo
 2012 540,213\$ 101 \$ 540,213\$ 1,354 \$ 2,608 \$ 101 \$ 2011 411,180 130 411,180 1,624 2,971 130
 2010 283,061 112 283,061 1,256 2,274 112 2009 338,408 78 338,408 829 1,688 78 2008 468,981
 167 468,981 1,363 2,751 167 2007 468,500 262 468,500 2,500 2,500 262 2006 368,500 218
 368,500 1,970 1,960 218 2005 386,900 183 386,900 1,800 1,800 183 2004 317,800 176 317,800
 1,600 1,600 176 2003 286,600 117 286,600 1,200 1,200 117 2002 309,500 128 309,500 1,300
 1,300 128 2001 486,700 147 486,700 1,400 1,400 147 2000 520,200 178 520,200 1,600 1,500
 178 1999 407,200 178 407,200 1,400 1,300 178 1998 271,600 222 271,600 1,400 1,400 222 1997
 198,000 172 198,000 1,100 1,100 172 1996 138,000 162 138,000 1,000 1,000 162 1995 98,400
 104 98,400 748 750 104 (\$ in millions) Source: Russell Investments Russell 3000 Russell 1000
 Russell 2000 The above exhibit shows that from 1995 until 2007, the separation between the
 Russell 1000 and Russell 2000 was discernible: the low range of the Russell 1000 matched very
 closely to the high range of the Russell 2000 for each respective year. For example, in 2006, the
 Russell 1000 contained companies with a market cap of \$36 8.5 billion at the upper end and
 \$1.97 billion at the lower end. The Russell 2000 then contained companies with market caps of
 \$1.96 billion at the upper end and \$218 million at the lower end. These are the types of ranges
 one would expect.

However, following the introduction of banding, the market cap ranges between the Russell 1000 and Russell 2000 began to overlap. During 2008, the Russell 1000 contained companies with a market cap of as little as \$1.36 billion. Meanwhile, the Russell 2000 held companies with market caps of as much as \$2.75 billion. In other words, the small cap index included companies that were almost twice as large as those in the large cap index. As a result, the indexes are no longer true to their original mandates.

This naturally leads to the question: “Why have they changed?”

Answer: Because they are no longer theoretical constructs; they are now substantial asset management businesses.

The Transformation of the Index into a Business Recall that Russell cites three specific goals for its indexes: 1. to act as performance standards for active managers 2. to serve as proxies for asset allocation purposes, and 3. to become purchasable and replicable vehicles for passive investment strategies.

Historically, the index industry fulfilled the three purposes to varying degrees, with the greatest emphasis on being a performance standard for active managers. This purpose is self-explanatory: without a credible benchmark, it is difficult for investors to evaluate their

investment managers. As well, the Russell indexes were utilized as proxies for asset allocation purposes. It is for this reason that Russell offers multiple indexes: by combining groups of equities with different characteristics (i.e. small cap and large cap; domestic and foreign), diversification can presumably be achieved, and in doing so, returns can be enhanced and risk reduced for the entire investment portfolio.

However, the index industry has increasingly evolved from one of measurement into one of providing investment products—the third leg of the index business “stool.” Although passive investing has existed for many years, such as through index mutual funds, over the last several years it has virtually exploded with the rise of ETFs. According to the Investment Company Institute, in only five years (2006-2011), assets under management in equity ETFs essentially doubled, from \$387.3 billion to \$754.4 billion, representing a compound annual growth rate of 14.2%. The benchmarks are now vigorously competing for investment dollars with active managers. As the following table will illustrate, the strategy is winning.

Exhibit 3: Net Cash Flows (2006-2011)

Year Equity Mutual Funds Equity ETFs 2011 (129,650) \$ 68,585 \$ 2010 (36,910) 80,031 2009 (138) 42,086 2008 (228,498) 143,644 2007 74,151 128,116 2006 148,494 59,792 (\$ in millions)

Source: Investment Company Institute Net New Cash Flows With this transformation, distortion of the index itself is unavoidable, because indexation as a business is a low-fee endeavor. The expense ratio of the iShares Russell 2000 Index (ticker IWM), for instance, with \$15 billion of AUM, is all of 0.2%. Accordingly, in order to remain a viable business, the indexes must be highly scalable. A given ETF's asset gathering capacity, though, will be limited by its least liquid member.

Let's illustrate with a simplified example: an ETF with, say, 100 constituents, all equally weighted at 1%, its smallest constituent having a \$150 million market cap, and limited to owning not more than 4.9% of a constituent's shares. The allowable limit for each holding would then be \$7.35 million. This ETF will be able to accept only \$735 million of assets.

If the next-largest constituent has a \$300 million stock market cap, and if the rule set can be altered to exclude the smallest company, then the asset gathering limit rises to \$1.5 billion. Alternatively, if that smallest company has inside ownership of 50%, and if the ETF adopts a float-adjusted weighting methodology, then it can be reduced from a 1% to 0.5% position, and the asset gathering limit again rises to \$1.5 billion. Thus, it is the liquidity issue that influences index construction, and the fixed mechanical rules are not really fixed anymore.

What is the result? Those companies that belong at the top of the Russell 2000, but which have limited float, are instead placed at the bottom of the Russell 1000. In this way, the weighting of the security (and thus the amount of stock required to be traded) is reduced greatly, since the indexes are capitalization weighted. Whereas the top 100 companies in the Russell 2000 typically have an average weighting of approximately 19 basis points, the bottom 100 companies in the Russell 1000 have weightings of less than one basis point.

In return, companies that would normally belong at the bottom of the Russell 1000 but which have ample liquidity are moved into the top of the Russell 2000, where the weighting is some 20 times larger than if they were in their proper place in the Russell 1000. But even this does not fully capture the potential rate of expansion, since there is far more money allocated to the Russell 2000 than the Russell 1000. Judging by the respective iShares ETFs, the differential is 6 \$150 million market cap x 4.9% ownership = \$7.35 million. 7 \$7.35 million per holding x 100 members = \$735 million.

on the order of 2.5-to-1. Effectively, then, the dollar volume of demand for companies placed at the top of the Russell 2000 is some 40x larger than if placed at the bottom of the Russell 1000.

The following table highlights a few of the companies whose placement appears to be biased in favor of the Russell 1000, even though their float-adjusted market caps would argue against their inclusion in that index.

Exhibit 4: Sample Companies in Russell 1000

Float-adjusted Company Market Cap Owned by % Ownership Market Cap U.S. Cellular (USM) 3,092.0 \$ Telephone & Data Systems 84.3% 507.2 \$ Echostar (SATS) 2,936.9 Charles Ergen 50.6% 1,330.0 Kronos Worldwide (KRO) 1,905.4 Valhi 50.5% 340.3 Titanium Metals (TIE) 2,883.9 Various Harold Simmons Entities 54.3% 1,316.0 Clear Channel Outdoor (CCO) 2,443.4 Clear Channel Communications 88.7% 273.6 (\$ in millions) Source: Bloomberg

It can be seen that these companies have substantial insider ownership, and therefore the float-adjusted market caps are substantially less than the stated market caps. Indeed, several of these (U.S. Cellular, Kronos Worldwide, Clear Channel Outdoor) have float-adjusted market caps of \$500 million or less.

If these securities were instead included in the Russell 2000, it would be very difficult to execute the index along the lines of the current methodologies, since the scalability of the index would be limited by the float of these companies and others like them. However, because these securities are overshadowed by the likes of Apple and ExxonMobil, which occupy the lion's share of the aggregate market cap of the Russell 1000, they are assigned to the lower reaches of

the Russell 1000, where they have essentially zero weight. Even one hundred of such companies, and even if they were all rapidly expanding and very successful, would make no statistically meaningful impact upon the performance of that index (although, within the Russell 2000, such a cohort could have a dramatic impact).

Compare the above with the following constituents of the Russell 2000:

Exhibit 5: Sample Companies in the Russell 2000

Float-adjusted Company Market Cap Market Cap Pharmacyclics (PCYC) 4,380.0 \$ 3,476.4 \$
Two Harbors Investment Corp. (TWO) 3,376.4 3,329.6 athenahealth Inc. (ATHN) 2,747.2
2,648.8 Dril-Quip Inc. (DRQ) 2,838.5 2,304.5 Coeur d'Alene Mines (CDE) 2,084.5 2,072.8 (\$ in
millions) Source: Bloomberg Comparing Exhibit 4 with Exhibit 5, one can easily observe the
inherent flaw in the actual implementation of the indexes. There is no reason—other than the
attempt to accommodate massive demand—why a company such as Two Harbors, with a
market cap of \$3.4 billion and a float-adjusted market cap of \$3.3 billion should be included in
the small cap index while a company such as Kronos Worldwide, which has a market cap of
\$1.9 billion and a float-adjusted market cap of only \$340 million, should be in the large cap
index.

In the process of filling that demand, the fund flows will ultimately influence the price of the
underlying stocks in significant ways. Take, for example, the following table, which contains
data on the bottom 50 companies in the Russell 1000 and the top 50 companies in the Russell
2000.

Exhibit 6: Comparative Valuation Metrics Difference Total Market Cap.: 109,984 \$ 135,540 \$
Total Book Value: 71,182 \$ 56,118 \$ Total Sales: 94,849 \$ 68,936 \$ Total Net Income: 5,062 \$
6,595 \$ Price/Book Value: 1.55 x 2.42 x 56% Price/Sales: 1.16 x 1.97 x 70% Forward P/E: 21.73
x 20.55 x -5% (\$ in millions) Source: Bloomberg Bottom 50 of the Russell 1000 Top 50 of the
Russell 2000 Collective Data for the

When comparing the Russell 1000's bottom 50 companies with the top 50 of the Russell 2000,
we see that the aggregate market cap of the top 50 Russell 2000 companies actually exceeds
that of the bottom 50 in the Russell 1000. As we said before, this is quite extraordinary given the
market cap-based weighting rules the indexes purportedly follow.

In terms of equity valuations, the top 50 Russell 2000 companies clearly command a valuation
premium. For instance, the aggregate price to book value multiple for the Top 50 is 2.4x, while
the Bottom 50 trades at a much lower multiple of 1.5x. The price to sales measure is equally
lopsided: the Top 50 companies have a price to sales multiple of almost 2x, which is much
higher than the Bottom 50's multiple of only 1.1x. Only on the forward P/E measure are the two
groups comparable, owing to a number of companies with negative earnings in the Bottom 50.
If we were to exclude these unprofitable companies (since negative P/Es distort the aggregate
valuation), the valuation discrepancy becomes even clearer, as shown below:

Exhibit 7: Comparative (Adjusted) Valuation Metrics

Difference Total Market Cap.: 84,904 \$ 132,690 \$ Total Book Value: 63,292 \$ 55,895 \$ Total
Sales: 77,324 \$ 68,732 \$ Total Net Income: 5,635 \$ 6,650 \$ Price/Book Value: 1.34 x 2.37 x 77%
Price/Sales: 1.10 x 1.93 x 76% Forward P/E: 15.07 x 19.95 x 32% *Adjusted to exclude companies
with negative earnings (\$ in millions) Source: Bloomberg Collective (Adjusted) Data for the
Bottom 50 of the Russell 1000 Top 50 of the Russell 2000*

The issue of index construction is not merely academic. Companies that should be in the top of
the Russell 2000 but which have fallen to the bottom of the Russell 1000 because of liquidity
concerns are heavily penalized. Were they to remain in their proper location, that is, the top of
the Russell 2000, their share prices would be higher, all else being equal. Conversely, companies
that are fortunate enough to be placed at the top of the Russell 2000 solely because of their
available float have an unfair advantage. The reason for this occurrence is that the Russell must
accommodate the needs of the ETF industry.

Unintended Consequences Aside from the problem of style drift, the way in which the Russell
indexes are currently constructed also introduces other unanticipated and unintended
consequences that would adversely impact returns. We highlight five that are especially
relevant.

Indexes vs. Insiders The purchase or sale of company stock by an insider, such as the chief
executive officer, chairman, or founder, is a closely watched affair. Those companies in which
insiders make significant purchases are considered to be attractive candidates for investment.
Conversely, when insiders sell, outside observers may infer that the shares might be overvalued
or that the company might be encountering problems.

This is a rational strategy that attempts to profit from asymmetric information: insiders are
much more knowledgeable about the prospects of their companies than the average outside
minority investor, and this information is manifest in their stock transactions. As well, one can
assume that wealthy entrepreneurs and owner-operators who have personal capital at risk in
their companies, tend to make a distinguished series of risk/reward decisions (i.e. capital
deployment) that ultimately should result in excess share price returns over time relative to

various broad market benchmarks. There is ample statistical support of this phenomenon.

According to this theory, large insider purchases should be a positive signal that argues for purchase of the shares. Yet, by the rules of the float-adjusted weighting system, the opposite would occur. As insiders buy shares, the available float declines, prompting a reduction in the company's weighting in the index, all else equal. This is counter-rational behavior and, we suspect, is not even noticed by the index investor.

At the other extreme, when companies issue more shares, it is typically considered to be negative for existing shareholders, as dilution reduces the intrinsic value of their equity. However, an increase in a company's available shares that leads to a higher float-adjusted market cap will induce the indexes to purchase more shares, even as informed insiders sell.

Ultimately, as a result of the index methodology, the emphasis is not on the qualitative features of the index members, but simply on their liquidity.

Small Cap vs. "Small Cap"

In light of the evidence presented thus far, one might legitimately ask, "What is a small cap stock?" This question is a serious one because, if a small cap stock is merely an enterprise that has the same characteristics as a large cap stock except that it has fewer shares outstanding and is, therefore, less liquid, then the method of indexation as practiced by Russell (and others) is entirely appropriate. However, if one views small capitalization stocks as smaller businesses and, consequently, as a source of incremental optionality (as we discuss below), then it means indexation as currently practiced is far from appropriate.

In this discussion, optionality refers to the phenomenon of investing in small cap companies which become extraordinarily successful. Indeed, the allure of holding small companies lies in the possibility of producing a return on the order of 10, 20, 30, or even 40 times one's invested capital. The whole concept of small capitalization investing, therefore, is to develop a structure in which it is possible to take the risk of allowing one or more companies to flourish.

However, if the structure of the index emphasizes market cap as dictated by available float, then the same dynamic would not apply. The index would include "small caps," which would only be small caps by the narrow definition of indexation, but which are really large (or larger) companies without the same potential for expansion as true small cap stocks, as dictated by the law of large numbers. Wal-Mart has \$445 billion of annual revenue, Microsoft \$74 billion; it has been a long time since they were capable of a sustained double-digit rate of revenue expansion. The index-altered definition is not small cap investing in the traditional sense; it's something else, and it's not likely to produce a very robust rate of return.

Gresham's Law and IPOs The investment banking community is beginning to realize that an initial public offering ("IPO") issued with sufficient liquidity and market cap will be included in a variety of indexes. Therefore, one effectively has a mathematically certain aftermarket bid, which is a phenomenon that had not previously existed, since the construction rules, as one may recall, allow for the addition of IPOs outside of the reconstitution time frame.

Gresham's law is an economic principle that states that when a government compulsorily overvalues one type of money and undervalues another, the undervalued money will leave the country or disappear from circulation into hoards, while the overvalued money will flood into circulation. It is commonly stated as: "Bad money drives out good."

If this pattern continues, it is mathematically possible that the Russell 2000 will gradually, but inexorably be populated by IPO companies. An IPO-type of Gresham's Law will crowd out the smaller cap, more reasonably valued small positions within the index.

Since index funds must follow their respective index construction rules (i.e. avoid any tracking error), they will be required to purchase such IPOs. Yet, not only would they have no information about the company in question, they would have no desire for any information other than its market cap and float. In other words, these funds would be the ultimate uninformed buyer.

Turnover as a Source of Risk

What does it mean for investors when the liquidity—the dollar volume of buying and selling—of a derivative instrument such as an ETF is greater than that of its constituent parts?

The average daily trading turnover for IWM is equal to roughly 20% of its market cap; phrased differently, annual turnover is on the order of 5,000%, which is, frankly, absolutely astounding. For 2011, ETFs alone were responsible for over \$20 trillion of trading, which means that turnover for the entire group was 20x their \$1 trillion of assets under management, or about 2,000%. As a result, there is greater trading volume in IWM than in many of the companies comprising it. It has been calculated that if IWM had to create a sufficient number of new units (provide liquidity) for the aggregate volume of short positions in that fund, it would take the ETF more than 180 days to buy all the component securities if it limits its buying to 10% of the average daily volume of each holding.⁹ This extreme does not apply to the more liquid Russell 2000 holdings, and there are other mechanisms for providing liquidity in a continuous market, such as derivatives, but clearly the daily liquidity needs of the ETF in creating new or destroying old units cannot always be met in the open market.

Because the liquidity and daily trading volume of some of the largest ETFs now exceed the liquidity of the stocks that comprise their underlying indexes, this phenomenon has reached a critical point: indexes, which are meant to measure the performance of, and provide exposure to groups of stocks, have come to distort the prices of the stocks they are meant to measure. Paradoxically, the trauma of the 2008 financial crisis, which instigated a surge by investors into these instruments as a tool to diversify their holdings in order to reduce overall portfolio volatility has, by the magnitude of these efforts, actually contributed to the increasingly lockstep movements among indexes.

Hard Indexes vs. Soft Indexes, the Nasdaq 100 Index, and the Top-Heaviness Problem

Hard indexes and soft indexes are terms invented for this paper. A “hard index” is one with fixed mechanical rules for inclusion and weighting, such as the Russell indexes or the Nasdaq 100 Index (“Nasdaq 100”). The Nasdaq 100, it should be remarked, is deceptively simple: the 100 largest domestic and international non-financial securities (based on market cap) listed on the Nasdaq.

9 Primary source: J.P. Morgan Equity Derivatives Strategy; cited by Harold Bradley and Robert E. Litan, for the Ewing Marion Kauffman Foundation in their October 19, 2011 prepared testimony before the U.S. Senate Committee on Banking, Housing and Urban Affairs, Subcommittee on Securities, Insurance and Investments. However, the ETF Industry Organization has estimated ETFs’ proportion of total notional U.S. equity volume at 33% (<http://www.etf-ia.com/sites/default/files/Notional-Volume.pdf>).

A “soft index” is one that allows for human judgment. The Dow Jones Industrial Average and the S&P 500 are examples of soft indexes because apart from any mechanical rules, there are committees that can decide to include or exclude given securities for those two indexes.

There is an inherent problem with hard indexes: they eventually become top-heavy. One of the constituents within the hard index is going to be the best investment, even if the index itself does not rise in value. In the case of the Nasdaq 100, that constituent would be Apple Inc. So successful was Apple that it was necessary to limit that company’s weight in the Nasdaq 100. At present, its weighting in the index is approximately 16%. Were an active manager to report a 16% concentration in a single stock, it would almost certainly attract due diligence attention.

To have a single company representing 16% of an index that purports to be a diversified portfolio representing the largest companies that trade on Nasdaq is reason enough for concern. However, that’s only the beginning.

It should be noted that Microsoft Corp. has a 7.5% weighting. Intel, the semiconductor company, accounts for 3.4% of the Nasdaq 100. Including Apple, these three companies are battling for dominance in the personal electronic device market. Other companies in the Nasdaq 100 that also compete in the personal computer industry include Adobe Systems (0.62% weight), Dell Inc. (0.61%), Symantec Corp. (0.44%), Seagate Technology (0.37%), NVIDIA Corp. (0.26%), Research in Motion (0.24%), and Micron Technology (0.23%).

What is the implication? It means that the Nasdaq 100, with the above companies accounting for one third of the index, is effectively a personal computer index, even though it’s supposed to have much broader representation. Thus, it negates the basic conception of indexation, which is that a panoply of companies will be owned such that no single element can unduly affect the investment result. However, it is obvious here that one single investment element not only affects the result, but determines the result in every conceivable scenario. No other industry in the Nasdaq 100 is as large as that of personal computers.

Conclusion The theory behind indexation is very simple: asset managers generally do not beat the indexes, and securities markets are generally efficient in any case. With fees and transactional friction, active managers might even subtract value.

ETFs enable investors to invest in passive indexes in an efficient manner. The advantages are well known: they have low expense ratios, are easily traded (with liquidity available throughout the day), provide broad diversification, and are tax efficient. However, much like any other investment product, buyers should conduct due diligence before purchasing.

While the concept of indexes is straightforward, implementation is another matter. There is ample evidence to suggest that, for business reasons, indexes have been manipulated in ways that belie their stated goals. However, because they are taken at face value, investors do not invest the time needed to fully understand the compromises that must inevitably be taken in order to make the product viable. In other words, what one intends to buy may not be what one actually gets, and that may mean the difference between meeting or missing one’s investment goals.

DISCLOSURES: This material is for information purposes only. Opinions and estimates offered constitute our judgment and are subject to change without notice, as are statements of financial market trends, which are based on current market conditions. Under no circumstances does the information contained within represent a recommendation to buy, hold or sell any security, and it should not be assumed that the securities transactions or holdings discussed were or will prove to be profitable or that a future investment decisions will be

profitable or will equal or exceed the past investment performance of the securities listed. All material presented is compiled from sources believed to be reliable, but no guarantee is given as to its accuracy. The following applies to all of the indices used in this document: Indices are unmanaged, may or may not be investable, have no expenses and reflect reinvestment of dividends and distributions, and may not be comparable either in composition or level of risk. This information should not be used as a general guide to investing or as a source of any specific investment recommendations, and makes no implied or expressed recommendations concerning the manner in which an account should or would be handled, as appropriate investment strategies depend upon specific investment guidelines and objectives. This is not an offer or solicitation to any person in any jurisdiction in which such action is not authorized or to any person to whom it would be unlawful to make such offer or solicitation. The information contained herein has been obtained from sources believed to be reliable; however, no representation is made regarding its accuracy or completeness. The Russell 2000® Index, a subset of the Russell 3000® Index, measures the performance of the small-cap segment of the U.S. equity universe and includes approximately 2000 of the smallest securities based on a combination of their market cap and current index membership. The index is unmanaged and cannot be invested in directly. Russell Investments is the owner of the trademarks, service marks and copyrights related to its indexes. Past performance is not a guarantee of future results. No part of this material may be: a) copied, photocopied, or duplicated in any form, by any means; or b) redistributed without Horizon Kinetics' prior written consent.

Original PDF includes Russell/ETF disclaimers, disclosure block, and reproduction restrictions.

Next: Chapter 11 — Not Your Grandfather's S&P.

Chapter 11 — Not Your Grandfather's S&P

Source. *Under the Hood: What's in Your Index?* (An Ongoing Series, July 2015). Horizon Kinetics LLC © 2015.

PDF in vault. 20160323071835_6bf4d9d1_Not_20Your_20Grandfather_s_20_20SP_FINAL.pdf (canonical row in `curated_works_book_yes.csv`; duplicate hash crawl also on disk).

Extract used for assembly.

book/build/text/20160323071835_6bf4d9d1_Not_20Your_20Grandfather_s_20_20SP_FINAL.txt

Presentation edits: flattened multi-column layout into linear paragraphs; merged line-break splits; table of 30 largest S&P revenue changes preserved from extract.

Not Your Grandfather's S&P

While much of the recent market commentary offered by the financial media outlets is focused on the reasonableness (or not) of the market P/E of about 17x and about the reasonableness of that P/E in relation to “consensus” estimates of corporate profit growth, whether 3% or 6%, it is interesting to note that the largest corporations in the S&P 500 Index do not actually appear to be growing much, if at all. Of the 30 largest companies in the S&P 500, only one-third experienced any revenue growth in 2014, and only one-half had revenue growth of 2% or more. A traditional growth company such as Coca-Cola was negative, Merck was negative, as were IBM, Cisco Systems and General Electric. The aggregate sales growth of these 30 largest U.S. companies was only 4%; exclude just Facebook and Google, and growth was barely above 1%. Perhaps the top tier of the S&P is now too large to grow at a meaningful rate, which has implications for investors in the largest companies; or perhaps it was just a bad year.

There is another question, though, more subtle yet with deep implications for any investor who selects equities, makes asset allocation decisions, or conducts retirement planning or pension plan funding projections. What if the historical returns of the S&P 500, which form the basis for projecting future stock market returns, and therefore powerfully influence the proportion of a pension plan or 401k that will be allocated to stocks or bonds, are not actually representative of what investors are purchasing if they invest in the index today?

The nature of the S&P 500 today — the way the performance is measured — is different than it was before 2005. The difference came about this way: in order to accommodate greater quantities of assets for management in index-based investment products linked to the S&P 500 — that is, provide greater liquidity — Standard & Poor's shifted to a float-adjusted weighting system for the index constituents. The necessity grew out of the way indexes are constructed: all positions were weighted based on their total, or gross, stock market values. Once indexation began to be truly popular, though, with truly large inflows of funds, the problem for the index manufacturer was that if 40%, say, of the shares of a company were owned by insiders, there might be an insufficiency of publicly available shares, or float, to accommodate the volume of assets destined to buy those shares without inflating the share price and valuation. In practical terms, float-adjustment means that a company like Oracle or Microsoft would have its weighting in the index reduced by the proportion of inside ownership. The higher the inside ownership, the smaller the weighting would be set, so that there would be a proportionately lower index-based demand for the shares.

The entrepreneurial companies of today, those with substantial inside ownership (such as Tesla or Google) are accorded lower weights in the S&P 500 than would have been the case prior to 2005. Likewise, their impact upon the Index's returns will be lower than they would have been prior to 2005, when they would have been given full weight based upon their stock market capitalization. Therefore, the historical returns of the S&P 500 – upon which our notions of historical rates of return are based – are no longer continuous with or representative of what the S&P 500 is today.

Accordingly, how big would the impact have been on the S&P 500 had the entrepreneurial companies of the past been float-adjusted, as they are today? Here are two examples known to all: Wal-Mart and Microsoft, which span two generations of innovative service and technology business development in the U.S.

Wal-Mart was added to the S&P 500 in August 1982. Its initial weighting was a mere 0.25%. Sam Walton died in the first week of April 1992. During that decade, the Wal-Mart stock price appreciated by 28.0x, which translates to over 40% per year, and its Index weighting expanded to 2.0%. Inside ownership in 1982 was 38%, as it was after Sam Walton died. Had the S&P 500 applied its current float-adjusted rules back then, 38% of that truly spectacular return would not have been part of the historical S&P 500 return.

Just two years later, in 1994, Microsoft was added to the S&P 500, at a 0.85% weighting. Insiders owned about 40% of the shares. When Bill Gates announced his departure a dozen years later in 2006, the shares had risen in value 8.3-fold, or by over 19% per year (and the S&P 500 by 9.1% per year, inclusive of dividends). Microsoft's weighting in the Index had expanded to 2.0% (and was as high as 2.6% in 2004). A large part of this return would also have been lost from the S&P 500 had it been float-adjusted at the time. The liquidity-motivated float-adjustment mechanism, even if applied only to these two companies, would have had a measureable negative impact upon the Index's returns. What of the others: the Apples, the Intels, the Hewlett-Packards, the Nikes, the Starbucks, the Schwabs? We believe that the S&P 500 today cannot provide what it did in the past, to the degree that any great future entrepreneurial ventures will find their way into the Index.

As an aside, there is something interesting about the pattern of the Microsoft share ownership. When added to the S&P 500, the share price was about \$2.41. At January 1999, when the stock was \$44.75, insiders still owned 31% of the shares. By September 1999, near the peak of the Great Technology Bubble, when the shares were \$47.50 (when Microsoft's weight in the S&P 500 exceeded 4%), inside ownership had been reduced to 26%, and by early September 2000, the very threshold of the collapse of that bubble, when the stock was \$35, inside ownership had dropped to 19%. Today, 15 years later, the shares are only \$44.37 – a 1.6% annualized return – and were quite a bit lower than \$35 only two years ago. Whether by fortune or perception, insiders were dramatically reducing their holdings going into a decade-plus period of decline and stagnation. What did outsiders do? Had the float-adjusted index weighting method been in place at the time, then as insiders sold, such that the float increased, the Index rules would have increased the Microsoft weighting, and mutual funds and other index investors would have been buying more— more of what the insiders were selling.

Although most invest in indexes on the basis of the name on the wrapper, the name doesn't really tell you what's inside. It's not your grandfather's S&P 500.

12-Month change in revenue, 30 largest S&P 500 companies

Company	Change	Company	Change
Apple Inc.	6.72%	Coca-Cola	-2.85%
Microsoft	11.69%	Merck	-6.89%
Exxon Mobil	-8.09%	Citigroup	-3.30%
Johnson & Johnson	5.92%	Facebook, Class A	54.69%
Berkshire Hathaway	12.12%	Google, Class A	19.56%
General Electric	-1.69%	Google, Class C	19.56%
Wells Fargo	1.31%	Gilead Sciences	16.72%
Procter & Gamble	0.58%	PepsiCo, Inc.	1.41%
JPMorgan Chase	7.51%	IBM	-4.55%
Verizon	4.06%	Comcast	3.34%
Chevron	-4.88%	Walt Disney	8.37%
Pfizer Inc.	-12.55%	Cisco Systems	-3.01%
Intel Corporation	-1.19%	Wal-Mart	1.52%
AT&T Inc.	1.03%	Oracle	2.95%
Bank of America	1.96%	Philip Morris Int'l	-1.32%

Average change: 4.1%. Excluding Facebook and Google: 1.1%. Source: Company reports, Bloomberg (as in original PDF).

Next: Chapter 12 — Equity as a Bond Substitute.

Chapter 12 — Equity as a Bond Substitute

Source. Horizon Kinetics market commentary, June 2015. Horizon Kinetics LLC © 2015.

PDF in vault. 20160323071733_52444169_Bond_20Substitute_June_202014_FINAL.pdf (*filename encodes “June 2014”; body title is June 2015 — use PDF title for citation precedence.*)

Extract used for assembly.

book/build/text/20160323071733_52444169_Bond_20Substitute_June_202014_FINAL.txt

Presentation edits: removed running “MARKET COMMENTARY” headers; merged hyphenation across line breaks; Table 1 converted to prose-friendly layout from the extract’s monospaced block.

Equity as a Bond Substitute

It is quite understandable, in the world of low interest rates, that income-dependent investors would be open to bond alternatives as a means of generating income. One such popular alternative is income-producing common stocks as, for instance, those held in the iShares Core High Dividend ETF (HDV). This fund has \$4.7 billion in assets under management, a very reasonable 12-basis-point fee, and a trailing 12-month yield of 3.4%.¹⁵

In early June, the 10-year Treasury bond was yielding 2.19%; hence, it is not difficult to see the allure. HDV has produced more than a 13% compound annual rate of return in the three years ended 03/31/2015, and an 8.6% rate of return in the last year. In contrast, in the three years ended March 31, 2015, the iShares 7-10 Year Treasury Bond ETF (IEF) produced an annual rate of return of about 3.5% (though the 1-year return of 8.8% is comparable to that of HDV).

How much does this return data really inform you? Are these statistics sufficient, in the absence of any qualitative information, to suggest that the future returns will reprise those of the past? It is important to consider the HDV holdings. There are 74, many of which are important positions in the S&P 500, as is evident in Table 1. Consequently, the performance of this fund has relevance to the performance of the S&P 500 Index.

Since the index is not very different from the S&P 500, the concept of income stocks as a bond substitute is not very different from the concept that stocks can be used interchangeably with bonds. In this framework, ExxonMobil, often the largest position in HDV, is a bond substitute. By contrast, using this logic, British Petroleum (BP), which is a European company belonging in the asset class of European equities, is not a bond substitute. In fact, British Petroleum, despite a dividend yield twice that of ExxonMobil, is not even a member of the iShares International Select Dividend ETF (IDV).

HDV is considered a relatively low-risk investment. This is simply because its equity beta is 0.80, that is, its share price has varied about 80% as much as the stock market during some measurement period. The dividend yield is 3.4% and its trailing 12-month P/E ratio is 17.87x, according to iShares. Consider, however, that 24.5% of the fund is invested in the energy sector, and that the price of oil declined by approximately 25% in the six months ended May 31, 2015—it is no surprise that the energy companies reported lower earnings in the first quarter of 2015, which lower earnings may well persist for the remainder of 2015. Consequently, HDV is at least potentially composed of fairly high-P/E stocks, depending on what happens to the price of oil. In any case, the ultimate P/E may well be substantially more than the current measure of 17.87x trailing earnings, unless the energy companies are somehow able to revert to their 2014 profit levels despite far lower oil prices. The energy companies, in aggregate, reported earnings above analyst estimates, but below earnings last year.

Of course, the most important question pertaining to index-trackers such as HDV relates to the risk relative to that of owning bonds. The index that HDV tracks, after all, was created as a bond substitute even though a stock, unlike a bond, has no maturity date and is, in fact, a long-duration asset.

An investor who thinks of the iShares Core High Dividend Index as a long-term bond might consider several factors. The iShares 20-Plus Year Treasury ETF (TLT) has produced a total return in the last 36 months of slightly more than 8% on a compound annual basis. By contrast, HDV has produced a return roughly 1.6 times that amount. That performance might tempt one to assert that the companies comprising HDV have been growing their businesses significantly during those 36 months, with this growth accounting for the substantial rate of return. However, the revenue growth for the year ended December 31, 2014 was a modest 2.7%. If oil prices don’t increase substantially, we know what’s going to happen in 2015. The three-year rate of revenue growth of the 50 companies is, cumulatively, 9.7%, on average, or 2.1% annualized.

There seems to be little doubt that HDV, essentially, is populated with mature firms that

dominate their industries yet collectively have little prospect of expanding their businesses much beyond the rate of inflation. Of course, these businesses are subject to cyclical forces, such as changes in the price of oil. They are also potentially subject to permanently disruptive forces, such as the growing impact of solar energy on electric utilities.

Table 1 (as in original extract layout). Revenue growth for HDV constituents; source line per PDF.

Year-over-Year	3-year cumulative	Year-over-Year	3-year cumulative
1 XOM Exxon Mobil Corp.	-6.35%	41 AEE Ameren Corp.	3.68%
2 T AT&T Inc.	2.87%	42 WEC Wisconsin Energy Corp.	2.78%
3 GE General Electric	1.74%	43 COH Coach Inc.	-9.77%
4 VZ Verizon Communications Inc.	5.42%	44 MXIM Maxim Integrated Products Inc.	-0.96%
5 CVX Chevron Corp.	-7.38%	45 CA CA Inc.	-3.31%
6 PFE Pfizer Inc.	-3.84%	46 SCG Scana Corp.	10.14%
7 PG Procter & Gamble	-2.50%	47 RSG Republic Services Inc.	4.41%
8 PM Philip Morris International Inc.	0.10%	48 POM PEPCO Holdings Inc.	4.54%
9 MO Altria Group Inc.	0.23%	49 MCHP Microchip Technology Inc.	12.27%
10 COP ConocoPhillips	-4.69%	50 HAS HASBRO Inc.	4.78%
11 MCD McDonalds Corp.	-2.37%	51 CINF Cincinnati Financial Corp.	9.14%
12 KMI Kinder Morgan Inc.	15.32%	52 PNW Pinnacle West Corp.	1.07%
13 LLY Eli Lilly	-15.13%	53 LNT Alliant Energy Corp.	2.23%
14 OXY Occidental Petroleum Corp.	-4.25%	54 LEG Leggett & Platt Inc.	8.77%
15 DUK Duke Energy Corp.	5.14%	55 ORI Old Republic International Corp.	-0.67%
16 SO Southern	8.08%	56 OGE OGE Energy Corp.	-14.47%
17 KRFT Kraft Foods Group Inc.	-0.07%	57 WR Westar Energy Corp.	9.70%
18 LMT Lockheed Martin Corp.	0.53%	58 WDR Waddell & Reed Financial Corp.	16.64%
19 D Dominion Resources Inc.	-5.21%	59 STR Questar Corp.	-2.54%
20 EMR Emerson Electric	-0.83%	60 FMR FirstMerit Corp.	8.62%
21 NEE Nextera Energy Inc	12.45%	61 VR Validus Holdings Ltd.	-3.29%
22 KMB Kimberly Clark Corp.	0.83%	62 PNY Piedmont Natural Gas Co.	14.95%
23 CTL CenturyLink Inc.	-0.35%	63 FII Federated Investors Inc.	-2.16%
24 BAX Baxter International Inc.	11.39%	64 CBRL Cracker Barrel Old Country Store	4.26%
25 GIS General Mills Inc.	-2.49%	65 TRMK TrustMark Corp.	2.85%
26 AEP American Electric Power Inc.	10.83%	66 SJI South Jersey Industries Inc.	21.34%
27 SE Spectra Energy Corp.	6.98%	67 NWN Northwest Natural Gas Co.	-0.66%
28 PPL PPL Corp.	-1.89%	68 GHJ Greenhill & Co.	-4.18%
29 RAI Reynolds American Inc.	2.85%	69 FI Frank S International NV	6.96%
30 NOV National Oilwell Varco Inc.	11.54%	70 EE El Paso Electric Co.	3.15%
31 IP International Paper	0.57%	71 IPHS Innophos Holdings Inc.	-0.59%
32 WM Waste Management Inc.	0.09%	72 CCOI Cogent Communications Holdings Inc.	9.20%
33 XEL Xcel Energy Inc.	7.06%	73 AVX AVX Corp.	-4.34%
34 MAT Mattel Inc.	-7.11%	74 OB OneBeacon Insurance Group Ltd.	1.85%
35 NYCB New York Community Bancorp Inc.	-2.31%	Average	2.69%
36 NUE Nucor Corp.	10.78%	Annualized	2.13%
37 K Kellogg	-1.45%		
38 PAYX Paychex Inc.	9.33%		
39 DTE DTE Energy	27.33%		
40 CLX Clorox	-0.66%		

Source: iShares, Company reports, Bloomberg. As of December 2014 (or closest month for companies with a different reporting cycle) — per original.

In summary, then, viewed as a bond substitute, the companies that comprise HDV are subject to various business risks. Furthermore, the ETF as a whole is subject to the valuation risk that inures to any perpetuity when interest rates rise, as eventually will be the case. As a bond substitute, HDV is, in effect, a super-convexity index. Its historical volatility characteristics—upon which investors have in large measure based their investment decisions—do not yet include periods of rising interest rates. (Indeed, there was no such index in the 1970s.) HDV's standard labeling statistics—the equity beta of 0.80 and its standard deviation of 9.1%—upon which buyers rely, do not reflect the risk characteristics of the index.

Moreover, since the firms held in HDV account for a substantial portion of the S&P 500, it follows that the potential volatility characteristics of even that index are poorly understood. The basic idea of asset allocation is that there are three basic asset classes — bonds, equities, and cash-equivalents— that have different levels of risk and return, meaning that they should behave differently over time. The creation of equity bond substitutes, and the manner in which these are valued of late, runs contrary to the historical practice.

Indexation is said to be based on modern portfolio theory. But the providers of these products don't follow modern portfolio theory. They simply say that they do.

Chapter 13 — International Diversification: Bet You Don’t Know How Much You’ve Got

Source. *Under the Hood: What’s in Your Index?* (An Ongoing Series, June 2015). Horizon Kinetics LLC © 2015.

PDF in vault. 20160323072107_00f409dc_InternationalDiversification_June2015.pdf (larger on-disk copy in the 00f409dc hash cluster; alternate crawl:

20150714204413_22ac304f_InternationalDiversification_June2015.pdf).

Extract used for assembly.

book/build/text/20160323072107_00f409dc_InternationalDiversification_June2015.txt

Presentation edits: merged words split across line breaks from PDF extraction; recreated allocation tables from the opening pages; preserved numerical claims and punchlines; light punctuation only.

International Diversification – Bet You Don’t Know How Much You’ve Got

The central tenet of modern diversification is to eliminate the unsystematic risk events in a portfolio, so that the positive performance of some securities will offset the negative performance of others. The policy is to own many investments, as opposed to simply owning a few, and estimate future rates of return by extrapolating past rates of return. This seemingly scientific and statistically based reasoning is extended beyond the individual security level to categories of investments. It is therefore also the practice to establish some meaningful international exposure as a risk diversifier for a U.S. equity portfolio. Amongst institutions, this orthodoxy has already been accomplished. It is also becoming the norm for individual investors, with software-generated asset allocation models (“robo advisers”) suggesting optimal mixes of ETFs. The now standard mode for accomplishing this is via indexes. Even sophisticated institutions like a large state pension fund might manage three-quarters of their internally managed equities (that portion not assigned to outside advisers) through indexes.

Displayed below is a typical asset allocation model for a large-scale pension plan that could be responsible for tens or hundreds of billions of dollars of assets, as well as one suggested by the American Association of Individual Investors and another by a robo adviser.

Table 1: NY State & Local Retirement System, recent asset allocation

Segment	Share
U.S. stocks	38%
Foreign stocks	17%
Bonds, cash, mortgages	21%
Inflation-indexed bonds	6%
Alternatives (private equity, real estate, other)	18%
Total	100%

Source: New York State & Local Retirement System, Annual Financial Report (as cited in original).

Table 2: Recommended asset mix — American Association of Individual Investors*

Segment	Share
U.S. stocks	50%
Foreign stocks	20%
Intermediate bonds	30%
Total	100%

*For moderate-risk investor: age 35–55, 20+ year investment horizon. *Source: American Association of Individual Investors (as cited in original).*

Table 3: Recommended asset mix — robo adviser*

Segment	Share
U.S. stocks	33%
Dividend stocks (also U.S.)	6%

Segment	Share
Total U.S.	39%
Foreign stocks	27%
Natural resources	5%
Municipal bonds	29%
Total	100%

*For a risk tolerance of 5.0 on a scale of 10. *Source: A top-10 robo adviser (as cited in original).*

The institutional portfolio, shown in Table 1, is allocated about 55% to common stocks; 31% of the stocks are allocated to foreign equities. Bonds and cash are about 21% and, having the sophistication and scale, the institutional account also has 18% of the portfolio in a variety of alternative asset classes like private equity and real estate. Despite the differences in resources between, say, a \$175 billion pension fund with institutional trading desks and teams of analysts, and a \$100,000 IRA owner, the asset allocation output is essentially similar.

The individual investor, at least in the case of a self-defined profile for moderate risk and return objectives, will be advised to employ various ETFs to achieve about a 66% to 70% weighting in equities, of which about 30%–40% will be in foreign stocks, as in Tables 2 and 3. AT&T, as an example of a major corporate pension fund, currently has 38% of its equity investments in international stocks.

So, those are portfolio construction rules based on statistical attributes. Now for a qualitative inquiry. Let's start with the leading domestic equity index, the S&P 500, which would be the building block for any of these allocation models. Its largest 50 positions, being only 10% of all the companies in the index, nevertheless account for 45% of the market value of the entire index; we'll presume that this is substantially representative of the balance. This exercise is truncated to pay some respect to the 80/20 rule of efficiency.

As to the S&P's true geographic diversification, even a domestic index is not purely domestic, in that a large multinational company such as Coca-Cola, despite being a truly all-American and U.S.-domiciled business, certainly sells a lot of its product outside our borders. So what do we have?

On average, **44%** of the revenues of these 50 companies are derived from outside the U.S. Even if surprising, this is actually a conservative figure inasmuch as some companies only list "North American" revenues, which would include Mexico and Canada, as opposed to U.S. revenues. Other companies, like AT&T, do not state their foreign revenues, so AT&T is counted in the exercise as zero, despite having significant non-U.S. activities. One piece of information AT&T does provide is that it has \$26 billion of foreign currency swaps (notional value) on its balance sheet, which may be compared with \$129 billion of total revenues—presumably, its non-U.S. revenue is not trivial.

The largest 100 companies in the S&P 500 Index, 20% of the list by number, comprise 61% of the market value; their average proportion of foreign revenues is **39%**. There is a different way to calculate this: not on a simple average basis for each company, but weighted by each company's market value; in this instance the average value-weighted foreign revenue contribution is **26%**. If weighted, instead, by asset value, the figure might well be higher.

But those fine points don't matter, do they? We now know that investors, whether major institutions that, effectively, advise themselves, as well as individuals who are advised by institutions, are being told to allocate roughly 30% to 40% of their stock investments, which might otherwise be in the S&P 500 or essentially similar indexes, to non-U.S. companies. **Yet, by holding the S&P 500 those investors are already fully allocated internationally.** Say what???

One of the paradoxes here is that the major U.S. companies are quite globally diversified in terms of sources of business, if anyone would care to read their annual or quarterly reports. Historians will one day be baffled as to how well-educated people were convinced to invest billions, and perhaps hundreds of billions of dollars, in companies of which they know nothing, often located in places they would be afraid to visit, all in search of diversification that was already achieved by the American firms that were being sold to fund the international diversification effort in the first place.

As a final note, without going into any further detail, would it be surprising to learn that the **10 largest holdings** in the basic international equity index that one is recommended to purchase in order to internationally diversify the already internationally diversified S&P 500 Index, namely the **MSCI EAFE** Index, get **30%** of their revenues from the U.S.? There must be a better way. Step 1, perhaps, is to understand that investing and economics are social sciences and cannot be reduced to simple or programmatic mathematical and statistical methodologies.

Upcoming Under the Hood reviews (as teased in the original): What's in your small cap index? What's in your emerging markets index? How safe is your utility index?

Chapter 14 — Further Basic Principles of Croupier Investing

Source. *Contrarian Research Report*, Horizon Research Group, April 27, 2006. © Horizon Research Group, 2006.

PDF in vault. 20160512031900_7a5f9d04_Further_20Croupier_20Basics_20-_20_204-27-06.pdf

Extract used for assembly.

book/build/text/20160512031900_7a5f9d04_Further_20Croupier_20Basics_20-_20_204-27-06.txt

Research staff (as listed on cover). Steven Bregman, Michael Gallant, Vivek Rattan, Murray Stahl, Peter Doyle, Naveen Kumar, Peter Schweitzer, Arul Thareja, Thomas Ewing, Tina Larsson, Eric Sites, Fredrik Tjernstrom, Steven Tuen.

Presentation edits: removed page banners and duplicate copyright lines; normalized blank lines; light hyphenation cleanup only.

Studies in Absurdity — Subject: Further Basic Principles of Croupier Investing

I — The Fundamental Theorem of the Croupier

Given any economic growth rate for a society, it appears that the socio-economic inter-relationships among the various constituent members of that society multiply more rapidly than the economic growth rate. If true, this would be a function of specialization; of course, it is specialization that makes possible productivity improvement that, in turn, makes possible the expansion of the production possibility frontier that, in turn, makes possible a higher standard of living. In a specialized society, the failure of even a small portion of the constituent members to perform economic functions as expected or required can entail widespread and possibly unpredictable consequences for the economic systems. The consequence for securities markets is that the reasons for market value fluctuation of any individual security or set of securities is multi-causal and probably theoretically infinite. Thus, alternatively expressed, the risks to a set of securities cannot be completely enumerated. One can only more completely enumerate the risks than might have been possible hitherto. This yields that which will be called the Fundamental Theorem of the Croupier: Given any risk control strategy for a portfolio, it should always be possible to devise a more precise and refined risk control strategy.

II — The Fundamental Premise of the Fundamental Theorem

It will be observed that the fundamental theorem is based, in part, upon a notion that might be referred to as the fundamental premise. This is stated at the onset of this paper in Section I and can be more briefly formulated as follows: The complexity of a given society grows more rapidly than the given society itself. It is perhaps for this reason that the trading volumes of both the 2-year and 5-year U.S. Treasury future on the Chicago Board of Trade each exceeded \$10 trillion in 2005. The entire U.S. national debt is only at the level of \$8.9 trillion. Hence, if the U.S. national debt were only comprised of 2 and 5 year notes (which is obviously false) and there were no trading in the actual underlying instruments (which is also obviously false), the turnover rate on U.S. Treasury debt would be at the level of 224%. The turnover rate is actually much greater. In terms of risk, the 2- and 5- year Treasury Notes may be considered to be one of the lower risk security types that exist in the world. The turnover figure should be compared with NYSE 2005 turnover of 102%. (source: NYSE Factbook). It seems reasonable to suppose that there is a greater need to control risk for the companies listed on the New York Stock Exchange than there would be for the 2- and 5-year U.S.

¹ Source: CRB Commodity Yearbook 2005

Treasury securities. In fact, according to the U.S. Bureau of Public Debt, the total amount of marketable U.S. debt outstanding as of March 31, 2006 is \$4.34 trillion². Approximately \$1.042 trillion of this amount is Treasury Bills that mature in less than one year. Treasury notes of all maturities are only outstanding at the level of \$2.4 trillion. Thus, viewing only the 2- and 5-year Treasury Futures traded on the Chicago Board of Trade and without reference to any trading in the underlying instrument, it is fair to say that turnover in these instruments easily exceeds the annual figure of 900%. One might wonder why relatively shorter term U.S. Treasury securities generate as much futures activity as has been the case. The answer can only be reasoned supposition. The analytical tools used for risk control in the field of bonds are far more precise than those used for equities. For instance, a bond portfolio with a given convexity and a given duration can be expected to generate a predictable return in accordance with the interest rate environment. One can know the effect of small changes in portfolio duration. The corresponding equity tools such as net long/short exposure or effective beta are far less precise. Ergo, it seems that portfolio activity varies in proportion to the precision of the

available risk control tools rather than in proportion to the risk of the underlying instruments. Hence, one can state the following working supposition: Working Supposition – A croupier can stimulate trading activity by making available risk control tools. Clearly, the complexity of a society, or in this case the complexity of the banking system, is a significant factor. The modern bank is a large holder of U.S. Treasury Securities. It answers not only to shareholders but also to stakeholders such as depositors, borrowers, regulators, correspondent banks as well as legislators. These stakeholders will tolerate very little volatility of shareholders' equity. The failure of the bank to properly function can be disastrous for wide swathes of a given society. Given interest and principal payments as well as the continual passage of time, a bank portfolio is constantly changing. Therefore, much activity is required merely to maintain convexity and duration at a constant level.

III — The Logical Consequences of the Working Supposition

- A. Activity Stimulation in Commodities Due to Competition The New York Mercantile Exchange, known as NYMEX, is the dominant North American crude oil contract trading venue. On April 21, 2006, the NYMEX set an all time daily volume record of 1,420,734 futures and options contracts. Of course,

- 2 Over \$4 trillion of the U.S. national debt is in the form of non-marketable securities held by the Social Security Administration, in the form of U.S. Savings Bonds or held by various state treasuries as escrow for bond redemption.

NYMEX trades other commodities aside from oil. In addition, recent record energy prices have contributed to interest in commodity trading. However, in recent years the Intercontinental Exchange (NYSE - ICE) has created competition for NYMEX. The Intercontinental Exchange has become a significant energy futures exchange. This is partly the result of the Intercontinental Exchange purchase of the International Petroleum Exchange in 2003. On April 5, 2006, the Intercontinental Exchange reported its third consecutive month of record volume with volume for March 2006 at a level that was 115% greater than March 2005. Over 50% of ICE volume is in the Brent Crude Contract. This contract directly competes with the primary NYMEX contract which is based upon Light Sweet Crude 1,000 barrels per contract delivery point Cushing, Oklahoma. The Brent contract traded on ICE is based upon light, sweet North Sea crude that is often thought of as an alternative to the NYMEX primary light sweet crude oil contract. Commentary with regard to the NYMEX and ICE usually makes reference to the competition between these two similar contracts. However, even such competition can present opportunity. For example, NYMEX now trades contracts based upon the spread between the West Texas and Brent contracts. If trading moves towards the ICE Brent contract, the resultant demand might create a price differential between the West Texas and Brent contracts and this would be reflected in a change in value in the spread contract. NYMEX now trades spread contracts on the differential between the light sweet crude oil futures contract and the Canadian Bow River crude at Hardisty, Alberta. NYMEX also trades contracts based upon the differential between light sweet crude oil and the following four domestic U.S. grades of crude oil: Light Louisiana Sweet, West Texas Intermediate, West Texas Sour and the so-called MARS blend. Thus, the differences between the NYMEX and ICE futures as well as the differences between NYMEX and Canadian Futures Exchanges actually have become the basis for new futures contracts. The competition actually creates the possibility of new trading securities. It should be noted that there are significant differences between NYMEX and ICE. NYMEX is primarily a futures exchange. A futures exchange trades commodity contracts with standard uniform characteristics. In contrast, nearly one half of ICE is an OTC or over the counter exchange. OTC contracts have non-uniform or customized terms. Thus, there can be many different variations of an OTC Brent contract. Moreover, a futures contract may ultimately be settled by actual delivery of an underlying commodity. An OTC contract is always by cash payment. Each contract is a non-standardized arrangement between two counterparties. Furthermore, many OTC contracts are actually differential or spread contracts. For instance, an OTC contract may be based upon the spread differential of energy product between two delivery points such as Henry Hub and Cushing. Roughly 45% of the ICE business is OTC. The OTC business in notional amount is growing at 235% per annum. Hence, ICE is

not merely an institution devoted to the capture of market share from NYMEX. It appears that the mere availability of new trading instruments is a stimulus to usage. The fact that each exchange can continually devise new trading instruments is an illustration of the Fundamental Theorem of the croupier as stated in Section I. B) Activity Managed Exchange Traded Funds According to the Mutual Fund Factbook 2005 edition published by the Investment Company Institute, there were 8,044 U.S. mutual funds in existence in 2004. This accounted for over \$8 trillion of net assets. There are over \$12 trillion of purchases of mutual funds as well as over \$12 trillion of redemptions each year. If the mutual funds industry were an exchange it would record 150% annual turnover and be larger than the NYSE. Naturally, none of this activity takes place upon an organized exchange. It is processed by a handful of custodian banks. These figures do not include the rapidly growing ETF's or exchange traded funds that actually do trade upon organized exchanges. According to the Investment Company Institute, Exchange Traded funds have (as of February 2006) net assets that total approximately \$315 billion. The comparable figure for U.S. mutual funds in February 2006 was \$9.2 trillion of net assets. Hence, the ETF's represent only 3.4% of total mutual fund assets. If conventional mutual funds could trade during the trading day as do the ETF's, it is obvious that an enormous quantity of

activity would be created that would be of direct benefit to various croupier companies. In fact, entirely new types of transactions would take place. It would become possible to create basis trades among managers. One could conceivably establish a long position in a fund with good management and a short position in a fund with poor management. One could further refine the trade by limiting the selection universe to managers of similar styles. In fact, it is not absurd to imagine that manager basis trades might evolve into an entirely new asset class. It would certainly have less risk than pairs trading since there is no possibility of an acquisition at a premium of the fund held in short position. Consequently, this type of basis trade would be at the very least viewed as a pairs trade with no exogenous event risk. The funds would almost certainly trade at net asset value, as does any conventional exchange traded fund, since there would never be a shortage of supply given that demand would be satisfied by the open-end nature of the actively managed exchange traded fund. In fact, manager skill level basis trades are not the only hedging or risk control that could be envisaged. In principle, a fund could be long or short one or more of its competitors. It is already possible to do this with passive exchange traded funds. Of course, croupiers would enjoy an enormously expanded level of business activity. The reader perhaps experiences at this point the feeling of futility that often accompanies any exercise in pure fantasy. However, this is not an exercise in fantasy.

The creation of actively managed Exchange Traded Funds is under consideration by the Securities and Exchange Commission. Interested readers should consult SEC release No. IC-25258; File No S7-20-01 entitled Actively Managed Exchanged-Traded Funds. The release is a request for public comment. Passive Exchange Traded Funds were only authorized in 1993. The first such fund was based upon the S&P 500 and is known as SPDR Trust. It trades on the American Stock Exchange. The SPDR Trust has \$52 billion of net assets and recently has been trading in excess of 70,000,000 shares per day. Daily trading value therefore exceeds \$9 billion, or 17.6% turnover every day. Therefore, over the course of a year, given perhaps 245 trading days, there should be the astonishing turnover figure of 4,330%. This is clearly an important product for the American Stock Exchange. In order to comprehend this turnover figure it may be useful to examine the trading characteristics of ExxonMobil which is the largest company in the S&P 500. ExxonMobil has an average trading volume per day of 18.4 million shares. This results, at current prices, in an average recent daily trading value of \$1.17 billion. During the course of a year at this rate ExxonMobil will have experienced 72% turnover. Similar calculations can be performed for all of the leading S&P 500 companies and a similar result will be obtained. For instance, General Electric daily trading value is only \$993 million. This would result in an annual turnover rate for the GE share of 68.7%. If one were to ignore double counting of volume on NASDAQ and accept without adjustment the reported Google volume figure, then Google has daily trading value that is equivalent to approximately \$5.7 billion. If continued at the current rate for a trading year, this would generate a turnover rate of 971.5%. In reality, the true turnover rate is lower. Yet this is higher than GE and Exxon. It is very small in relation to the SPDR turnover rate. A useful comparison of Google without adjustment for NASDAQ double counting of volume is to compare these figures with the comparable unadjusted volume figures for the NASDAQ 100 Trust, otherwise known as the QQQQ. This trust has a share price of 42 and has recent daily volume, as reported by NASDAQ, of 130 million shares. Thus daily trading value equals \$5.46 billion. The net assets of the NASDAQ 100 are, according to NASDAQ, roughly \$21.4 billion. Thus the NASDAQ 100 would, in principle, generate an annual turnover rate of 6,150%. It must be emphasized again that due to NASDAQ double counting of volume, neither the Google nor the NASDAQ turnover rates are as high as enumerated for the purposes of this exercise. However, since double counting affects both instruments the ratio between the Google and the NASDAQ 100 turnover figures may be examined to obtain a sense of the differences in activity levels between the two instruments. NASDAQ 100 seems to have over 6x the activity of Google. This is similar to the U.S. Treasury futures activity level observed in relation to the activity level of the New York Stock Exchange (section II). In fact, the turnover rate of

the SPDR Trust traded on the American Stock Exchange is far greater than the turnover rate of the New York Stock Exchange even though most of the constituent members of the SPDR (S&P 500) actually trade on the NYSE. As a consequence another working supposition may be stated as observed in Section II of this paper: Working Supposition # 2 – It seems as if trading activity is a function of the perceived precision of the instrument rather than of the risk of the underlying instrument. Readers may object to this supposition upon the basis that trading activity is really a function of liquidity. Yet, liquidity is merely a synonym for trading activity. Thus, one would say a given security trades at great volume because it is liquid and it is liquid because it trades at great volume. The more interesting point is that ExxonMobil, with a market value of \$396 billion, will perhaps generate trading activity of \$1.17 billion, whereas the SPDR with a market value of \$52 billion will generate trading activity with a value that exceeds \$9 billion. One might therefore expect that an actively managed exchange traded fund will generate much trading activity. The reader might wonder why anyone might propose the creation of an actively managed exchange traded fund for any reason apart from the generation of mere trading activity. In fact, many well reasoned arguments can be advanced. One such argument relates to proper expense allocation. As noted earlier in this section, the turnover of open end mutual fund assets, based upon the figures made available by the Investment Company Institute, suggest a turnover rate that easily exceeds 150%. In order to accomplish the purchase and redemption of shares in a mutual fund, the fund itself must engage the services of a custodian to keep shareholder records. In any fund, there are those who trade often, which means frequently purchase and redeem, and those who are simply content to be long term

investors. Since the expenses associated with purchase and redemption are paid by the fund, these costs are shared by all fund participants. The most significant costs are never included in the expense ratio. These are the commission and trading friction costs that the fund experiences when the fund itself trades to accommodate admission and withdrawals. Although the operating costs of the fund must necessarily be shared by all fund participants, an actively managed exchange traded fund would provide a mechanism whereby investors could purchase and sell units outside of the fund on an exchange and thus reduce activity within a fund much to the benefit of those long term investors who merely wish to be continually invested in a given fund. Naturally, in order to accomplish frequent purchase and sale, the investors who wish to engage in this activity would need to make use of brokers, perhaps electronic brokers and exchanges. In other words, new business would be generated by a croupier. Of course, it could be argued that the increase in trading in the actual ETF would be exactly balanced by the reduction in investor induced turnover at the mutual fund level. Ergo, there should be no net change to systemic activity levels. However, this does not

appear to be the actual ETF experience given the data presented regarding the SPDR and the NASDAQ 100. Actively managed exchange traded funds might soon become a reality. If this does occur, it should prove to be a most entertaining laboratory for the inventive and imaginative faculties of the croupier employees. The proposal envisages that each actively managed mutual fund would create an ETF class. C) One Chicago and Single Stock Futures

One Chicago is the exchange upon which trade single stock futures as well as futures upon so-called narrow-based indexes. One Chicago is jointly owned by the Chicago Mercantile Exchange, the Chicago Board of Trade and the Chicago Board Options Exchange. Trading in single stock futures formally commenced on November 8, 2002. On that day 3,000 contracts were traded. A single stock future typically represents 100 shares of underlying stock. In November 2002 there were single stock futures on 21 stocks. The exchange currently lists 206 single stock futures. In principle, the need for single stock futures is not different from the need for commodity futures. For instance, a pension fund or a foundation may have knowledge of future cash flow or transactions and wish to guarantee the price of a security or group of securities. In the case of a foundation, it may have been awarded a significant amount of a single stock upon the demise of a benefactor. Yet, the foundation cannot actually receive the shares until the estate of the benefactor emerges from probate. The foundation wishes to guarantee the price of the stock some number of months in the future. It can accomplish this by the sale of the stock future. Of course, this could be accomplished as well by short sale of the actual stock. However, single stock futures have a margin requirement of only 20%. This is in comparison to the customary 50% margin requirement. Consequently, futures on single stocks are merely a device to own or control the underlying instruments with more leverage than is ordinarily available. It should be evident that this can become a very dangerous activity. Consequently, one should not be surprised to discover that this area has one of the most rapid growth rates of any of the newly created financial instruments. In March 2006, 395,988 such contracts traded on One Chicago. The compound annual growth rate of this activity since inception has been 88.1%. The compound annual growth rate for the past 36 months has been over 70.8%. Open interest has essentially doubled in the past 12 months. It is evident that single stock futures are more attractive than margin debt. Margin debt, as calculated by the NYSE was at the level of \$222 billion in February 2006, which is the most recent available figure. This is consistent with underlying equity value of \$444 billion, assuming a 50% margin requirement. This figure can be translated into an equivalent open interest figure for One Chicago if all margin debt

became single stock futures open interest. In such a circumstance, One Chicago open interest would be roughly 10 million contracts as opposed to the current open interest of roughly one million contracts. It would entail a 10-fold expansion at One Chicago. At the current rate of growth this could be accomplished within 48 months. At the current rate of activity, open interest turnover is at an annual rate of 540%. This is roughly 5 times the level of current New York Stock Exchange turnover. Most participants in single stock futures trading assert that the various instruments are being used for hedging purposes and not for speculative purposes. It is merely interesting to observe that hedging or risk control activity seems to generate more turnover than ordinary investment activity. However, this evidence is entirely consistent with the experience of bond hedging activity conducted at the Chicago Board of Trade as noted in Section II of this paper.

IV — The Theory of Hedging Equivalence

It may seem that the continued use of the term “croupier” in preference to the other less pejorative and more obvious terms betrays the bias that the development of new alternative trading or hedging securities is not necessarily a long term positive development for financial markets. Indeed, it should be manifest that possibility for widespread abuse with more unfortunate economic consequences does exist. Moreover, the emergence of a “croupier” class of entrepreneurs whose only function is to encourage financial market participants to engage in ever more exotic and larger financial transactions may serve to concentrate capital upon zero sum trading actions rather than wealth creation activities. Nevertheless, it can be asserted with equal vigor and perhaps even more persuasiveness that new “hedging” instruments are merely mechanisms with which investors can properly manage portfolio risk. The increased trading volume associated with such instruments only contributes to the depth and liquidity of financial

markets and as such is an element of enhanced market stability. Of course, it is also a source of enormous return on equity for the typical croupier. For instance, the Chicago Mercantile Exchange might well achieve a 34% return on equity in 2006. The expected return on equity for Google in 2006 should slightly exceed 25%. However, none of this is important for the purposes of this paper. If these new instruments enhance the risk/reward balance of portfolios and produce a positive good, then these will become universally used. If these new instruments are simply opportunities to engage in fruitless trading at best and serious abuse at worst, then ultimately these will fall into disuse. Thus, the world of portfolio management cannot remain only in part users of such devices. The field must either adopt universal usage or reject such usage. Therefore, another theorem can be stated as follows:

Risk control as currently practiced will either be universally accepted or universally rejected. If the instrumentation and impedimenta of modern risk control achieves universal acceptance, the already high returns on equity of many “croupier” companies will markedly expand from already elevated levels.

Afterthought

Whatever might occur in the future with regard to publicly traded exchanges will probably occur quickly and upon a very large scale. Consider these few recent developments:

1. The Chicago Board of Trade in cooperation with the Singapore Exchange plans to open a joint Asian Derivatives Exchange in the third quarter of 2006, to be hosted upon the CBOT electronic trading platform. If successful, the CBOT will effectively become a 22-hour Trans Pacific Derivatives Exchange.
2. The New York Mercantile Exchange (NYMEX) plans to combine with the Dubai government to create the Dubai Petroleum Exchange. NYMEX will also make its energy trading futures available for trading on the Chicago Mercantile Exchange Globex Platform.
3. The London Stock Exchange might merge with either NASDAQ, Euronext or possibly even the New York Stock exchange. The exchanges are establishing the ability to conduct business upon a far more expansive scale. Moreover, it appears that, at least among the primary global exchanges, there is a concerted effort to cooperate rather than to compete.

Original PDF includes Horizon Research Group disclosures and reproduction restrictions.

Next: Chapter 15 — A Review & Update of the Croupier Business Model (2008).

Chapter 15 — A Review & Update of the Croupier Business Model

Source. Interim market commentary, Horizon Asset Management, Inc., December 15, 2008.

PDF in vault. 20160703122128_6e323ac8_12-16-

08_20Review_20_20Update_20of_20The_20Croupier_20Business_20Model.pdf

Extract used for assembly. book/build/text/20160703122128_6e323ac8_12-16-

08_20Review_20_20Update_20of_20The_20Croupier_20Business_20Model.txt

Presentation edits: removed letterhead and page numbers; merged line-break splits; preserved disclaimer block at end.

A Review & Update of the Croupier Business Model

To review my definition of croupiers, they are facilitators of financial transactions. Some examples of croupiers are securities exchanges, custodians or processing companies. As a generalization, the croupier does not put its own capital at risk. They are analogous to a dealer in a gambling casino who does not place his or her own capital at risk. Croupiers are to be distinguished from financial services firms, whose capital, generally speaking, is always at risk. An example of such a company would be Citigroup. The fact that Citigroup is not a croupier should in no way lead one to conclude that I am recommending or not recommending Citigroup. I merely use it as an example of a financial services company that does not qualify as a croupier in the sense that I've just defined the term.

At the outset, it needs to be recognized that the world exists today in a pecuniary society. Pecuniary refers not only to its ordinary, narrow definition relating to money, but more broadly in terms of negotiable securities. It is not possible to have a modern industrial economy without a pecuniary society, though it is possible to have a good economy without a pecuniary society. In Carl Sandburg's biography of Abraham Lincoln, he describes how, as a young man, Lincoln lived for an entire year without ever seeing money.

During Lincoln's youth, it was possible to sustain oneself in the U.S. by hunting, fishing and farming without ever participating in the pecuniary society. Today, that would be impossible, because the industrial enterprises that sustain the standard of living we have today are beyond

the ability of even the wealthiest individuals to finance; for example, to build a utility power plant could conceivably cost many billions of dollars. Even the largest companies, in terms of market capitalization, would not allocate such a very large portion of their market value for investment in one utility project.

Such a move for a company, or for very wealthy individuals, would dramatically undiversify their portfolios, and they would be unlikely to make such a choice. As a result, it is necessary to have the ability to agglomerate the capital saved by many different individuals and businesses into one project. Representation of the value deposited by each can only be in the form of a negotiable instrument like a security.

To attract the savings of individuals and businesses, the invested savings must be available to be drawn upon as needed; in other words, those negotiable instruments must freely circulate, and be exchangeable; there is no alternative. In the current era, perhaps the most important reason for the apparent breakdown of the financial mechanisms is that the authorities permitted the evolution and construction of exchange mechanisms that were outside the domain of any reasonable rule system. This vast sphere of portfolio exotica, including interest rate swaps, credit default swaps and the like, exist outside of any organization.

The thrust of modern regulation is clearly and inexorably to bring transparency to those instrumentalities. That goal will likely be accomplished on some sort of organized exchange or clearinghouse. Beyond any question, announcements and implementation to that effect will take place within months, or possibly within weeks, though that shorter time frame is less likely. This situation represents the greatest opportunity that the croupiers could ever have had, because the scope of their activity is going to increase several fold. Beyond that, it will continue to increase exponentially, because there are vast regions of the world that are developing economically, and they will enter the pecuniary economies.

It's interesting to note that when the unregulated business of negotiable instruments begins to perform in a dysfunctional manner, it forces liquidation of securities in the regulated businesses as a way to fund the resultant collateral calls and collateral postings of the unregulated businesses. The result is a bizarre correlation amongst all sorts of asset classes that were never before correlated. It's a correlation that I think is almost at its end.

Let's use exchange traded funds (ETFs) as examples, which, incidentally, now embrace much more than the commonly understood stocks and bonds; they also embrace commodities. In the energy arena, for example, the ETF that contains the Dow Jones AIG Energy Index is down 66.8% since June 30, 2008. Looking at another commodity, in the same time period, the Dow Jones AIG Agricultural Index is down 50%. The Dow Jones AIG Industrial Metals ETF, yet another group of commodities, is down 58.3%.

Correlation can be found even when looking at subsets within the various ETFs; for example, the Dow Jones AIG Grains Index is down 54.3% since June 30, 2008. Grain is clearly a subset of the agricultural index. This situation is almost unheard of in the world of commodities, because one would imagine that the supply and demand imbalances, to the extent that there are imbalances, should logically be different at various points in time between energy commodities, agricultural commodities, and industrial metal commodities. There should also be differences among the agricultural commodities, but they are not.

The collapse of the unregulated, pecuniary economy acts like a suction pump that drains liquidity from all sources. One would have thought that, in the midst of such a liquidity crisis, the demand for the exchange of securities in the regulated pecuniary economy should be lessening, as it did in prior cases, but that is definitely not the case at the moment. For example, ProShares has Ultra Long and Ultra Short indexes that are weighted to perform 2x the underlying indexes on any given day (that's the definition of "ultra"). In the last month, a company known as Direxion created the innovation of a 3x, or super-ultra, ETF. At the moment, there are the Direxion Large Cap Bull 3x ETF (BGU), the Direxion Large Cap Bear 3x ETF (BGZ), the Direxion Small Cap Bull 3x ETF (TNA) and the Direxion Small Cap Bear 3x ETF (TZA).

In one month, the following volumes have been achieved: in the Large Cap Bull 3x ETF the volume is now at 13.9 million shares daily. The volume at the end of one month for the Large Cap Bear 3x ETF is 6.9 million shares daily. The Small Cap Bull 3x ETF, also one month old, has achieved the volume of 9.3 million shares daily, and the Small Cap Bear 3x ETF, the laggard of the group, has volume of 3.5 million shares daily.

In this connection, it is worthwhile noting that the original ProShares 2x Ultra has had undiminished volume throughout; if anything, it has increased. The appetite among market participants for risk may well have decreased, but the demand for liquidity has increased, and it's just at its beginning. Another example is the Lehman 20+ year Treasury ETF that now trades 4.5 million shares daily, an amount that is historically unprecedented for that particular instrumentality. The S&P SPDR (SPY) has 50% turnover daily. For the PowerShares QQQ (QQQQ) which is, of course, the NASDAQ 100, volume is recorded in a double counting mode; nevertheless, it trades over 200 million shares daily.

The market has spoken. It demands liquidity at every hour of the day on a 24 hour basis. The croupiers are merely processing companies; they are no more, and no less. They do not risk their

capital other than in the plant and equipment necessary to facilitate liquidity transactions. They earn copious amounts of cash flow which, in the fullness of time, will increase. Over the last year, credit markets have essentially become frozen; they basically refuse to function. In the world of commodity exchanges, however, interest rate futures trading has diminished only somewhat vis-à-vis last year, which is an incredible phenomenon. While the entire securitization market has disappeared, including mortgages, student loans, credit card receivables, and any type of securitization, the interest rate futures volume is down only slightly.

At the moment, various governments of the world are creating more fixed income securities than have ever been created in the history of civilization. Though the United States is a leader in that effort, it is but one among many. This creation of fixed income securities is being done at record low levels of interest rates. Is it not reasonable to assert that, in some number of months, the market participants, the owners, or in some cases the former owners, of these securities will wish to hedge themselves by the purchase or sale of interest rate futures? Is it not likely that that market, which is already very large, will grow enormously? The likelihood of that scenario is the market opportunity that stands before the croupier.

In my humble opinion, the croupiers are on the verge of a golden era that is just now starting. Of course, given the low valuations of croupiers, most market participants would disagree. Time will tell which of us is right. As of the most recent reckonings, the available empirical data suggests that the croupiers' business is growing, not diminishing.

Disclaimer (as in original). For existing clients of Horizon only — not for prospect use. This information is intended solely to report on investment strategies as reported by Horizon Asset Management, Inc. Opinions and estimates offered constitute our judgment and are subject to change without notice, as are statements of financial market trends, which are based on current market conditions. The holding information presented is for illustrative purposes only. Actual account holdings and performance will vary depending on the size of an account, cash flows within an account, and restrictions on an account. Portfolio holdings are subject to change daily. Under no circumstances does the information contained within represent a recommendation to buy, hold or sell any security and it should not be assumed that the securities transactions or holdings discussed were or will prove to be profitable.

Next: Chapter 16 — Q&A on trading volume, innovation, and Treasury / Fed tactics (January 2009).

Chapter 16 — Q&A: Trading Volume, Innovation, and the Treasury & Federal Reserve (January 2009)

Source. Horizon Asset Management, Inc., January 5, 2009.

PDF in vault. 20150506062604_608d7d68_1-5-09_20Q_A-_20Croupier_20_20Fed_20Tactics.pdf

Extract used for assembly. book/build/text/20150506062604_608d7d68_1-5-09_20Q_A-_20Croupier_20_20Fed_20Tactics.txt

Editorial note. Transcript-style Q&A; **answers reproduced as Stahl's voice** from the original document.

Presentation edits: removed letterhead and page numbers; merged line-break splits; preserved disclaimer at end.

Trading volume trends

Q: With respect to your enthusiastic view of trading volume trends, and hence what the future holds for exchanges, do you have an opinion on whether or not those volumes currently have a natural ceiling? If they do, what are the constraining factors? Do they include computing power, regulatory angst, contract/instrument types, etc.?

A: I don't believe the trading volumes have any natural constraint. Historically, processing power was one of those constraints because, up until the 1960's, everything was done by paper. At the moment, the new evolving standard, which will be with us in no more than several months, is the ability to transact at the sub-millisecond level. It's worthwhile reviewing that a millisecond is one-thousandth of a second. Whenever the ability to transact more frequently has combined with increased information reported by the markets, the result has been more frequent transactions; it breeds more transactional activity. The more venues available for trading activity in like instruments, the greater the probability that there will be an exploitable, if slight, differential in price between them. That price difference is essentially what drives the trading volume.

I think it's not an exaggeration to say that the London Stock Exchange has more business now than it had prior to the inception of the Chi-X platform, even though the Chi-X platform has

obtained significant market share from the London Stock Exchange. As a matter of fact, when the first securities were introduced to Chi-X on a test basis, for example Lloyds TSB, even in those first few days, there was an increase in Lloyds TSB trading on the London Stock Exchange.

I believe it is true that on days when the London Stock Exchange is closed, Chi-X, which in a regulatory sense could be open, chooses to be closed. In my opinion, Chi-X exists as a potential arbitrage point of venue relative to the London Stock Exchange; the greater the volume in Chi-X, the greater the volume will be in the London Stock Exchange. If there were ten other exchanges like the Chi-X, I think that the volume would be multiplied accordingly.

The regulators are interested in liquidity, securitization and allowing investors to exit asset classes or individual securities that the owners consider problematic; for example, auction rate preferreds. Taking even the greatest security disasters of the last year that include Bear Stearns, Lehman Brothers, Fannie Mae, Freddie Mac, and AIG, in each and every case, the fixed income instruments in those securities were difficult to trade. The causes of the decline in securities prices, which were largely in the form of derivative instruments in the securities inventories, were found to be absolutely untradeable even though the underlying instruments were, in many cases, AAA rated mortgage backed securities. In terms of market scope those underlying securities are many, many times the size of the putative market values of Bear Stearns, et al.

Let's reflect on that statement. There is a \$14 trillion mortgage market in the United States of which at least \$7 trillion has been securitized. That \$7 trillion market became absolutely untradeable; yet, a stock like Bear Stearns, say what one will about it, never stopped trading. The owners were very displeased with the prices obtained, but there was always liquidity available. Therefore, it seems logical that the regulators would encourage the listing of many more items.

One of the great problems in the economy at the current moment is that the credit market is frozen, because there's no liquidity. There's no liquidity, because there's no price discovery; no one knows what the price is, and there is no contact between those who want to sell and those who might be interested in buying the securities. The entire housing market in the United States is not functioning, because liquidity doesn't exist. I think that a remedy for that liquidity problem is the listing of those securities. In any event, it appears fairly likely that the regulators are proceeding along those lines.

Golden age of financial innovation

Q: Is it possible that we are at the end of a Golden Age of financial innovation that began 30 years ago with the advent of personal computers and programs like Lotus 1-2-3? If so, would this situation alter your view on the desirability of exchanges as investments at the current prices?

A: The answer is in the question. If we were at the end of a Golden Age of financial innovation, obviously, croupiers would not be the wonderful investment opportunities that I believe them to be. In fact, I believe that we're at the beginning of an age of financial innovation. Let's take municipal bonds as just one of the many examples I could use.

Municipal bonds always existed, and they were always very difficult to trade, because the natural owners of them, insurance companies, for instance, would buy large portions of issues and hold them to maturity. If an individual wished to restructure a portfolio, add or subtract to an exposure, or change the nature of a portfolio ladder, it was extraordinarily difficult and created transactional friction, which is very expensive.

Recent months have seen the evolution of Exchange Traded Funds (ETFs) that own diversified portfolios of municipal bonds. In theory, these ETFs make it possible to buy one share of a diversified municipal bond portfolio. In an ideal world, that portfolio should function not very differently from a bank account, by avoiding the denominational problem, which is the minimum lot size at which it's necessary to buy or sell. Typically, in the world of municipal bonds, a \$100,000 face amount is considered to be an inadequate lot size; round lots, as a practical matter, are in the millions of dollars.

A person who wishes to withdraw \$5,000 or \$10,000 from a municipal bond account doesn't want to have to wait for an interest payment, and ETFs make it possible. If one wishes to own securities in foreign markets, it's extraordinarily difficult to transact except for in the most liquid securities, especially in the Asian markets. It's much easier to transact in an ETF. I believe that there are well over 500 ETFs in registration at the moment. Consider the number of publicly traded securities that exist in the world (which I think well exceeds 50,000 specimens), and imagine the many combinations and permutations possible with 50,000 specimens of securities incarnated as ETFs. That explanation is but one way to support the case that we're at the beginning of an age of financial innovation.

Another argument to support that case is that many markets in the world are still developing. A deep and rich publicly traded debt market does not exist anywhere in the world. The debt markets are all dealer markets in which the quotations, price discovery and debt are almost inscrutable. The debt market is many times the size of the equity market, and it's going to become listed and tradable.

In addition, there are the many companies in emerging markets that are not yet listed. In terms

of financial innovation, the emerging markets stand the way the American financial market stood 100 years ago. The access to business activity is clearly shifting to the Pacific Basin and not a great quantity of the Chinese GDP is listed anywhere, or is even tradable. The same might be said of the Indian GDP, and the GDPs of those countries are likely to increase dramatically. Similar comments could be made about the Russian GDP, or about the Eastern European markets.

The changing tactics of the U.S. Treasury & the Federal Reserve

Q: Do the dark forces for evil, i.e. the bear raiders, exist? If they do, have the Treasury and the Fed, via their actions with the GSEs and AIG, made a terrible mistake by underwriting the bear raiders' playbook vis-à-vis finding a large leveraged company with opaque assets and assumed counterparty risk, short the stock, fire up the rumor mill, and ramp up the CDS price, spook the rating agencies and counterparties, and wait for the Treasury and the Fed to wipe out the equity?

A: The first way to respond to the question is that there are always going to be short sellers, and the short sale thesis, as outlined in the question, is not an unreasonable one for a short seller to engage in. Should they be called "the dark forces for evil?" I don't know if that's appropriate. A short seller is merely a market participant who tries to make a profit and, in particular, to profit from the short sale of securities.

It's very clear that sometime in mid-September, the government made an enormous change in its tactics. From the advent of the Bear Stearns crisis in early March 2008 to the AIG crisis in September 2008, the government was willing to lend assistance to various enterprises but, to establish moral hazard, the government exacted a terrible price from the equity shareholders, and that price was massive dilution.

In the case of Bear Stearns, it wasn't dilution on the part of government action, it was merely that J.P. Morgan was able to acquire the company at a large discount to book value. In the case of Fannie Mae, Freddie Mac and AIG, it was massive dilution. In the case of Lehman Brothers, although the government did render assistance initially, it did not act to preserve the enterprise. These are the documented actions of the government. It was not unreasonable for the short sellers to have presumed that the government would act in the same manner with all the other companies. However, the government very quickly changed its tactics and began to provide capital on fairly reasonable terms, given the environment.

The government can exact moral hazard against some market participants, but it's not reasonable to impose moral hazard against all participants in the marketplace. That approach would be tantamount to destroying the financial system, which the government must act to preserve. I don't think that the short sellers fully understand the implications of the government's actions. The government's now assisting the various firms by any means possible. Providing capital is not the only action the government will take; it is merely the first.

Through the Federal Reserve, the government is enormously increasing the money supply. From an arithmetical perspective, that's a debasement of the currency, but it's designed to assist those market participants who suffer from excess indebtedness. Such a policy has limited effectiveness in the short run but, in the long run, though it has many aftereffects that are not very pleasant, it will have the desired effect, which is to help the indebted companies.

In addition, the government plans a massive intervention in various types of markets, not the least of which is the government's plan to buy over \$600 billion of agency paper with a view to orchestrating the mortgage rate to be somewhere around the 4.5% level. The two GSEs, which probably jointly account for 21% of all the foreclosures in the United States, are rapidly restructuring defaulted mortgages, and the banks are doing likewise. There is massive assistance being rendered, and I think that there is more on the way. The government will act to preserve the system. The announcements literally say (and these are not merely approximations of the words; these are the words), "We will do whatever we must to defend the system." In my opinion, that's not a good sign for the short sellers, who may not have appreciated that fact yet, but I think they will soon.

Investing alongside the Treasury & Fed

Q: At this stage, how wise is it to become a minority shareholder alongside the Treasury and the Fed in Fannie, Freddie or AIG? In theory, at the right odds, any bet is a good one, but do you think that the Treasury and the Fed will be predisposed to a future trampling on the minorities as a tonic for moral hazard?

A: The government always has the option of trampling on the minorities, but let's not forget about the majority. The government has made, and is making, a substantial investment in Fannie Mae and Freddie Mac. The government has a vested interest in seeing to it that the value of its investment is maximized. At the end of the day, all entities act in their own self-interest. It would be senseless to create moral hazard by destroying the value of the government's own investment in the GSEs. I think that the point about moral hazard has already been made in the

case of Fannie Mae and Freddie Mac, so I don't think that anything would be accomplished by further trampling on the rights of the minorities.

At the moment, there are no viable substitutes for Fannie Mae and Freddie Mac. The original plan announced by the authorities in August was that, over time, Fannie Mae and Freddie Mac would liquidate. We've seen that, if those GSEs were allowed to liquidate, the consequences for the mortgage market would be disastrous. The private market has neither the means nor the willingness to provide an adequate substitute for Fannie Mae and Freddie Mac; therefore, according to the recent announcement, those entities will need to buy many more mortgages. Despite what might've been announced in August, Fannie Mae and Freddie Mac are going to, in my view, enormously increase the balance sheet assets, both in the actual mortgage book, and in the guaranteed book.

I believe that Fannie Mae has on its mortgage book and its guarantee book, probably about \$6 trillion of assets. At the moment, it has a market capitalization of about \$4 billion. Let's assume that on a \$6 trillion book, Fannie Mae could earn one basis point, or \$600 million. If Fannie Mae, as an entity, could merely earn one basis point on its book then it would be trading for, roughly, 6.7x earnings. If the company could earn two basis points on its book, which is not an extraordinary amount of money to earn, then Fannie Mae would be trading at 3.3x earnings. If Fannie Mae could earn 10 basis points on its book, which would be much less than its historical experience, it would be trading for less than 1x earnings.

If one were to assert that Fannie Mae could earn no more than 10 basis points on its book, which is probably a very poor assumption, and if Fannie Mae were not going to grow its book which, in light of recent announcements from the government, I think is also a poor assumption, then Fannie Mae is now trading at well below 1x earnings. Let's say that Fannie Mae were now trading at 0.6x earnings; if that were to increase to 6.6x earnings, an investor would make 10x his or her money. Even if it took five years to happen, the compounded annual rate of return would be 58.5%.

It shouldn't be difficult to imagine what might happen if the return on book, measured in basis points, were higher, and what would happen if the book itself became larger. One would see some truly extraordinary rates of return. Of course, since the government owns 80% of Fannie Mae, it would achieve 80% of those returns.

The government does not wish to own all of the mortgages in the United States. If it had wished to, it would've completely nationalized Fannie Mae, that is, owned 100% of the entity. However, the government did not wish to do so, because it would have become the effective guarantor of all the Fannie Mae debt. Even though there's no explicit guarantee, analysts viewing the U.S. Government's balance sheet would begin to include the agency debt in their calculations of the U.S. debt. That situation could affect the ability of the United States government to obtain funding.

One would think that the U.S. Government should logically wish to avoid having the agency funding added to the U.S. debt. That reason, and no other, is why 20% of Fannie Mae is still in the public domain.

If the minorities are trampled upon, then where is the moral hazard? If the government essentially obliterated the minority interest in Fannie Mae by issuing a sufficient number of warrants, since it would be disastrous for Fannie Mae to place its bondholders in moral hazard, where would be the moral hazard? It can only exist if there are publicly traded stockholders that are at risk; therefore, the minority interest serves a useful social function from the point of view of the government. I don't think that the minority interest is likely to be trampled upon; I think it's much more likely that it is going to make a great deal of money.

Disclaimer (as in original). For existing clients of Horizon only — not for prospect use. This information is intended solely to report on investment strategies as reported by Horizon Asset Management, Inc. Opinions and estimates offered constitute our judgment and are subject to change without notice, as are statements of financial market trends, which are based on current market conditions. The holding information presented is for illustrative purposes only. Actual account holdings and performance will vary depending on the size of an account, cash flows within an account, and restrictions on an account. Portfolio holdings are subject to change daily. Under no circumstances does the information contained within represent a recommendation to buy, hold or sell any security and it should not be assumed that the securities transactions or holdings discussed were or will prove to be profitable.

Next: Chapter 17 — Exchanges: Less Talk, More Figures (August 2009).

Chapter 17 — Exchanges: Less Talk, More Figures

Source. Horizon Asset Management, Inc. commentary, August 2009.

PDF in vault. 20160703154353_b8f56eae_8-21-

09_20Croupiers_20Less_20Talk_20More_20Figures.pdf

*Presentation edits: removed letterhead and page footers; reconstructed select tables from the PDF extract; left the long historical options volume grid as a **verbatim extract block** where column alignment was unreliable in OCR/text extraction.*

The exchanges — less talk, more figures

Profitability

First, as to profitability, we wrote frequently last year, as the credit crisis wreaked havoc upon the financial system broadly and the equity markets specifically, that the daily transactional activity of the exchanges appeared to remain fairly robust. There were, of course, somewhat mixed results reported from the exchanges, and a certain degree of earnings diminution was inevitable, particularly in Europe and Asia, as exchanges in these regions adopt the ad valorem method of pricing, which bases fees on the market value of the transaction. Clearly, as equity markets fell by 50% in nearly every region of the world last year, such companies would be expected to report a larger degree of revenue diminution than would an exchange that bases its fee structure on volume alone. This latter pricing method is almost exclusively used by the U.S. exchanges.

With over one year of data in hand, the draconian margin compression scenario so widely expected to occur last year, and especially this year during recession, has not appeared. The most recent profit margins of the exchanges in the Core Value strategy are presented below.

Operating profit margin

Exchange	Fiscal 08	Fiscal 07
CME Group (1)	42.5%	39.4%
Hong Kong Exchange	78.5%	85.7%
London Stock Exchange (2)	50.4%	52.9%
Nasdaq OMX Group	17.5%	15.0%
NYSE Euronext	18.1%	16.3%

1. Before amortization of purchased intangibles & impairment of long-term investment.
2. Fiscal March 09 and 08. Before amortization of purchased intangibles and goodwill impairment charge.

Undoubtedly, these companies mostly reported lower earnings versus year ago periods. Yet, the current and reduced level of profitability still remains extraordinarily high relative to general corporate profit margins, perhaps the highest of any other industry. If it were remarked last year that a group of companies dependent on the health of the capital markets would be able to sustain 20%–75% operating margins throughout the credit crisis and any recessionary effects that would follow, this most likely would have been met with abundant laughter (or derision).

Given their resiliency throughout one of the worst operating periods for financial companies in history, as well as our expectation for their future prospects, we maintain a large weighting in this sector.

Competition

Among the most frequent questions, if not criticisms, is that of competition among the exchanges. Yet by a universal standard, there is much less competition today than there was only a few years ago. What would be the typical reaction if an investor could somehow be given foreknowledge that the constituents of a given industry would undergo a massive consolidation, to the degree that perhaps 20 or more competitors would be absorbed into five? It has happened already in the North American securities exchange market. Three years ago, a long list of independent venues existed—including, among others, the NYSE, Amex, CBOT metals complex, Archipelago, ISE, NASDAQ, PHLX, Boston, IDCG, CME, CBOT, NYMEX, COMEX, Swapstream, ICE, NYBOT, Winnipeg, Toronto, Montreal, BOX, and Oxen (Alberta power)—and the commentary tracks how **76%** of those names were ultimately absorbed by **four** North American exchanges and **one** European exchange (with additional NYSE Euronext and Nasdaq OMX activity abroad noted in the original).

A more pointed criticism of exchanges as an investment is the loss of market share by major exchanges, such as by the London Stock Exchange, NYSE Euronext and others. Such figures are reported monthly, along with other statistics, by many of the exchanges. And it is true. There is unquestionably a loss of market share by many exchanges, sometimes of large magnitude, and it has been occurring almost without surcease for a long period of time. Witness the severe loss of market share by the New York Stock Exchange in stock trading volume during the past several years; it is virtually monotonic in its decline, and since 2005 amounts to about **38** percentage points, from over **80%** to under **45%**:

NYSE handled volume — market share (% of consolidated NYSE volume)
(Fifth column: 2009 where reported in original; sixth column on June row: point change 2005–2009 per extract.)

Month	2005	2006	2007	2008	2009
December	77.8%	69.1%	59.7%	47.9%	—
November	78.3%	69.8%	60.1%	47.4%	—
October	79.7%	70.9%	60.3%	47.7%	—
September	79.8%	71.2%	62.2%	46.7%	—
August	80.0%	71.8%	65.5%	46.5%	—
July	80.1%	74.2%	66.3%	48.3%	—
June	81.5%	76.5%	68.7%	51.0%	43.5%
May	81.0%	76.9%	68.1%	51.8%	42.9%
April	82.9%	75.5%	68.5%	53.6%	43.4%
March	81.7%	76.1%	69.9%	56.2%	44.7%
February	81.6%	77.5%	69.9%	58.0%	46.5%
January	82.3%	76.6%	69.0%	58.4%	46.9%

NYSE handled volume (millions of shares) — verbatim from the extract (same source file):

2004	2005	2006	2007	2008	2009
December	33,222	34,546	34,669	39,241	52,300
November	32,357	36,467	40,848	52,835	54,921
October	33,313	40,956	41,763	47,107	77,383
September	28,438	36,986	36,732	38,123	66,768
August	27,924	34,670	38,115	64,319	42,443
July	30,335	30,886	36,737	50,009	60,956
June	29,370	34,698	44,981	48,271	50,215
May	30,574	32,894	44,579	46,458	42,329
April	32,780	36,907	34,464	41,273	46,508
March	34,764	38,265	40,828	50,004	53,429
February	28,815	30,882	32,102	38,890	45,877
January	34,204	33,360	39,760	41,869	60,780
Total	376,096	421,517	465,579	558,400	653,910
% Change	12.1%	10.5%	19.9%	17.1%	15.1%

Witness also the climb in handled volume alongside the share decline: the narrative conclusion in the piece is that **market share loss and volume growth are not mutually exclusive**—competition is associated with increased volumes.

We described this phenomenon in a *Studies in Absurdity* review in September 2006. Lest this seem a product of an anomalous few years, the abbreviated table below describes the share volume growth at the New York Stock Exchange over the past nearly 50 years since 1960. On a decade-by-decade basis, the result is remarkably consistent at nearly **15%** per annum (the subdivision of the 2000 to 2008 period is due to the change in tabulation method between the NYSE and Euronext upon their merger).

NYSE annual share volume expansion

Period	Annualized
1960 – 1970	14.4%
1970 – 1980	14.5%
1980 – 1990	13.3%
2000 – 2003	10.3%
2004 – 2008	14.8%

Options and “Securities Exchange Moore’s Law”

This odd behavior among securities exchanges is even starker in an entirely different security type, options. The original piece presents long-run **total U.S. options industry volume** and **CBOE** share and growth statistics. The machine extract preserves rows but not perfect column alignment; the following block is **verbatim** from book/build/text/20160703154353_b8f56eae_8-21-09_20Croupiers_20Less_20Talk_20More_20Figures.txt (page 7 of the extract).

1980 - 1990	8.1%	1980 - 1990	9.4%
1990 - 2000	13.2%	1990 - 2000	9.7%
2000 - 2008	22.0%	2000 - 2008	17.6%
2005 - 2008	42.5%	2005 - 2008	36.6%
2007 - 2008	36.9%	2007 - 2008	26.4%
Year			

Total Industry				
Options	Volume	Year	Total	Volume
Yr/Yr				
Change				
% of				
Industry				
Vol				
1973	1,119,245	1973	1,119,177	100%
1974	5,682,907	1974	5,682,907	408% 100%
1975	18,103,018	1975	14,431,023	154% 80%
1976	32,373,925	1976	21,498,027	49% 66%
1977	39,637,328	1977	24,838,632	16% 63%
1978	57,231,018	1978	34,277,350	38% 60%
1979	64,264,863	1979	35,379,600	3% 55%
1980	96,728,546	1980	52,916,921	50% 55%
1981	109,405,782	1981	57,584,175	9% 53%
1982	137,306,205	1982	75,735,739	32% 55%
1983	150,056,075	1983	82,468,750	9% 55%
1984	196,437,361	1984	123,273,736	49% 63%
1985	232,910,547	1985	148,889,091	21% 64%
1986	289,211,135	1986	180,357,774	21% 62%
1987	305,168,935	1987	182,112,636	1% 60%
1988	195,948,591	1988	111,784,045	-39% 57%
1989	227,016,660	1989	126,765,253	13% 56%
1990	209,922,748	1990	129,500,018	2% 62%
1991	198,801,600	1991	121,689,918	-6% 61%
1992	201,995,757	1992	121,467,604	0% 60%
1993	232,662,095	1993	140,348,955	16% 60%
1994	281,382,402	1994	183,934,483	31% 65%
1995	287,296,909	1995	178,533,465	
-3%	62%			
1996	294,797,702	1996	173,944,877	-3% 59%
1997	353,823,118	1997	187,243,741	8% 53%
1998	406,343,198	1998	206,865,991	10% 51%
1999	507,891,483	1999	254,331,851	23% 50%
2000	726,727,939	2000	326,359,531	28% 45%
2001	492,615,219	2001	306,667,851	-6% 62%
2002	457,717,950	2002	267,616,496	-13% 58%
2003	535,143,977	2003	283,946,495	6% 53%
2004	815,899,814	2004	361,086,774	27% 44%
2005	1,234,309,674	2005	468,249,301	30% 38%
2006	1,705,689,052	2006	674,735,348	44% 40%
2007	2,611,445,719	2007	944,471,924	40% 36%
2008	3,574,445,814	2008	1,193,355,070	26% 33%
Annualized GrowthAnnualized Growth				
Total CBOE	Options	Contract	Volume	
1973 - 2008				
Historical Options Industry Volume				
1973 - 2008				

The essay’s conclusion in prose: between **2000** and **2008**, CBOE **market share dropped dramatically** while **transaction volume rose dramatically**, on the order of **36%** per year; in **2008**, volumes rose by **26%**. The same sorts of results might be produced for trading in financial futures or commodities. Whatever one might say about securities exchanges, the empirical data, whether employing a multi-decade time frame or examining the tumultuous past two years, does not support the notion that market share and volume growth are mutually exclusive; if anything, the results are consistent with the opposite viewpoint, that competition is associated with increased volumes.

Perhaps there is a special form of Moore’s law that operates for securities exchanges, a sort of **Securities Exchange Moore’s Law**. (Intel co-founder Gordon Moore noted in a 1965 paper that since the invention of the integrated circuit in 1958, the number of transistors that could be economically placed on an integrated circuit had increased exponentially, doubling roughly every two years. That phenomenon has continued for the almost one-half century since that paper’s publication.)

What could be the various enabling factors for exchange volumes, analogous to the process, physical chemistry and engineering advancements that facilitate the operation of Moore’s law? First, it should not be ignored that the mere expansion of the world economy, business activity, corporate profits and private savings will result in an expansion of the market capitalization of companies and the number of shares that can be traded, and of the volumes of commercial exposure that must be hedged via futures contracts, and so forth. Beyond that, it would appear that virtually any development in regulations, technology, portfolio or trading practice, or product innovation that lowers a barrier to trading, whether it is via increased transaction speed, lower transaction costs, better or more diverse tools for hedging, diversifying, reducing risk or acquiring risk, will invite more trading. Accordingly, one could hardly list all of the variables that have historically contributed to trading volume expansion in the absence of a comprehensive study.

As to future innovations, some of these can only be imagined — and they will be imagined. We believe that the pace of innovation has accelerated and that securities exchanges are now at the doorstep of a golden age. We might examine a few that are right now in the process of explosive

development, in a future communication.

While the current economic environment is not without its challenges, we continue to believe the current investment environment offers a once in a generation opportunity for the long-term investor. The prior comments attempt to demonstrate there are companies that have only experienced share price compression in the past year, not fundamental diminution of value. Accordingly, we continue to monitor our businesses, as well as search for new ideas. We remain committed to our belief that the patient, long-term investor will ultimately be rewarded with superior returns.

Disclaimer (as in original). For existing clients of Horizon only — not for prospect use. This information is intended solely to report on investment strategies as reported by Horizon Asset Management, Inc. Opinions and estimates offered constitute our judgment and are subject to change without notice, as are statements of financial market trends, which are based on current market conditions. The holding information presented is for illustrative purposes only. Actual account holdings and performance will vary depending on the size of an account, cash flows within an account, and restrictions on an account. Portfolio holdings are subject to change daily. Under no circumstances does the information contained within represent a recommendation to buy, hold or sell any security and it should not be assumed that the securities transactions or holdings discussed were or will prove to be profitable.

Next: *Part III — inflation, royalties, and real assets (01_TABLE_OF_CONTENTS.md).*

Part VIII — Video and interviews

Editorial rule for this edition: interviews include **Murray Stahl’s answers only**, with publication context in the headnote. Video chapters use **official Horizon Kinetics pages** and Vimeo embeds where available.

Video — official pages and embeds

Source list: `_stahl_video_embed_urls.txt` in the vault root.

Event / title (short)	Page (Horizon Kinetics)	Vimeo embed (where captured)
Murray Stahl at the Project Punch Card Conference	https://horizonkinetics.com/videos/murray-stahl-at-the-project-punch-card-conference/	https://player.vimeo.com/video/826950692
Murray Stahl at Hedgeye Macrocosm — Big Opportunities in Global Market (May 5, 2023)	https://horizonkinetics.com/videos/murray-stahl-at-hedgeye-macrocosm-big-opportunities-in-global-market-may-5-2023/	https://player.vimeo.com/video/827365338
Murray Stahl at Hedgeye Fall 2024 Investing Summit — Inflation and Hard Asset Investing (Oct 9, 2024)	https://horizonkinetics.com/videos/murray-stahl-at-hedgeye-fall-2024-investing-summit-inflation-and-hard-asset-investing-october-9-2024/	https://player.vimeo.com/video/1020004863
Murray Stahl on Bloomberg <i>Inside Active</i> — “Horizons: Stahl on Letting Winners Run”	https://horizonkinetics.com/videos/murray-stahl-on-bloomberg-inside-active-horizons-stahl-on-letting-winners-run/	(No static Vimeo URL in the vault list; use the page player or capture embed after load.)

Transcript placeholders

For each row above, add a subheading and transcript **after** you transcribe audio. Suggested stub:


```
### [Title] – transcript (Stahl only)
```

```
**Recorded.** [date if known]
```

```
**Source URL.** [page link]
```

```
**[Stahl]:** ...
```

Print interviews and profiles

Assign from your non-PDF sources (e.g., *GuruFocus*-style Q&A, magazine profiles). Each piece gets the same headnote pattern as numbered chapters: outlet, date, interviewer, URL, and note on cuts (answers only).

Done when every embed row has either a transcript file linked here or an explicit “deferred” note with reason.

Subject index (stub)

Add entries as you draft Parts II–VII. Suggested groupings:

- Artificial asset class / passive / index concentration / Russell / float / dividend ETFs / international overlap → Ch. 9–13
- Croupiers (exchanges, clearing, liquidity, volume) → Ch. 14–17
- Inflation, royalties, real assets
- Bitcoin, monetary architecture
- Exchanges, MIAX, ICE, private markets
- Japan, JAPN, geography
- Oil services, energy
- Diamonds, crisis wealth

Each line can later link to chapter anchors in the exported master.

Chronological bibliography (vault book_candidate=yes)

Machine-generated from `book/sources/corpus_inclusion_manifest.csv`. Sorted by **filename** (vault crawl timestamps often prefix `YYYYMMDDHHMMSS_`). Tiers are audit labels, not legal attributions.

Bytes	Tier	PDF filename
106764	commentary	20120417091135_a2c79b40_HorizonKinetics_Commentary_June2011.pdf
203555	commentary	20120417091139_2a384828_HorizonKinetics_Commentary_Sept2011.pdf
219395	commentary	20120417091210_5ffabb18_Horizon_Commentary_December_2010.pdf
190743	commentary	20120522230132_69b2d23a_2012Q1_commentary.pdf
159386	commentary	20120526113704_7acbe50f_Contrarian_20_Compendium_December_2011.pdf
291138	commentary	20120526113717_be68055e ETF_20Compendium_20July_202010.pdf
188991	commentary	20120526113735_3afb1ca5_Fixed_Income_Contrarian_Compendium_December_2011.pdf
295151	commentary	20120526125939_948c4bfe_Bits___Pieces_March_2012.pdf
305240	commentary	20120526125950_d456f4bf_Contrarian_20Compendium_20June_202010.pdf
79251	commentary	20120526125954_0a72aa33_Devil_s_20Advocate_20Compendium_20February_202009.pdf
170703	commentary	20120526125959_87ca3642_Devil_s_20Advocate_20Compendium_20January_202012.pdf
81987	commentary	20120526130004_893b3e23_Devil_s_20Advocate_20Compendium_20July_202009.pdf
123583	commentary	20120526130016_9cde293d_Global_20Contrarian_20Compendium_20Part_20I_20April_202010.pdf
77204	commentary	20120526130020_00e623c4_May_2012_commentary_Bond_Market_Panic.pdf
31317	thematic_essay	20120526130026_e66d8f86_ModernInvestmentLanguage_20Feb02.pdf
95056	commentary	20120617014745_b84d7b8a_Devil_s_20Advocate_20Compendium_20November_202008.pdf
86717	commentary	20120617014753_63e1a5d8_Devil_s_20Advocate_20Compendium_20Supplement_20May_202008.pdf
137153	commentary	20120617015829_798ea18f_Scratch_20Compendium_20August_202008.pdf
84015	commentary	20120813085750_f1285060_Commentary_Interest_Rates.pdf
424827	commentary	20120905172039_d26c8497_2Q-Market_Commentary_HAM.pdf
210693	research_report	20121115121104_866059d3_Stahl_Report_Compendium_May_2012.pdf

Bytes	Tier	PDF filename
198552	commentary	20130512204058_98641d4d_Q1_2013_Core_20Value_20Commentary.pdf
178739	commentary	20130920021546_2b4485aa_Q4_2012_Commentary.pdf
311699	commentary	20130920030109_5b37c62d_Q3_2012_Commentary.pdf
105725	commentary	20131021170946_e82dc85c_Liquidations_September_2013_Commentary.pdf
183573	commentary	20131107150707_7e705583_Q3_2013_Commentary.pdf
140190	commentary	20131207124510_acaf0864_May_2013_Commentary_Long_Product_Lifecycle.pdf
764556	commentary	20140125061958_a5bd9963_Q4_2013_Commentary.pdf
151340	commentary	20140209173757_02d4e808_Spin_Commentary_September2012.pdf
183338	commentary	20140211161609_25e7bcad_December_Commentary_Canadian_REIT.pdf
227922	commentary	20140211173858_c8404d5c_Contrarian_Compendium_April_2013.pdf
279763	commentary	20140327074823_ee38eb5c_HorizonKinetics_Commentary_May2011.pdf
145903	commentary	20140602201231_a13a6b63_Horizon_Commentary_4thQuarter_2010.pdf
418105	commentary	20140630174110_0ecdbe9f_May_202014_20Commentary_FINAL.pdf
337536	commentary	20140801121217_4f9c6c03_2011Q4_commentary.pdf
35823	thematic_essay	20140817001750_4efeda8f_TexasPacificLandTrust_20TPL_20CRRMay95.pdf
268662	commentary	20140824195453_d92f0a90_Q1_2014_Commentary.pdf
213650	commentary	20140824195511_ba1eef34_Q2_2012_Commentary.pdf
221135	commentary	20140824195524_11f5e40e_Q2_2013_Commentary.pdf
237078	commentary	20140912122500_3a2b8372_July2014Commentary_FINAL.pdf
418482	commentary	20141013213122_934ae9d1_Sep2014Commentary_FINAL.pdf
450624	commentary	20141111104317_fc619f87_Mar_2014_Commentary_FINAL.pdf
375806	commentary	20141221234412_09ecdeca_Feb_2014_20Commentary_FINAL.pdf
210540	commentary	20150122021600_b464e304_Q4_202014_20Commentary_FINAL.pdf
275856	commentary	20150206040134_064105d6_Q3_2014_Commentary_FINAL.pdf
322435	commentary	20150306030933_6eb6f39d_Contrarian_Compendium_June_2014.pdf
729006	commentary	20150430230114_85d44bec_Q1_202015_20Commentary_FINAL.pdf
47502	commentary	20150506062604_608d7d68_1-5-09_20Q_A-_20Croupier_20_20Fed_20Tactics.pdf
212501	commentary	20150506064724_7c1778c7_March_2012_Web_commentary_Diversification_Index_Final.pdf
162341	thematic_essay	20150714204413_22ac304f_InternationalDiversification_June2015.pdf
218003	commentary	20150726150135_d6eb36b9_Q2_202015_20Commentary_FINAL.pdf
280122	commentary	20151106120300_687f1bba_Q3_202015_20Commentary_FINAL_PRINT.pdf
278854	commentary	20151123134022_e8a7713c_Q3_202015_20Commentary_FINAL_footnote.pdf
242834	commentary	20151211050836_12c2df56_Global_20Contrarian_20Compendium_20June_202015_.pdf
148242	commentary	20151223141426_0cef613a_Fixed_20Income_20Contrarian_20Compendium_20May_202015.pdf
185320	commentary	20151223142544_4ab46900_Spin-Off_20Compendium_20April_202015.pdf
148451	commentary	20151227091529_c6b51998_2011Q3_commentary.pdf
296321	commentary	20151227092248_7d8a3eae_2Q11_HK_Commentary_Final.pdf
1013404	commentary	20151227092912_582b1b94_Asia_20Opportunity_20Commentary_2Q2015_FINAL.PDF
1333202	commentary	20151227092927_0bb5d4fd_Asia_20Opportunity_20Commentary_3Q2015_FINAL.pdf
185853	commentary	20151227092928_60e0fbe4_Bits_and_Pieces_Feb__202013.pdf
173640	thematic_essay	20151227092942_1d7f4ea5_Dormant_Assets_December.pdf
244912	commentary	20151227093003_1e5a8694_Horizon_Commentary_1stQuarter_2011.pdf
210245	commentary	20151227093012_8e4ba44f_Horizon_Commentary_February_2011.pdf
138347	commentary	20151227093017_4c2c560c_Horizon_Commentary_March_2011.pdf
281849	commentary	20160127171710_9222c5dc_Q4_202015_20Commentary_FINAL.pdf
212816	commentary	20160306203003_e77b0f5a_q3_2013_commentary.pdf
283869	thematic_essay	20160321033708_7a7ab2ea_Owner_Op_Paper.pdf
823267	thematic_essay	20160321033713_a234e824_HK_20Alternative_20Income_20Investing_White_20Paper_4-2-13_Final.p
194709	commentary	20160321033814_afc9ab2e_Fixed_Income_Contrarian_Compendium_November_2012.pdf
210772	commentary	20160321033930_bd3b2943_Devil_s_20Advocate_20Compendium_20March_202015.pdf
229072	commentary	20160321034032_564a97d8_Spin-Off_20Compendium_20November_202014.pdf
244320	commentary	20160321034155_b71f071a_Contrarian_Compendium_October_2013.pdf

Bytes	Tier	PDF filename
224682	commentary	20160321034202_8bd3d800_Spin-Off_20Compendium_20February_202014.pdf
294339	commentary	20160321034232_6152a400_Fixed_20Income_20Contrarian_20Compendium_20December_202013.pdf
280112	commentary	20160321034530_d0de0b70_Devil_s_Advocate_Compendium_September_2012.pdf
169215	commentary	20160321035122_58c2b522_Fixed_Income_Contrarian_Compendium_August_2012.pdf
236025	commentary	20160321035247_027223b8_Global_20Spin-Off_20Compendium_20September_202014.pdf
352884	thematic_essay	20160321035257_1cdcd84f_R2000_20Index_20Construction_20White_20Paper_Dec2013.pdf
55064	thematic_essay	20160321035458_b179b0af_SpinOff_WhitePaper.pdf
241142	commentary	20160321035503_c61fdc4c_Contrarian_20Compendium_20December_202014.pdf
249767	commentary	20160321035518_0d79678c_Fixed_Income_Contrarian_Compendium_September_2013.pdf
213775	commentary	20160321035806_014d59c9_Spin-Off_Compendium_July_2014.pdf
508896	commentary	20160321092209_9df974f0_Fixed_Income_Contrarian_Compendium_May_2014.pdf
155210	thematic_essay	20160323071733_52444169_Bond_20Substitute_June_202014_FINAL.pdf
330816	thematic_essay	20160323071835_6bf4d9d1_Not_20Your_20Grandfather_s_20_20SP_FINAL.pdf
285196	thematic_essay	20160323072107_00f409dc_InternationalDiversification_June2015.pdf
329881	commentary	20160508172527_242067f0_Q1_202016_20Commentary_20Final.pdf
63442	commentary	20160512031900_7a5f9d04_Further_20Croupier_20Basics_20-_20_204-27-06.pdf
271981	commentary	20160512141032_56cd5cc4_Stahl_Spinoffs_Going_20Separate_20Ways_Value_20Investor_20Insight_
129613	commentary	20160512141243_95001167_Contrarian_20Compendium_2011-08.pdf
357858	commentary	20160615083834_e243b3cd_Fixed_20Income_20Contrarian_20Compendium_20December_202015.pdf
61903	commentary	20160703102751_9fadf612_12-31-08_204th_20Quarter_20Commentary.pdf
104359	commentary	20160703103346_878d0b91_Horizon_Commentary_EuroDebt_May2010.pdf
50876	commentary	20160703110258_382b1797_9-30-09_203rd_20Quarter_20Commentary.pdf
47932	commentary	20160703120337_0c457f3c_3-31-09_201st_20Quarter_20Commentary.pdf
70306	commentary	20160703120433_746b4fe8_6-30-09_202nd_20Quarter_20Commentary.pdf
51485	commentary	20160703121650_1b7fcbb3_12-31-09_204th_20Quarter_20Commentary.pdf
36226	commentary	20160703122128_6e323ac8_12-16-08_20Review_20_20Update_20of_20The_20Croupier_20Business_20M
46376	commentary	20160703122136_3bb813aa_10-30-08_20Expanding_20on_20The_203rd_20Quarter_20Commentary.pdf
52551	commentary	20160703154353_b8f56eae_8-21-09_20Croupiers_20Less_20Talk_20More_20Figures.pdf
175790	commentary	20160703164719_4c1a3503_Horizon_Commentary_ChinaProperty_June2010.pdf
288684	commentary	20160703164728_c12c4c87_Horizon_September_2010_Commentary.pdf
184920	commentary	20160703164747_c0852683_Horizon_Commentary_November_2010.pdf
229200	commentary	20160703164803_62dc2d81_Horizon_Commentary_May_2010.pdf
316838	commentary	20160703164807_daf8978b_Horizon_Commentary_1stQuarter_2010.pdf
133399	commentary	20160703164813_6870ce21_Horizon_Q3_2010_Commentary.pdf
910603	commentary	20160803104648_9dc29e69_Q2_202016_20Commentary_20FINAL.pdf
1258550	commentary	20160910064825_a7787580_Asia_20Opportunity_20Commentary_2Q2016_FINAL.pdf
365036	commentary	20161130105919_a406fc2a_Contrarian_20Research_20Report_HRB_Y_DISH_ETM_.pdf
262947	commentary	20161202035613_301123b1_Q3_202016_20Commentary_20FINAL.pdf
1352716	commentary	20170115121926_b7207202_Asia_20Opportunity_20Commentary_4Q2015_FINAL.pdf
987745	commentary	20170115154708_4a139fcf_Asia_20Opportunity_20Commentary_1Q2016_FINAL.PDF
174451	thematic_essay	20170115182625_6bf4d9d1_Not_20Your_20Grandfather_s_20_20SP_FINAL.pdf
188596	core_musings	20201216134009_cb410767_TheAchievementofDiversificationNovember2020.pdf
1551233	commentary	20201226173916_bad9472c_Q3-CVALUE-Review_FINAL.pdf
585902	core_musings	20210116203441_c1999a83_Fixed-Income-and-Cryptocurrency-Compendium-August-2020.pdf
585392	research_report	20210116210043_7587ef02_London-Stock-Exchange-European-Contrarian-6-25-19.pdf
345726	research_report	20210116210703_e41adafe_Penske-Auto-Contrarian-Report-1-10-20.pdf
515809	research_report	20210116211249_50f75403_Carnival-Corp-Stahl-Report-7-9-20.pdf
308984	research_report	20210116211646_73139c81_msg-spin-off-report-february-2019.pdf
579201	research_report	20210116212320_e8a55fcc_AP-Moller-Maersk-April-2-2019.pdf
1048576	commentary	20210116213437_bad9472c_Q3-CVALUE-Review_FINAL.pdf
1816301	commentary	20210126023451_86f6291a_Q4-2020-Review_Final_Approved.pdf
994467	commentary	20210207195706_60691f1d_Quarterly-Commentary_Q2-2020.pdf

Bytes	Tier	PDF filename
847749	commentary	20210210012230_021ea65c_Q1-2020-CVALUE-Review_FINAL.pdf
187558	thematic_essay	20210211205259_c4239bdc_Revisiting-an-Old-Friend-_E2_80_93-The-Thesis-for-a-Defensive-De-M Investment-in-Bitcoin_Feb-2021.pdf
262016	thematic_essay	20210325003922_c485e1a8_Outlawing-Cryptocurrency_Feb-2021_final-clean.pdf
1501600	commentary	20210424015609_ff69eadf_Q1-2021-Review_FINAL.pdf
362313	thematic_essay	20210509095550_c485e1a8_Outlawing-Cryptocurrency_Feb-2021_final-clean.pdf
1048576	commentary	20210509101542_ff69eadf_Q1-2021-Review_FINAL.pdf
1048576	commentary	20210509103854_86f6291a_Q4-2020-Review_Final_Approved.pdf
1003880	thematic_essay	20210511221915_684933d9_Horizon-Kinetics-Japan-Special-Opportunity-White-Paper_April-2021_5.pdf
947303	commentary	20210622192511_d45f7d39_Q2-2019-Horizon-Kinetics-Commentary-_Final-1.pdf
108294	letters	20210622194432_be2bb41f_Letter-from-our-Founders_03252020.pdf
40737	thematic_essay	20210622195755_29f44b80_June-2016_The-Rise-of-the-Artificial-Asset-Class.pdf
1048576	commentary	20210622200151_5546c698_Q3-2019-CVALUE-Review_FINAL.pdf
1048576	commentary	20210622205904_b651e898_Q4-2019-Horizon-Kinetics-Commentary_FINAL.pdf
1048576	commentary	20210622205942_6d15fac8_Q1-2019-CVALUE-Review.pdf
2144978	commentary	20210725074629_7b732ff2_Q2-CVALUE-Review_FINAL.pdf
1850521	commentary	20210729054313_5546c698_Q3-2019-CVALUE-Review_FINAL.pdf
2143184	commentary	20210729054356_b651e898_Q4-2019-Horizon-Kinetics-Commentary_FINAL.pdf
1210089	commentary	20210729054817_6d15fac8_Q1-2019-CVALUE-Review.pdf
465600	commentary	20210729055106_9519fa9b_Asia-commentary-1Q2021_APPROVED.pdf
1004012	thematic_essay	20210804112555_684933d9_Horizon-Kinetics-Japan-Special-Opportunity-White-Paper_April-2021_5.pdf
1048576	commentary	20210922060417_7b732ff2_Q2-CVALUE-Review_FINAL.pdf
2098222	commentary	20211022155918_917690ad_Q3-CVALUE-Review_FINAL-1.pdf
1812706	commentary	20220115231144_86f6291a_Q4-2020-Review_Final_Approved.pdf
2536764	commentary	20220124230931_30e9e19f_Q4-2021-Quarterly-Review_FINAL.pdf
2368978	commentary	20220328062400_7b732ff2_Q2-CVALUE-Review_FINAL.pdf
1340265	commentary	20220504194523_0d9d0d67_Q1-2022-Roundtable-Discussion-for-Website_FINAL.pdf
51173	core_musings	20220510031223_1a9e945c_TheTransformationOfTheFixedIncomeAssetClass.pdf
319492	commentary	20220516192711_f817989f_Interim-Q1-Q2-Commentary-2022_APPROVED.pdf
357494	commentary	20220705214729_e64a2bb9_Asia-commentary-1Q2022_APPROVED.pdf
1652632	commentary	20220723173454_7e9ece3a_Q2-2022-Review_FINAL_2.pdf
1652628	commentary	20220729165435_5e3fc16c_Q2-2022-Review_FINAL.pdf
1048576	commentary	20220813060605_7e9ece3a_Q2-2022-Review_FINAL_2.pdf
1048576	commentary	20220813065932_5e3fc16c_Q2-2022-Review_FINAL.pdf
193657	core_musings	20220813070400_348c5522_InflationandEconomicThought_Nov2020.pdf
1652632	commentary	20220902105301_7e9ece3a_Q2-2022-Review_FINAL_2.pdf
311244	thematic_essay	20221005012007_35400d7c_MINING-ECONOMICS-WHAT-DRIVES-THE-BITCOIN-PRICE_Sep-2022_Final-1.pdf
1615989	commentary	20221021150448_e54e6284_Q3-2022-Review-1.pdf
400028	commentary	20221102201744_207c6b89_INFL-Fundamentals-Q3-2022-Final.pdf
299234	thematic_essay	20221204013850_82573773_Mining-Economics-What-Drives-the-Bitcoin-Price_Sep-2022_Final.pdf
1048576	commentary	20221204030647_e54e6284_Q3-2022-Review-1.pdf
1377194	commentary	20230128111302_163baf82_Q4-2022-Commentary_FINAL.pdf
1501600	commentary	20230217024320_ff69eadf_Q1-2021-Review_FINAL.pdf
1340265	commentary	20230227072517_0d9d0d67_Q1-2022-Roundtable-Discussion-for-Website_FINAL.pdf
1551233	commentary	20230227072517_bad9472c_Q3-CVALUE-Review_FINAL.pdf
1615989	commentary	20230227072517_e54e6284_Q3-2022-Review-1.pdf
565063	thematic_essay	20230319230440_ac6000d0_Bitcoin-and-Legal-Tender_December-2017.pdf
1048576	commentary	20230326191840_163baf82_Q4-2022-Commentary_FINAL.pdf
2302317	commentary	20230504220203_689b5ef1_Q1-2023-Commentary_FINAL.pdf
224465	thematic_essay	20230523111445_2def7d8d_COEXISTENCE-OF-FIAT-AND-CRYPTOCURRENCIES.pdf
1377194	commentary	20230531095812_163baf82_Q4-2022-Commentary_FINAL.pdf

Bytes	Tier	PDF filename
161806	core_musings	20230531095935_91881140_Spin-Off-Compendium-October-2022_The-Importance-of-the-Oil-Service-Industry.pdf
1048576	commentary	20230602144125_689b5ef1_Q1-2023-Commentary_FINAL.pdf
2302317	commentary	20230606043711_689b5ef1_Q1-2023-Commentary_FINAL.pdf
1652628	commentary	20230606044412_5e3fc16c_Q2-2022-Review_FINAL.pdf
2536764	commentary	20230606053316_30e9e19f_Q4-2021-Quarterly-Review_FINAL.pdf
319596	thematic_essay	20230630134022_5c84bf44_Under-the-Hood_June-2023.pdf
2487998	commentary	20230801084049_2c86d07a_Q2-23-Commentary_FINAL.pdf
2487998	commentary	20231013015731_2c86d07a_Q2-23-Commentary_FINAL.pdf
1048576	commentary	20231201030746_a0b21484_Q3-Commentary-FINAL.pdf
178026	thematic_essay	20231201032448_f9d677a5_VII_Fertile-Ground_July-2023.pdf
186988	letters	20240114124053_9a65a3e7_Horizon-Kinetics-Letter-from-Founders_Jan-2024_APPROVED.pdf
1048576	commentary	20240303121118_dfd39310_Q4-2023-Commentary_FINAL.pdf
1048576	commentary	20240303125629_3e1613ec_Q3-Commentary-FINAL.pdf
1672041	commentary	20240308002456_dfd39310_Q4-2023-Commentary_FINAL.pdf
621204	thematic_essay	20240329003752_1cf4eac6_VII_Worth-The-Time_Stahl_Feb-2024.pdf
3098088	commentary	20240509222013_1402e3d3_Horizon-Kinetics-Q1-2024-Commentary.pdf
1048576	commentary	20240522092508_1402e3d3_Horizon-Kinetics-Q1-2024-Commentary.pdf
3024416	commentary	20240715095340_3e1613ec_Q3-Commentary-FINAL.pdf
3098088	commentary	20240718092512_1402e3d3_Horizon-Kinetics-Q1-2024-Commentary.pdf
3998156	commentary	20240808131536_e5f35473_Q2-2024-Horizon-Kinetics-Commentary_FINAL.pdf
830371	letters	20240808144918_28e1d19d_Horizon-Kinetics-Corporate-Update-2024_08022024_Final.pdf
1048576	commentary	20240813015008_e5f35473_Q2-2024-Horizon-Kinetics-Commentary_FINAL.pdf
4043762	commentary	20240825192543_e5f35473_Q2-2024-Horizon-Kinetics-Commentary_FINAL.pdf
6383336	commentary	20241105220421_17214d6d_Horizon-Kinetics-Q3-2024-Commentary_Final.pdf
1812706	commentary	20241202233811_86f6291a_Q4-2020-Review_Final_Approved.pdf
380547	letters	2025- INFBN-Annual-Letter-Final.pdf
6383307	commentary	20250119020326_17214d6d_Horizon-Kinetics-Q3-2024-Commentary_Final.pdf
334310	letters	20250130083656_9c4c3ba5_Horizon-Kinetics-Founders-Letter-2025_Final.pdf
3339037	commentary	20250204143736_11f6c5ca_Horizon-Kinetics-Q4-2024-Commentary_Final.pdf
3024416	commentary	20250219042006_a0b21484_Q3-Commentary-FINAL.pdf
3407745	commentary	20250504131037_7f75300b_Q1-2025-Horizon-Kinetics-Commentary_Final.pdf
3339037	commentary	20250528193622_11f6c5ca_Horizon-Kinetics-Q4-2024-Commentary_Final.pdf
2420283	commentary	20250823223649_d0492631_Horizon-Kinetics-Q2-2025-Commentary_Final.pdf
704893	thematic_essay	20250914113336_351813fb_JAPN-Portfolio-Highlight-6544JP-Japan-Elevator-Service-Holding_Aug 2025_FINAL.pdf
3930765	commentary	20251031062933_a3f8a8eb_Horizon-Kinetics-Q3-2025-Commentary-Final.pdf
6805699	thematic_essay	830350067-Compilation-of-Murray-Stahl-s-Writings.pdf
319480	letters	BCDF-2023-Annual-Letter.pdf
288451	letters	BCDF-Annual-Letter-2022-Final-APPROVED.pdf
309731	letters	BCDF-Annual-Shareholder-Letter-2025-Final.pdf
265573	thematic_essay	BITCOIN-CO-EXISTING-WITH-FIAT-CURRENCY-AND-THE-BANKING-SYSTEM_Nov-2020_FINAL.pdf
220617	thematic_essay	BITCOIN-ELECTRICITY-CONSUMPTION_April-2023_Final.pdf
503931	commentary	Bitcoin-Supply-Over-Time-Commentary_June-2020_final-clean.pdf
360902	thematic_essay	Bitcoin-newsletter_October-2017.pdf
297920	thematic_essay	CFTC-Testimony_Feb-2018_Final.pdf
370094	thematic_essay	CRYPTOCURRENCY-BY-THE-NUMBERS_November-2025_Final.pdf
192490	core_musings	Contrarian-Compendium-December-2022_Diamonds-as-an-Investment.pdf
1201549	research_report	Credit-Acceptance-Corp-Devils-Advocate-Report-December-2-2019.pdf
210279	thematic_essay	Cryptocurrency-Mainstream-Legitimacy_A-Brief-Survey-of-Institutional-Acceptance_April-2020
352227	thematic_essay	Cryptocurrency-and-Exchanges_July-2025_final.pdf
35230	thematic_essay	December-2015_Cryptocurrencies-An-Emerging-Asset-Class.pdf
189684	core_musings	Devils-Advocate-Compendium-February-2023_Trends-in-Healthcare.pdf

Bytes	Tier	PDF filename
181945	core_musings	Devils-Advocate-Compendium-November-2022_Bitcoin-vs.-Bonds.pdf
170872	core_musings	European-Contrarian-Compendium-January-2023_Wealth-and-Crisis_The-Role-of-Diamonds-in-Asset-Allocation.pdf
405929	letters	Founders-Letter-2026_Final.pdf
875014	thematic_essay	GIRO_4.24.26_His-stream-of-mind-was-perpetual.pdf
523135	thematic_essay	GIRO_5.28.21_Reveille-in-Japan.pdf
547114	thematic_essay	GIRO_9.13.24_Inflation-has-legs.pdf
216002	letters	HK-Letter-from-the-Board_April-2026_Final.pdf
334310	letters	Horizon-Kinetics-Founders-Letter-2025_Final.pdf
3308027	commentary	Horizon-Kinetics-Q1-2026-Commentary-Final.pdf
2420283	commentary	Horizon-Kinetics-Q2-2025-Commentary_Final.pdf
3930765	commentary	Horizon-Kinetics-Q3-2025-Commentary-Final.pdf
3339037	commentary	Horizon-Kinetics-Q4-2024-Commentary_Final.pdf
4155467	commentary	Horizon-Kinetics-Q4-2025-Commentary-Final.pdf
441814	thematic_essay	Horizon-Kinetics_Royalty-As-an-Asset-Class_April-2023-4.pdf
384118	letters	INFBN-Semi-Annual-Letter-2023-1.pdf
311082	letters	INFL-2024-Annual-Letter_Final_Feb2025.pdf
495920	letters	INFL-2025-Annual-Shareholder-Letter_Final.pdf
863679	letters	INFL-2025-Semi-Annual-Letter.pdf
465983	thematic_essay	INFL-5-Year-Anniversary-Update_Jan-2026.pdf
200413	letters	INFL-Annual-Letter-FINAL_2021.pdf
656032	thematic_essay	INFL-Portfolio-Positioning-Piece_Dec-2024_Final.pdf
387482	letters	INFL-Semi-Annual-Letter-2023_APPROVED.pdf
602984	letters	INFL_2023_Annual-Letter.pdf
611313	letters	INFL_2024-Semi-Annual-Letter_Final.pdf
641928	letters	INFL_Annual-Letter_2022_Final-Approved.pdf
574945	commentary	INFL_SemiAnnual-Commentary_FINAL.pdf
193657	core_musings	InflationandEconomicThought_Nov2020.pdf
420732	commentary	Interim-Q1-Q2-Commentary-2022-with-TPL-addendum_FINAL.pdf
337915	commentary	NVIR-Semi-Annual-2024-commentary_Final.pdf
3407745	commentary	Q1-2025-Horizon-Kinetics-Commentary_Final.pdf
187558	thematic_essay	Revisiting-an-Old-Friend---The-Thesis-for-a-Defensive-De-Minimis-Investment-in-Bitcoin_Feb
229192	commentary	SPAQ-Semi-Annual-2024-Commentary_Final.pdf
161806	core_musings	Spin-Off-Compendium-October-2022_The-Importance-of-the-Oil-Service-Industry.pdf
188596	core_musings	TheAchievementofDiversificationNovember2020.pdf
51173	core_musings	TheTransformationOfTheFixedIncomeAssetClass.pdf
986642	thematic_essay	Why-JAPN_November-2025_Final.pdf

Extract exceptions

See `book/sources/extraction_missing_yes.txt` for PDFs without a usable `.txt` under `book/build/text/` after the `pypdf` → `PyMuPDF` → `duplicate-repair` pipeline.

Editorial consistency pass (checklist)

Use this file as a **single pass** before treating the manuscript as “frozen.” Check off items as you complete them.

Names, tickers, and institutions

- ☐ Horizon Kinetics / affiliate names match originals in each headnote.
- ☐ Ticker symbols and company names: pick one spelling (U.S. vs U.K.) per entity across chapters; align with the **original publication date** (do not silently modernize tickers that changed).
- ☐ “Murray’s Musings” vs “Murray’s musing” / section headers: match original casing in headnotes.

Citations and cross-references

- ☐ **In-text cross-links:** bond convexity, index concentration, inflation, diamonds, Bitcoin, oil services — add “see Chapter ...” only where it aids the reader (optional for v1).
- ☐ **External URLs:** verify Horizon pages still resolve; Wayback links noted in 97-note-on-textual-variants.md where used.

Style

- ☐ One dash style (em dash vs hyphen) per chapter after paste-from-PDF cleanup.
- ☐ Tables recreated from originals: spot-check numbers against PDF for any chapter you intend to publish commercially.

Part VIII

- ☐ Every video row: transcript or explicit deferral.
- ☐ Bloomberg item: confirm embed or transcript path.

Extraction exceptions

- ☐ `book/sources/extraction_missing_yes.txt` — should list only items needing **manual** remediation. Current known blockers: password-protected PDF (see file).

Last reviewed: edition metadata in `book/EDITION.md`.

Note on textual variants

Some compendium PDFs exist in **multiple vault filenames** (different crawl timestamps) for the same underlying hash suffix. For drafting, this edition prefers **UTF-8 extracts** under `book/build/text/`, using the **manifest filename stem**; where a vault copy was truncated or empty, text may have been **repaired from a duplicate PDF** with the same content hash (see `book/scripts/repair_extract_from_duplicates.py`).

Web vs PDF: if you recovered a document from the Wayback Machine or a live site, say so in the chapter headnote.

OCR: none of the Part I Musings chapters in this batch were OCR-only; if you add scanned letters, label OCR lineage in the headnote.

Encrypted PDFs: if `extraction_missing_yes.txt` lists a file, the PDF may be print-restricted or passworded — supply a clean export or OCR rather than guessing a password.

1. At the time of this writing, a 30-year Treasury bond yields 2.22%. If the yield of that bond increases to 5%, it would produce 2.25x more yield. Increasing the 3.31% yield of the Bank of America bonds by 2.25x produces a yield of 7.45%. A bond with a 4% coupon that yields 7.45% would trade below par.↵[□](#)
2. The source compendium uses the ticker symbol “APPL”; Apple’s ordinary listing on Nasdaq is **AAPL**.↵[□](#)
3. Robert Skidelsky, *John Maynard Keynes*, Vol. 1: *Hopes Betrayed, 1883–1920* (New York: Viking, 1983); Vol. 2: *The Economist as Savior, 1920–1937* (New York: Viking, 1983); Vol. 3: *Fighting for Freedom, 1937–1946* (New York: Viking, 2001).↵[□](#)
4. Milton Friedman & Anna Jacobson Schwartz, *A Monetary History of the United States, 1867–1960* (Princeton, N.J.: Princeton University Press for the National Bureau of Economic Research, 1963).↵[□](#)
5. Friedrich A. Hayek, *Denationalisation of Money: The Argument Refined – An Analysis of the Theory and Practice of Concurrent Currencies* (London: The Institute of Economic Affairs, 1976).↵[□](#)
6. John Kenneth Galbraith, *Money: Whence It Came, Where It Went* (Boston: Houghton Mifflin, 1975).↵[□](#)
7. See minneapolisfed.org.↵[□](#)
8. John F. Wasik, *Keynes’s Way to Wealth: Timeless Investment Lessons from the Great Economist* (New York: McGraw Hill Education, 2014).↵[□](#)
9. [Statista.com](https://www.statista.com)↵[□](#)
10. Security Industry Financial Markets Association (SIFMA).↵[□](#)

11. Bitinfo.com↵□
12. Energy Information Administration.↵□
13. Norman Angell, *The Story of Money* (New York: Frederick A. Stokes, 1929), 110–112.↵□
14. As cited in the original: www.economist.com/content/global_debt↵□
15. As in original: <http://www.ishares.com/us/products/239563/ishares-high-dividend-etf> (June 10, 2015).↵□